

Characters of p -adic groups

Cheng-Chiang Tsai
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0. NOTATIONS THROUGHOUT THE COURSE

Let p be a prime number. Let $\mathbb{Q}_p$ be the p -adic completion of $\mathbb{Q}$. Let F be a finite extension of $\mathbb{Q}_p$. We will denote by $\mathcal{O} = \mathcal{O}_F$ the ring of integers in F , by $\mathfrak{m} = \mathfrak{m}_F \triangleleft \mathcal{O}_F$ the maximal ideal and by $k = k_F := \mathcal{O}_F/\mathfrak{m}_F$ the residue field. We denote by $q := \#k$. Sometimes we take an algebraic closure $\bar{F}/F$. Let $\mathbb{G}$ be a connected reductive linear algebraic group over F (see §2 for reviews). We will denote the identity element(s) by $e \in \mathbb{G}$, and $e \in G$, and in any group. Let $\mathcal{C}$ be any field of characteristic **different** from p . More assumptions on $\mathcal{C}$ will be added in individual sections, and for a quick reading it might be convenient to assume that $\mathcal{C}$ is algebraically closed.

Let $G = \mathbb{G}(F)$, in particular a p -adic Lie group. We write $\mathfrak{g} := T_e G = (T_e \mathbb{G})(F)$ the Lie algebra. It has a Lie bracket map

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$$

see §2 for a review on this. We will denote by $\mathfrak{g}^* := \text{Hom}_F(\mathfrak{g}, F)$ the linear dual space of $\mathfrak{g}$. There is an adjoint action $\text{Ad} : G \rightarrow \text{End}_F(\mathfrak{g})$. We denote by Ad^* the composition $\text{Ad}^* : G \rightarrow \text{End}_F(\mathfrak{g}) \cong \text{End}_F(\mathfrak{g}^*)$.

1. INTRODUCTION

To begin with, most of the material in this course originates from [HC99], with partial revisions in [Kot05], and with further adaptations of our own, partly inspired by [VW01]. We thank Stephen DeBacker for extremely helpful discussions on the subject. We thank Gemini and ChatGPT for many useful suggestions.

Let us begin with a finite group H . We consider a complex representation $\pi \in \text{Rep}(H)$, i.e. there is an underlying complex vector space V , and $\pi : H \rightarrow \text{Aut}_{\mathbb{C}}(V)$ is a group homomorphism. It is known that if π is irreducible then it is finite-dimensional. If it is finite-dimensional, we can define its character

$$\begin{aligned} \Theta_\pi : H &\rightarrow \mathbb{C} \\ h &\mapsto \text{tr}(\pi(h)). \end{aligned}$$

Characters Θ_π for isomorphism classes of irreducible representations (π, V) form an orthonormal basis of over H . Every finite-dimensional representation (π, V) of H is the direct sum of a finite number of irreducible representations, and therefore the isomorphism class of (π, V) is determined by Θ_π . Similar statement is true if we replace “finite group” by “compact Hausdorff topological group,” and “representation” by “continuous representation.”

Now we begin with a p -adic reductive group $G = \mathbb{G}(F)$; if you have not seen them, you are encourage to think of the example

$$G = \text{GL}_n(\mathbb{Q}_p) = \{g \text{ is an } n \times n \text{ matrix in } \mathbb{Q}_p \mid \det(g) \neq 0\}.$$

equipped with the topology inherited from $(\mathbb{Q}_p)^{n^2}$. In particular G is (always) a totally disconnected locally compact Hausdorff topological group. We would like a representation theory of G similar to that of compact Hausdorff groups. The colloquial way of saying it is that

p -adic (reductive) groups try very hard to behave like finite (reductive) groups.

In some sense the goal of the course is to address “How?” from the point of view of characters. To begin with, let us ask what kind of continuity is reasonable. Any neighborhood of $\text{id} \in G$ contains an open (pro- p) subgroup of G . Using it we can prove

Exercise 1. *Let H be a p -adic Lie group. Let $\mathcal{C} = \mathbb{C}$ or $\mathcal{C} = \overline{\mathbb{Q}_\ell}$ with its usual topology. Let n be an integer and $\rho : H \rightarrow \text{GL}_n(\mathcal{C})$ be a continuous group homomorphism. Show that $\ker(\rho)$ contains an open subgroup.*

For the rest of the introduction, $\mathcal{C}$ is a fixed algebraically closed field of characteristic different from p .

Definition 1.1. *Consider a $\mathcal{C}$ -representation (π, V) (i.e. V is a vector space over $\mathcal{C}$, often infinite-dimensional) of a p -adic Lie group H . A vector $v \in V$ is **smooth** if its stabilizer $\text{Stab}_H(v) \subset H$ is open. A representation is called **smooth** if every vector is smooth.*

From now on, unless otherwise stated, representations will always be smooth $\mathcal{C}$ -representations of G . All maps on $\mathcal{C}$ -representations are assumed to be $\mathcal{C}$ -linear.

Definition 1.2. *A smooth representation (π, V) is called **admissible** if $\dim_{\mathcal{C}} \pi^J < +\infty$ for any open subgroup $J \subset G$.*

The next theorem is a big one. We will not attempt to prove it.

Theorem 1.3. *(Jacquet, Vignéras) Every irreducible smooth representation of G is admissible.*

Most of this course will be about irreducible admissible representations. Because of Theorem 1.3, they are the same as irreducible smooth representations, and thus are natural objects of study. Maybe this is a good time to pause, and ask why we care about representations of G . I can think of a few reasons:

- (1) A p -adic reductive group G acts on automorphic forms, with coefficient ring $\mathcal{C} = \mathbb{C}, \mathbb{Q}_\ell, \mathbb{F}_\ell, \mathbb{Z}_\ell$, etc.. Such automorphic forms live in representations that can be decomposed into irreducible admissible representations of (various) G .
- (2) There is a subject, of fundamental importance to mathematics, called Langlands program. It relates properties about representations of G to structures in number theory, algebraic geometry, and mathematical physics.
- (3) Representation theory of G are connected to that of affine Hecke algebras and double affine Hecke algebras, of quantum groups, and to the geometry of affine Grassmannians and various related moduli spaces.
- (4) Probably because of the above reasons, representation theory of p -adic reductive groups is itself rich and fun.

Despite so many fancy stuff mentioned, this course will be content to develop a character theory for a p -adic reductive group G , developed by Harish-Chandra and a few others mostly in the 70's. The role it plays is similar to the basic character theory of finite groups developed in the 1900's by Frobenius and others; a foundation for characters of irreducible representations is provided, but without a method to actually determine the characters. In particular we will see very little (if any) connection to the fancy stuff just mentioned.

Let's go back to characters. Already in the case of compact groups, e.g. $H = \mathbb{Z}_p$, to say that characters form an orthonormal basis of functions we need to choose a measure on the group.

Definition 1.4. We denote by $\Omega(G)$ be the set of open compact subgroups of G . We consider M_G the set of functions $\nu : \Omega(G) \rightarrow \mathbb{Z}[1/p]$ such that

$$J, J' \in \Omega(G), J \subset J' \implies \nu(J') = [J' : J] \cdot \nu(J).$$

Lemma 1.5. M_G is a free module of rank 1 over $\mathbb{Z}[1/p]$.

Exercise 2. Show that M_G is indeed a free module of rank 1 over $\mathbb{Z}[1/p]$. Show that any open compact subset $X \subset G$ can be written as a finite union of J -cosets $X = \bigsqcup_{i=1}^m g_i J$ for some $J \in \Omega(G)$. Show that the quantity $\nu(X) := m \cdot \nu(J) \in \mathbb{Z}[1/p]$ depends only on X and ν but not on the choice of g_i and J .

[CC: re-do this measure thing later.]

Let us fix $\nu \in M_G$ so that $M_G = \text{span}_{\mathbb{Z}[1/p]}(\nu)$, thought of our measure on G . It will be clear that what we do below do not depend on the choice. We will discuss some important property of the measures, mostly that $\nu(gJg^{-1}) = \nu(J)$, in later sections. Let X be any topological space. The following definition is most often applied to $X = G$.

Definition 1.6. Denote by $\mathcal{C}^\infty(X)$ the space of locally constant $\mathcal{C}$ -valued functions on X , and by $\mathcal{C}_c^\infty(X) \subset \mathcal{C}^\infty(X)$ the space of compactly supported locally constant $\mathcal{C}$ -valued functions on X .

Definition 1.7. The space of **locally constant compactly supported measures**, sometimes just **test measures**, on G is

$$M_c^\infty(G) := \mathcal{C}_c^\infty(G) \otimes_{\mathbb{Z}[1/p]} M_G = \mathcal{C}_c^\infty(G) \otimes_{\mathcal{C}} (\mathcal{C} \otimes_{\mathbb{Z}[1/p]} M_G).$$

For $\mu = f \otimes \nu$ where $f \in \mathcal{C}_c^\infty(G)$, we will talk about left, right and conjugation action of G on μ via that on f .

Remark 1.8. As the name suggest, any $\mu = f \otimes \nu \in M_c^\infty(G)$ is sort of a measure; for any $f' \in \mathcal{C}^\infty(G)$, we define

$$(1) \quad \mu(f') := \left(\sum_{g_i \in G/J} (ff')(g_i) \right) \cdot \nu(J) \in \mathcal{C}$$

where $J \in \Omega(G)$ is such that ff' is right J -invariant. The quantity $\mu(f')$ is independent of the choice of J .

Let π be a smooth representation. Whenever $X \subset G$ is open compact, we denote by $1_X \in \mathcal{C}_c^\infty(G)$ the function that takes the value 1 on X and 0 elsewhere. Consider any $\mu \in M_c^\infty(G)$ such that

$$\mu = \left(\sum_{i=1}^m c_i 1_{g_i J} \right) \otimes \nu$$

for some $J \in \Omega(G)$, $g_i \in G$ and $c_i \in \mathcal{C}$. We define

$$\pi(\mu)_J : \pi^J \rightarrow \pi, \quad \pi(\mu)_J v := \int_G \pi(g)v \cdot \mu(g) := \left(\sum_{i=1}^m c_i \pi(g_i)v \right) \cdot \mu(J)$$

One verifies that when $J \subset J'$, we have $\pi(\mu)_J|_{\pi^{J'}} = \pi(\mu)_{J'}$. Hence the maps can be glued to be $\pi(\mu) : \pi \rightarrow \pi$. For $g \in G$, denote by $l_g(f) \in \mathcal{C}_c^\infty(G)$ given by $(l_g(f))(h) := f(g^{-1}h)$, and $l_g(\mu) = l_g(f) \otimes \nu$. Then we have

$$l_g(f) = \sum_{i=1}^m c_i 1_{gg_i J} \implies \pi(l_g(\mu)) = \pi(g) \circ \pi(\mu) \in \text{End}(\pi)$$

Observe that there exists $J_\mu \in \Omega(G)$ such that $\mu = l_g(\mu)$ for any $g \in J_\mu$. Then $\pi(g)\pi(\mu)v = \pi(\mu)v$ for any $g \in J_\mu$, i.e. $\pi(\mu)v \in \pi^{J_\mu}$. Let us highlight

Lemma 1.9. *Suppose π is admissible. Then $\dim_{\mathcal{C}} \text{im}(\pi(\mu)) < +\infty$ for any $\mu \in M_c^\infty(G)$.*

Definition 1.10. *Suppose π is admissible. Then the $\mathcal{C}$ -linear functional*

$$\begin{aligned} \Theta_\pi : M_c^\infty(G) &\rightarrow \mathcal{C} \\ \mu &\mapsto \text{tr}(\pi(\mu)) \end{aligned}$$

*is called the **character** of π .*

Example 1.11. Suppose $\dim_{\mathcal{C}} \pi < +\infty$. For $\mu = f \otimes \nu$ we have

$$(2) \quad \Theta_\pi(f \otimes \nu) = \int_G \text{tr}(\pi(g))\mu(g) := \left(\sum_{g_i \in G/J} \text{tr}(\pi(g_i))f(g_i) \right) \cdot \nu(J) = \text{tr}(\mu(\pi))$$

where $J \in \Omega(G)$ is such that $\pi = \pi^J$ and that μ is left J -invariant and $\text{supp}(\mu) = \bigsqcup_i Jg_i$. In this case we usually say Θ_π is represented by the function $g \mapsto \text{tr}(\pi(g))$.

For $g \in G$ and $\mu = f \otimes \nu \in M_c^\infty(G)$ with $f \in \mathcal{C}_c^\infty(G)$, denote by $c_g(f) \in \mathcal{C}_c^\infty(G)$ the function $(c_g(f))(h) = f(g^{-1}hg)$ and $c_g(\mu) := c_g(f) \otimes \nu$.

Definition 1.12. *A **generalized function** on G is a linear functional $\Theta : M_c^\infty(G) \rightarrow \mathcal{C}$. An **invariant generalized function** is a generalized function $\Theta : M_c^\infty(G) \rightarrow \mathcal{C}$ such that $\Theta(c_g(\mu)) = \Theta(\mu)$ for any $g \in G$, $\mu \in M_c^\infty(G)$.*

In particular, when π is irreducible admissible, our character Θ_π is an invariant generalized function.

Remark 1.13. For $f \in \mathcal{C}_c^\infty(G)$ and $\mu = f \otimes \nu \in M_c^\infty(G)$, we have that $\text{supp}(f) = \text{supp}(\mu)$ is always open. Suppose $K \subset G$ is open compact, we denote by $f|_K \in \mathcal{C}_c^\infty(K) \subset \mathcal{C}_c^\infty(G)$ the obvious restriction, and also $\mu|_K := (f|_K) \otimes \nu$. When Θ is a generalized function, we denote by $\Theta|_K$ the “restricted” generalized function given by $\Theta|_K(\mu) := \Theta(\mu|_K)$.

Naturally, one expects that the theory for Θ_π for irreducible smooth (thus admissible) representations is important for the study of these representations. It is however

difficult to directly operate (invariant) generalized functions. Ideally, we would like a “function” t_π on G such that for any $\mu \in M_c^\infty(G)$ we have

$$\Theta_\pi(\mu) = \int_G t_\pi(g) \mu(g) = \mu(t_\pi)$$

in appropriate sense, generalizing (2). However, recall that for complex representation of finite groups, $\text{tr}(\pi(e)) = \dim_{\mathbb{C}}(\pi)$. Hence if $\dim_{\mathbb{C}}(\pi) = +\infty$, we are supposed to have $t_\pi(e)$ blows up to infinity.

In classical analysis, useful generalized function (e.g. those in $L^2(\mathbb{R})$) are often defined almost everywhere. Similar phenomenon happens here. We will define a non-zero analytic (in fact algebraic) function [\[CC: reference to a later section to be inserted\]](#)

$$D_G : G \rightarrow F$$

and

$$\mathfrak{g}^{\text{rs}} := \{g \in G \mid D_G(g) \neq 0\} \subset G$$

the **regular semisimple locus**, that is open, dense, and such that $G \setminus \mathfrak{g}^{\text{rs}}$ has measure zero. This gives an injection

$$\mathcal{C}_c^\infty(\mathfrak{g}^{\text{rs}}) \hookrightarrow \mathcal{C}_c^\infty(G)$$

Our first main goal is

Theorem A. (*Harish-Chandra*) [Corollary 5.37] *Suppose π is irreducible admissible. Then there exists a function $t_\pi \in \mathcal{C}^\infty(\mathfrak{g}^{\text{rs}})$ such that for any μ with $\text{supp}(\mu) \subset \mathfrak{g}^{\text{rs}}$ we have*

$$\Theta_\pi(\mu) = \mu(t_\pi)$$

in the sense of (1).

As discussed, at least when $\mathcal{C} = \mathbb{C}$ the function t_π is doomed to blow up near $e \in G$. Hence t_π should **not** extend to a locally constant function on G , and consequently $\mu(t_\pi)$ cannot be directly defined if $e \in \text{supp}(\mu)$. However, our second main goal is to show that

Theorem B. (*Harish-Chandra*) *Suppose $\mathcal{C} = \mathbb{C}$ and π is irreducible admissible. Then*

- (1) *The function $t_\pi \cdot |D_G|^{1/2}$ is **locally bounded**; for any compact subset $E \subset G$ there exists $C_{\pi,E} \in \mathbb{R}$ such that*

$$g \in \mathfrak{g}^{\text{rs}} \cap E \implies t_\pi(g) \cdot |D_G(g)|^{1/2} \leq C_{E,\pi}$$

- (2) *t_π **locally integrable**, meaning that for any $\nu \in M_G$, any open compact set $K \subset G$ and any sequence of increasing open compact subsets K_n with $K \cap \mathfrak{g}^{\text{rs}} = \lim_{n \rightarrow +\infty} K_n$, we have that*

$$\nu(t_\pi|_{K_n}) \in \mathbb{C}$$

converges absolutely. Consequently, for any $\mu \in M_c^\infty(G)$ we may define

$$\mu(t_\pi) := \lim_{n \rightarrow +\infty} (\mu|_{K_n})(t_\pi)$$

and it is independent of the choice of K_n .

(3) Moreover, t_π represents Θ_π , meaning that

$$\Theta_\pi(\mu) = \mu(t_\pi)$$

for any $\mu \in M_c^\infty(G)$.

Remark 1.16. Theorem B is not as analytical as it might seem. Modulo certain coefficients in $\mathbb{C}$ - in fact often algebraic numbers - what essentially appears in

$$\lim_{n \rightarrow +\infty} (\mu|_{K_n})(t_\pi) \rightarrow \Theta_\pi(\mu)$$

is a sequence of rational numbers converging to another rational number. It's often of the flavor

$$\frac{p}{p-1} = 1 + \frac{1}{p} + \frac{1}{p^2} + \dots$$

In this sense it is possible to make sense of this for any $\mathcal{C}$ with $\text{char}(\mathcal{C}) = 0$.

Exercise 3. Let $G = \text{GL}_2(\mathbb{Q}_p)$, which acts on $\mathbb{P}^1(\mathbb{Q}_p)$ in a natural way. Consider the space $\pi = \mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ on which G also acts. Show that π is generated by a single vector, i.e. there exists $v \in \mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ such that

$$\mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p)) = \text{span}_{\mathcal{C}} \langle g.v \mid g \in \text{GL}_2(\mathbb{Q}_p) \rangle.$$

Our proof for Theorem A, B and C will actually work with admissible representations generated by finitely many vectors. In the next exercise we motivate part of Theorem B.

Exercise 4. (*) Let $G = \text{GL}_2(\mathbb{Q}_p)$, which acts on $\mathbb{P}^1(\mathbb{Q}_p)$ in a natural way, which acts on $\mathbb{P}^1(\mathbb{Q}_p)$ in a natural way. Consider the space $\pi = \mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ on which G also acts. For $g \in \text{GL}_2(\mathbb{Q}_p)$ with distinct eigenvalues λ_1, λ_2 , we show that

$$t_\pi(g) = \begin{cases} 0 & \text{when } \lambda_1, \lambda_2 \notin \mathbb{Q}_p \text{ (but in a quadratic extension),} \\ \frac{|\lambda_1| + |\lambda_2|}{|\lambda_1 - \lambda_2|} & \text{when } \lambda_1 \neq \lambda_2 \in \mathbb{Q}_p. \end{cases}$$

See the exercise sheet for more detail.

To prove Theorem B, we will prove our third main goal that

Theorem C. (Howe, Harish-Chandra) Let $g \in G$. There exist a finite-dimensional space F_g of generalized functions on G , such that for any irreducible admissible representation π , there exist $\phi_{\pi,g} \in F_g$ and an open compact subset $U_{\pi,g} \subset G$ containing g such that

$$\Theta_\pi|_{U_{\pi,g}} = \phi_{\pi,g}|_{U_{\pi,g}}$$

The dimension $\dim_{\mathcal{C}} F_g$ is bounded by a constant that depends only on G but not on g .

Note that we don't need to assume $\mathcal{C} = \mathbb{C}$ nor assume $\text{char}(\mathcal{C}) = 0$ in Theorem C. The elements in F_g need only be defined on some neighborhood of g . On such a neighborhood we will perform **Fourier transform**, and show that we can represent the results as invariant distributions on some **dual nilpotent cone**, which is one point in the special case when $g \in \mathfrak{g}^{\text{rs}}$, thereby proving Theorem A.

After that we prove Theorem B by proving the corresponding property for these Fourier transforms, which will be another highly nontrivial task.

When $\text{char}(\mathcal{C}) = \ell > 0$ (recall that in this case we assume $\ell \neq p$), the classical theory of $\mathcal{C}$ -representations of finite groups was developed particularly by Brauer. More precisely, let $R := W(\mathcal{C})$ be the Witt ring over $\mathcal{C}$. Then a finite-dimensional semisimple $\mathcal{C}$ -representation of a finite group is determined by its Brauer character¹, which is an R -valued function on the finite group. For our p -adic reductive group G , we can also define a Brauer character $\tilde{\Theta}_\pi$, an R -linear functional on $R_c^\infty(G) \otimes_{\mathbb{Z}[1/p]} M_G$. We have

Theorem 1.18. (*Vignéras, Dat, Tsai*) *The same statement of Theorem C holds for $\tilde{\Theta}_\pi$. With any fixed choice of ring embedding $R = W(\mathcal{C}) \hookrightarrow \mathbb{C}$, the statements of Theorem B also holds for $\tilde{\Theta}_\pi$.*

When $\text{char}(\mathcal{C}) = 0$ and $g = e \in G$ in either Theorem C, or when $\text{char}(\mathcal{C}) > 0$ and $g = e$ in Theorem 1.18, the generalized function $\phi_{\pi,e}$ controls (some of) the so-called degenerate Whittaker models for π , in particular the Whittaker models for π . This is the important theorem of Rodier [Rod75], Mœglin–Waldspurger [MW87] and Varma [Var14]. We also hope to talk about this if we have time.

2. REVIEWS ON F , $\mathbb{G}$ AND $G = \mathbb{G}(F)$

2.1. $\mathbb{Q}_p$ and its finite extensions. We recommend the first two chapters in *Local Fields and Their Extensions* by Fesenko–Vostokov, <https://ivanfesenko.org/wp-content/uploads/2021/10/vol.pdf>, for a reference.

Recall that p is a fixed prime number. The field $\mathbb{Q}$ of rational numbers has a norm (with values in $p^{\mathbb{Z}} \sqcup \{0\}$) given by

$$|0|_p := 0, \quad |p^v \cdot \frac{a}{b}|_p := p^{-v} \text{ for any } v \in \mathbb{Z} \text{ and } a, b \in \mathbb{Z} \text{ with } \gcd(a, p) = \gcd(b, p) = 1.$$

This norm is called the p -adic norm on $\mathbb{Q}$. We define $\mathbb{Q}_p$, the field of p -adic numbers, to be the completion of $\mathbb{Q}$ under it (as a metric space). This makes $\mathbb{Q}_p$ a locally compact Hausdorff normed field, just like $\mathbb{R}$. We denote by $\mathbb{Z}_p$ the closure of $\mathbb{Z}$ in $\mathbb{Q}_p$.

Let $F/\mathbb{Q}_p$ be any finite extension. We will review properties about F . We have to be minimally very familiar with the case $F = \mathbb{Q}_p$, and in fact in this course all results are equally strong when assuming $F = \mathbb{Q}_p$; see Remark 2.29. Let $\mathcal{O} = \mathcal{O}_F$ be those elements in F that are integral over $\mathbb{Z}_p$. Then

- (1) $\mathcal{O}_F$ is a local ring. It has a principal maximal ideal $\mathfrak{m} = \mathfrak{m}_F$, and its residue field $k = k_F := \mathcal{O}_F/\mathfrak{m}_F$ is commonly called the residue field of F .
- (2) We have $p \in \mathfrak{m}_F$. The field $k_F \supset \mathbb{Z}/p$ is a finite field. We denote by $q := \#k_F$, a power of p . The quotient ring $\mathcal{O}_F/\mathfrak{m}_F^n$ is a finite ring of order q^n .
- (3) For $x \in F$, define $|x|_F := [x\mathcal{O}_F : \mathcal{O}_F]$ to be the relative index. More precisely, we define
 - (a) If $x = 0 \in F$, then $|x|_F := 0$.
 - (b) If $x \in \mathcal{O}_F$, then $|x|_F := \frac{1}{[\mathcal{O}_F : x\mathcal{O}_F]}$.
 - (c) If $x \notin \mathcal{O}_F$, then $x^{-1} \in \mathcal{O}_F$, and $|x|_F := [x\mathcal{O}_F : \mathcal{O}_F] = \frac{1}{[\mathcal{O}_F : x^{-1}\mathcal{O}_F]}$

¹If we have time to talk about this, we will give some review.

Then $|\cdot|_F : F \rightarrow q^{\mathbb{Z}} \sqcup \{0\} \subset \mathbb{R}_{\geq 0}$ defines a norm. The restriction of $|\cdot|_F$ to $\mathbb{Q}_p$ agrees with $(|\cdot|_p)^{[F:\mathbb{Q}_p]}$. If E/F is another finite extension, then $(|\cdot|_E)|_F = (|\cdot|_F)^{[E:F]}$.

We will often skip the subscript F and just denote the norm by $|\cdot|$. When we refer to topology on F , we always mean the topology induced by $|\cdot|$. The natural map

$$\mathcal{O}_F \rightarrow \varprojlim_n \mathcal{O}_F / \mathfrak{m}_F^n$$

is an isomorphism of topological rings. In particular $(\mathcal{O}_F, +)$ is a pro- p -group, and is totally disconnected. Likewise F is totally disconnected.

- (4) We may define a valuation $\text{val}_F : F \rightarrow \mathbb{Z} \sqcup \infty$ with $\text{val}(x) := -\log_q(|x|)$. This makes F a complete discrete valuation field. In particular $\mathcal{O}_F$ is a PID. Also $\text{val}_F(x, y) \geq \min(\text{val}_F(x), \text{val}_F(y))$ and equivalently $|x + y| \leq \max(|x|, |y|)$.
- (5) For any $n \in \mathbb{Z}$, we have

$$\mathfrak{m}_F^n = \{x \in F \mid \text{val}(x) \geq n\} = \{x \in F \mid |x| \leq q^{-n}\}.$$

- (6) $\mathcal{O}_F$ is open compact under $|\cdot|$. Hence any $\mathfrak{m}_F^n$ ($n \in \mathbb{Z}$) is also open compact.

Remark 2.1. Unlike $\mathbb{R}$, the field $\mathbb{Q}_p$ has (many) infinite algebraic extensions; any algebraic closure of $\mathbb{Q}_p$ has countable but infinite dimension over $\mathbb{Q}_p$.

Remark 2.2. For any algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$, the norms $(|\cdot|_F)^{\frac{1}{[F:\mathbb{Q}_p]}}$ on all finite extensions agree, so that we have a well-defined norm $|\cdot|_p : \overline{\mathbb{Q}_p} \rightarrow \mathbb{R}_{\geq 0}$.

2.2. F -analysis. We recommend [Ser06, Part II, Chapter I and II] for a reference for most results. We have the following analogues of the concepts in introductory real analysis²:

- (1) F^n is a normed space, with $|(x_1, \dots, x_n)| := \max(|x_1|, \dots, |x_n|)$. All norms defined on a finite-dimensional vector space are equivalent in the sense that any two differ by at most a constant, and therefore define the same topology.
- (2) A subset in F^n is compact iff it is sequential compact, iff it is closed and bounded.
- (3) Chain Rule for analytic functions from F^{n_1} to F^{n_2} ; a function is **analytic** if it is locally given by converging power series.
- (4) Inverse Function Theorem and Implicit Function Theorem, for F^n or F -analytic manifolds.
- (5) Definition of differential forms*, that are F -valued on F -analytic manifolds.
- (6) $\mathbb{R}$ -valued Lebesgue integrals of (absolute values of) differential forms using $|\cdot|$. (See however Lemma 2.10 for the version we actually use.
- (7) Fubini's theorem.

We also give a partial list of results that, to the best of my knowledge, has **no** simple analogue over F :

- (1) Connectedness of $\mathbb{R}$ and $\mathbb{R}^n$.
- (2) Riemann integrals.
- (3) Fundamental theorem of calculus, Stokes' theorem.

²the subject that is sometimes called “advanced calculus.”

2.3. F -analytic manifolds. We recommend [Ser06, Part II, Chapter III] for a reference for most results.

Definition 2.3. An n -dimensional F -analytic manifold is a second countable Hausdorff topological space X together with an open covering $X = \bigcup U_i$ and homeomorphisms $c_i : U_i \xrightarrow{\sim} \mathcal{O}_F^n$ such that $c_j|_{U_i \cap U_j} \circ (c_i|_{U_i \cap U_j})^{-1}$ is a converging power series, defined from an open subset of $\mathcal{O}_F^n$ to another.

There is an obvious notion of F -valued functions (locally given by converging power series) on X , and similarly for analytic maps between F -analytic manifolds. There is an obvious notion of the tangent space $T_x X$ for any $x \in X$, which is a vector space of dimension n over F .

Remark 2.4. The dimension is unique as part of this data. This can be quickly seen as follows: if we have two different charts with different dimensions, then we have an invertible analytic map from an open subset of $\mathcal{O}_F^n$ to an open subset of $\mathcal{O}_F^m$, for some $n > m$. This is impossible by the implicit function theorem.

Remark 2.5. However, unlike $\mathbb{R}$, the topological space $\mathbb{Q}_p$ is similar to a Cantor set; there exists a homeomorphism $\mathbb{Q}_p \cong \mathbb{Q}_p \times \mathbb{Q}_p$, or $\mathbb{Z}_p \cong \mathbb{Z}_p \times \mathbb{Z}_p$. (Meanwhile, I think non-trivial algebraic topology can be used to show that $\mathbb{R}^m \not\cong \mathbb{R}^n$ topologically for any $m < n$.) The thing is that such a homeomorphism cannot be given by an analytic map.

Definition 2.6. A (closed) sub-manifold $Y \subset X$ of dimension m for some $m \leq n$ is a closed subset that in local coordinates can be given by $\mathcal{O}_F^m \subset \mathcal{O}_F^n$.

Remark 2.7. Since X , like $\mathcal{O}_F^n$, is totally disconnected, it has many open sub-manifolds of the same dimension. For example $\mathfrak{m}_F^n \subset \mathcal{O}_F^n$ is an open (and also closed) sub-manifold, both of dimension n .

Fix an n -dimensional F -analytic manifold X for the rest of this subsection.

Definition 2.8. Fix $k \in \{0, 1, \dots, n\}$. A differential k -form on an open subset $U \subset X$ is an assignment

$$U \ni u \mapsto \sum_{1 \leq j_1 < \dots < j_k \leq n} f_{\vec{j}} dx_{j_1} dx_{j_2} \dots dx_{j_k}$$

where each $f_{\vec{j}}$ is an analytic F -valued function on a neighborhood of U , and $(x_1, \dots, x_n) : V \xrightarrow{\sim} \mathcal{O}_F^n$ is any coordinate of a chart on an open subset $V \subset X$ covering u .

For different coordinates/charts, differential forms are identified in the obvious way using Jacobians.

Definition 2.9. Suppose X is an n -dimensional F -analytic manifold, and ω is a nowhere-vanishing differential n -form on it. Consider any chart $c : U \xrightarrow{\sim} \mathcal{O}_F^n$ such that $c_* \omega$ is of the form $f dx_1 dx_2 \dots dx_n$ and that $|f|$ is constant, for which we define $|\omega|(U) := |f|$. Suppose $E \subset X$ is a disjoint union $E = \bigsqcup_{i=1}^m U_i$ for finitely many such U_i , then $|\omega|(E) := \sum |\omega|(U_i)$.

[CC: An extra factor $q^{n/2}$ is added on March 17th, and then moved to (5) on the 24th.]

Lemma 2.10. *Suppose we have a ring homomorphism $\iota : \mathbb{Z}[q^{-n/2}] \rightarrow \mathcal{C}$. Then the above definition is well-defined for any open compact subset $E \subset X$, giving $|\omega|(E) \in \mathbb{Z}[q^{-1/2}]$. Consequently, for any $f \in \mathcal{C}_c^\infty(X)$, we can write $f = \sum_j c_j 1_{E_j}$ and define $\int_X f |\omega| := \sum_j c_j \iota(|\omega|(E_j))$. Again the integral is well-defined.*

2.4. p-adic Lie groups. We recommend [Ser06, Part II, Chapter IV] for a reference for most results.

Definition 2.11. *A Lie group over F (of dimension n) is an F -analytic manifold H (of dimension n) together with a multiplication map $m_H : H \times H \rightarrow H$ that is analytic. A Lie subgroup is a closed subgroup that is also a sub-manifold.*

Remark 2.12. Lie groups over F are more often called a p -adic Lie group. When $F/\mathbb{Q}_p$ is finite of degree d , a Lie group over F of dimension n can be easily verified to be a Lie group over $\mathbb{Q}_p$ of dimension nd over $\mathbb{Q}_p$, so the notation is not quite an abuse of language.

Lemma 2.13. *Suppose H is a p -adic Lie group. Then there exists an open compact Lie subgroup $J \subset H$ such that*

- (1) *As a topological group, J is a pro- p -group, meaning that it is isomorphic to an inverse limit of J/J_m for some open compact normal Lie subgroups $J_m \subset J$ with index $[J : J_m]$ being a power of p .*
- (2) *In fact, possibly after shrinking J , we may take*

$$J_m = \{h^{p^m} \mid h \in J\}.$$

If H has dimension n , then $[J : J_m] = (\#(\mathcal{O}_F/p\mathcal{O}_F))^{nm}$.

- (3) *Consequently, any open subset of J contains a J_m -coset for some m . Same for any open subset of H .*

Sketch. We claim that with any local coordinate matching $0 \in \mathcal{O}_F^n$ with e , the analytic map m_H is of the form

$$m((x_1, \dots, x_n), (y_1, \dots, y_n)) = (x_1 + y_1 + f_1(x, y), \dots, x_n + y_n + f_n(x, y))$$

where each $f_i(x, y)$ has no constant nor linear terms. Indeed, any analytic function $m : \mathcal{O}_F^n \times \mathcal{O}_F^n \rightarrow \mathcal{O}_F^n$ satisfying $m(x, 0) = m(0, x) = x$ has this form. The formula basically says $m(x, y) = x + y + O(\max(|x|, |y|)^2)$. One then verifies that if we take J to be the subset $(\mathfrak{m}_F^N)^n$ for a sufficiently large N depending on f_i , all asserted properties in the lemma hold. $\square$

An alternative sketch. Alternatively, any group that we actually work with is a closed subgroups of $\mathrm{GL}_n(F)$, and the result follows from that for $\mathrm{GL}_n(F)$ which can be directly checked. $\square$

2.5. Varieties. A **variety** (always over F) will be defined as a geometrically reduced separated scheme of finite type over F . In this course, schemes are never needed, and the readers unfamiliar with varieties are advised to refer to various online resources for more friendly introductions to varieties. We will only sketch the concept below.

Consider $\bar{F}/F$ any algebraic closure. A lot of time we work with affine varieties, and $\mathrm{char}(F) = 0$, so an affine variety is the vanishing locus $\mathbb{X} = \mathbb{X}(\bar{F}) = (I = 0)$ in $\mathbb{A}^N := \bar{F}^N$ of some ideal $I \triangleleft F[x_1, \dots, x_N]$ for some $N \in \mathbb{Z}_{\geq 0}$, such that $F[x_1, \dots, x_N]/I$

is reduced. As $F[x_1, \dots, x_N]$ is Noetherian, I is generated by a finite number of elements, and usually we just say the variety is the vanishing locus of those generators. We write $\mathbb{X}(F) = \mathbb{X}(\bar{F}) \cap F^n$. When $I = 0$, the variety is denoted $\mathbb{A}^N$, for which the corresponding locus is the whole $\bar{F}^n$. A variety is glued from finitely many affine varieties in some way. A **morphism** between varieties over F is a map that is given locally by polynomial equations with coefficients in F .

A closed subvariety (over F) is something that is locally of the form $(J = 0) \subset (I = 0)$, where $J \supseteq I$ is a possibly bigger ideal. A (Zariski) closed F -subset of a variety is $\mathbb{Y}(\bar{F}) \subset \mathbb{X}(\bar{F})$ for some closed subvariety $\mathbb{Y} \subset \mathbb{X}$. A (Zariski) open subset is the complement in $\mathbb{X}(\bar{F})$ of a Zariski closed subset. A variety is said to be **irreducible** if it is not the union of two closed subsets, and **connected** if it contains a proper subset that is both Zariski closed and open.

Remark 2.14. One might be a bit confused about that we only defined closed F -subset, but not closed subset in general. I am mostly being lazy; every subset of a variety that we'll encounter in this course should be defined over F anyway.

A variety is **smooth** at $x \in \mathbb{X}(\bar{F})$, if Zariski locally (possibly after shrinking) it is of the form $x \in (f_1 = f_2 = \dots = f_n = 0) \subset \mathbb{A}^N$ such that the matrix $\left[\frac{\partial f_i}{\partial x_j}(x) \right]$ has rank $n \leq N$. In this case it is said to be smooth of dimension $N - n$. A variety is said to be smooth (of dimension n) if it is smooth at every $\bar{F}$ -point $x \in \mathbb{X}(\bar{F})$ (of dimension n).

Lemma 2.15. *For any variety $\mathbb{X}$ (over F) that is smooth at every point $x \in X := \mathbb{X}(F)$, the set X has a canonical structure of an F -analytic manifold. For any $x \in X$, we have a canonical isomorphism $T_x(X) \cong (T_x \mathbb{X})(F)$; here $T_x \mathbb{X}$ is the tangent space at x to X as a smooth variety. An algebraic differential k -form on $\mathbb{X}$ gives an (analytic) differential k -form on X .*

Sketch. For any $x \in X = \mathbb{X}(F)$, the variety is locally a closed subvariety of $\mathbb{A}_F^N$ cut out by $N - n$ equations with non-singular differential. By implicit function theorem, this is then an n -dimensional analytic sub-manifold of F^N . The rest are tautology. $\square$

Lemma 2.16. *Let $\mathbb{X}$ be a connected smooth variety over F with $\mathbb{X}(F) \neq \emptyset$. Then $\mathbb{X}(F)$ is (Zariski) dense in $\mathbb{X}$.*

Sketch. Suppose on the contrary the Zariski closure $\mathbb{Y}$ of $\mathbb{X}(F)$ in $\mathbb{X}$ is a proper closed subvariety. Since $\mathbb{X}$ is connected smooth, therefore irreducible, this implies that $\dim \mathbb{Y} < \dim \mathbb{X}$.

We then argue that any closed subvariety with $\dim \mathbb{Y} < \dim \mathbb{X} = n$ cannot contain a non-trivial n -dimensional F -analytic manifold, thus a contradiction. We can either argue that $\mathbb{Y}(F) \subset \mathbb{X}(F)$ has measure zero in Lebesgue sense, or cut $\mathbb{Y}$ into a finite number of smooth sub-varieties of smaller dimensions and argue that their union cannot contain an n -dimensional F -analytic manifold. This is essentially because an F -analytic manifold of dimension $< n$ cannot be an F -analytic manifold of dimension n , as in Remark 2.4. $\square$

Remark 2.17. The same proof works with $F = \mathbb{R}$, and in fact also for $F = \mathbb{F}_q((t))$; see [Poo17, Proposition 3.5.75].

Example 2.18. As non-examples, $\mathbb{X}_1 = (x^2 + y^2 = 3) \subset \mathbb{A}^2$ over $\mathbb{Q}_3$ is smooth but $\mathbb{X}_1(\mathbb{Q}_3) = \emptyset$. Meanwhile $\mathbb{X}_2 = (x^2 + y^2 = 0) \subset \mathbb{A}^2$ is not smooth, and $\mathbb{X}_2(F) = \{(0, 0)\}$ is just one point and not dense.

2.6. Linear algebraic groups and reductive groups. An **algebraic group** over F is a variety $\mathbb{G}$ over F together with morphisms $m_{\mathbb{G}} : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}$, $i_{\mathbb{G}} : \mathbb{G} \rightarrow \mathbb{G}$ and $e \in \mathbb{G}(F)$ that makes $\mathbb{G}(\bar{F})$ a group. A **(closed/algebraic) subgroup** $\mathbb{H} \subset \mathbb{G}$ is a Zariski closed variety $\mathbb{H} \subset \mathbb{G}$ that again satisfies $m_{\mathbb{G}}(\mathbb{H} \times \mathbb{H}) \subset \mathbb{H}$, $i_{\mathbb{G}}(\mathbb{H}) \subset \mathbb{H}$ and $e \in \mathbb{H}(F)$.

An algebraic group is said to be **connected**³ if it is connected as a variety. Any linear algebraic group over F is automatically smooth. Lemma 2.15 then implies

Lemma 2.19. *Let $\mathbb{G}$ be an algebraic group. Then $G = \mathbb{G}(F)$ is a Lie group over F .*

As a variety, GL_n is the closed subvariety of

$$\mathbb{A}^{n^2+1} = \{(g, x) \mid g \text{ is an } n \times n \text{ matrix with } \det(g) \cdot x = 1\}.$$

It is often more convenient to view it as an open subvariety $(\det(g) \neq 0) \subset \mathbb{A}^{n^2}$. It has the obvious multiplication map $m_{\mathrm{GL}_n}(g, g') = gg'$, an inverse map $i_{\mathrm{GL}_n}(g) = g^{-1}$, and an identity $e = \mathrm{Id}_n \in \mathrm{GL}_n(F)$. That makes it an algebraic group. A **linear algebraic group** is a closed subgroup of GL_n .

Example 2.20. Let $\mathbb{G}_a$ be the algebraic group that as a variety is $\mathbb{A}^1$, and with the group law given by addition. It can be identified

$$\mathbb{G}_a \cong \begin{bmatrix} 1 & * \\ & 1 \end{bmatrix} := \{g = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{GL}_2 \mid a = d = 1, c = 0\}.$$

Let B_2 be the group

$$B_2 \cong \begin{bmatrix} * & * \\ & * \end{bmatrix} := \{g = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{GL}_2 \mid c = 0\} \subset \mathrm{GL}_2.$$

Note that the $\det \neq 0$ condition forces $a, d \neq 0$. We have that $\mathbb{G}_a \subset B_2 \subset \mathrm{GL}_2$ are closed subgroups.

Let V be an n -dimensional vector space over F . The **general linear group** GL_V is the group identified with GL_n using any basis $V \cong F^n$.

The **special linear group** $\mathrm{SL}_n \subset \mathrm{GL}_n$ (or $\mathrm{SL}_V \subset \mathrm{GL}_V$) is the closed subgroup consisting of matrices of determinant 1. In particular $\mathrm{SL}_n := \mathrm{SL}_{F^n}$.

Suppose V be equipped with a symmetric bilinear form $(\cdot, \cdot)$. The **orthogonal group**⁴ is $\mathrm{O}_V \subset \mathrm{GL}_V$ is the closed subgroup consisting of orthogonal operators with $(gv_1, gv_2) = (v_1, v_2)$. The **special orthogonal group** $\mathrm{SO}_V \subset \mathrm{O}_V$ is the closed subgroup consisting of orthogonal operators with determinant 1. If $(\cdot, \cdot)$ is non-degenerate, then $\mathrm{SO}_V \subset \mathrm{O}_V$ is a connected component, and has index 2.

Suppose V be equipped with an anti-symmetric bilinear form $\langle \cdot, \cdot \rangle$. The **symplectic group** $\mathrm{Sp}_V \subset \mathrm{GL}_V$ is the closed subgroup consisting of symplectic operators with $\langle gv_1, gv_2 \rangle = \langle v_1, v_2 \rangle$.

³When an algebraic group is connected, it is automatically geometrically connected.

⁴Usually, when we say “orthogonal group”, “special orthogonal group” and “symplectic group” we assume that the underlying bilinear form is non-degenerate.

For the rest of this article, when we write O_V , SO_V or Sp_V we **assume that the bilinear form is non-degenerate unless otherwise stated**. In such non-degenerate cases, the groups GL_V , O_V and Sp_V , together with U_V introduced in Exercise 5, are called **classical groups**. This is probably the most well-accepted notion, but it is also common, depending on context, to include SO_V as a classical group, to include SL_n as a classical group, or to exclude GL_V from classical groups.

Definition 2.21. *An element $g \in GL_n(F)$ is called **semisimple** if it is diagonalizable after some finite extension. It is **unipotent** if all eigenvalues = 1.*

*Let $\mathbb{G} \subset GL_n$ be a linear algebraic group. An element is **semisimple** (resp. **unipotent**) if it is so in $GL_n(F)$. A Jordan decomposition of $g \in G = \mathbb{G}(F)$ is $g = su = us$ where $s \in G$ is semisimple and $u \in G$ is unipotent.*

For the following theorem see [Bor91, §4].

Theorem 2.22. *Let $\mathbb{G}$ be a linear algebraic group. The notion of being semisimple and unipotent does not depend on the embedding $\mathbb{G} \hookrightarrow GL_n$. Every element $g \in G = \mathbb{G}(F)$ has a unique Jordan decomposition in G , and that also does not depend on the embedding $\mathbb{G} \hookrightarrow GL_n$.*

Definition 2.23. *An algebraic group $\mathbb{U}$ is **unipotent** if it is a linear algebraic group and every element $u \in \mathbb{U}(\bar{F})$ is unipotent.*

Definition 2.24. *An algebraic group $\mathbb{G}$ is **reductive** if it is a linear algebraic group, and that every unipotent connected normal closed subgroup is trivial.*

Remark 2.25. A previous version of the note skipped the “connected” in “unipotent connected normal closed subgroup.” We thank Hao-An for pointing this out. The previous definition turns out to be equivalent due to Remark 2.28, but is not a good idea over a general field.

Remark 2.26. In old books it is common to require in definition that a reductive group is connected. It seems more common nowadays to not assume that.

Remark 2.27. In Example 2.20, the groups $\mathbb{G}_a$ and B_2 are not reductive. (They both have $\mathbb{G}_a$ as an unipotent normal subgroup.) The groups GL_V and SL_V are reductive. Without assuming that the bilinear form is non-degenerate, the groups O_V , SO_V and Sp_V are reductive if and only if the underlying symmetric / anti-symmetric form is non-degenerate or trivial. Probably the easiest way to check reductivity in these cases is to do some linear algebra and argue that the conjugates of any non-trivial unipotent subgroup generates some non-unipotent element.

Remark 2.28. We have the following facts:

- (1) Every unipotent subgroup is automatically connected. (This uses $\text{char}(F) = 0$. Roughly, over a field of characteristic 0, any closed subgroup that contains all powers of u must connect u to the identity e .)
- (2) If $\mathbb{H}$ is a linear algebraic group for which every $h \in \mathbb{H}(F)$ is unipotent, then $\mathbb{H}$ is unipotent. (Thanks to Lemma 2.16.)
- (3) A linear algebraic group is reductive if and only if it does not contain a normal closed subgroup isomorphic to $\mathbb{G}_a$.

- (4) A linear algebraic group $\mathbb{G}$ is reductive if and only if $\mathbb{G}(F)$ does not contain a normal closed (topological) subgroup isomorphic to $F = \mathbb{G}_a(F)$, or just $\mathbb{Q}_p$.

However, we remark that in our opinion the important thing about reductive groups is not their definition, but many consequential properties that we will gradually see (or you might have seen in related subjects).

Remark 2.29. Suppose $\mathbb{G}$ is an algebraic group over F . Then there exists an algebraic group $\text{Res}_{\mathbb{Q}_p}^F \mathbb{G}$ over $\mathbb{Q}_p$ that gives a canonical isomorphism $\mathbb{G}(F) \cong (\text{Res}_{\mathbb{Q}_p}^F \mathbb{G})(\mathbb{Q}_p)$ as p -adic Lie groups. (This is called the **Weil restriction**. The group $\text{Res}_{\mathbb{Q}_p}^F \mathbb{G}$ is linear/connected/reductive if and only if $\mathbb{G}$ is (this uses that thankfully $\mathbb{Q}_p$ is perfect). Therefore, in some sense we can always reduce to the case when $F = \mathbb{Q}_p$, though in my opinion that's generally not a good idea.

Exercise 5. Let E/F be any quadratic extension. For $x \in E$ denote by $\bar{x}$ the (unique) Galois conjugate of x over F . Let V be an n -dimensional vector space, on which we are given a hermitian form $\langle \cdot, \cdot \rangle : V \times V \rightarrow E$, i.e. that is bi-additive and such that

$$\langle c_1 v_1, c_2 v_2 \rangle = c_1 \bar{c}_2 \langle v_1, v_2 \rangle, \quad \forall c_1, c_2 \in E, v_1, v_2 \in V.$$

Construct an algebraic group $\mathbb{G}$ over F , such that as Lie groups over F we have

$$\mathbb{G}(F) = \text{group of unitary operators on } (V, \langle \cdot, \cdot \rangle).$$

Show that your $\mathbb{G}$ is reductive only if $\langle \cdot, \cdot \rangle$ is non-degenerate or identically zero.

Exercise 6. As a follow-up exercise, suppose that $\langle \cdot, \cdot \rangle$ is non-degenerate in Exercise 5. Compute $\mathbb{G}(E)$. Show that your $\mathbb{G}$ is reductive. This $\mathbb{G}$ is commonly denoted by U_V , called the **unitary group**

2.7. Lie algebra and exponential map.

Definition 2.30. Let H be a p -adic Lie group. Then $\text{Lie}(H) := T_e H$ is the tangent space at the identity. Likewise, when $\mathbb{H}$ is an algebraic group, we put $\text{Lie}(\mathbb{H}) := T_e \mathbb{H}$.

Example 2.31. The Lie algebra $\text{Lie}(\text{GL}_V(F))$ (resp. $\text{Lie}(\text{GL}_V)$) is canonically identified with the space of linear operators on V . In particular $\text{Lie}(\text{GL}_n(F))$ (resp. $\text{Lie}(\text{GL}_n)$) is the space of $n \times n$ matrices in F . We write $\mathfrak{gl}_n := \text{Lie}(\text{GL}_n(F))$ and We write $\mathfrak{gl}_V := \text{Lie}(\text{GL}_V(F))$.

Suppose $H \subset H'$ is a Lie subgroup. Then obviously $\text{Lie}(H) \subset \text{Lie}(H')$ is a subspace. For example, $\mathfrak{sl}_V \subset \mathfrak{gl}_V$ is the subspace of traceless $n \times n$ matrices, $\mathfrak{so}_V \subset \mathfrak{gl}_V$ is the space of anti-self-adjoint operators, and likewise $\mathfrak{sp}_V \subset \mathfrak{gl}_V = \{X \in \mathfrak{gl}_V \mid \langle Xv, w \rangle = -\langle v, Xw \rangle\}$.

Definition 2.32. Let H be a p -adic Lie group and $g \in H$. We denote by

$$\begin{aligned} c_g : H &\rightarrow H \\ h &\mapsto ghg^{-1} \end{aligned}$$

and $\text{Ad}(g) := dc_g|_e : \text{Lie}(H) \rightarrow \text{Lie}(H)$ its differential. We will sometimes also write $\text{Ad}(g)$ in terms of c_g .

Definition 2.33. Write $\mathfrak{h} := \text{Lie}(H)$. We also denote by

$$\begin{aligned} \text{Ad} : H \times \mathfrak{h} &\rightarrow \mathfrak{h} \\ (h, X) &\mapsto \text{Ad}(h)X \end{aligned}$$

It is easy to see that this map is analytic. We also write it as $\text{Ad} : H \rightarrow \text{GL}_{\mathfrak{h}}$, which is also analytic. They are called the **adjoint action** and the **adjoint representation**.

Definition 2.34. We define $\text{ad} : \mathfrak{h} \rightarrow \mathfrak{gl}_{\mathfrak{h}}$ as $\text{ad} = d(\text{Ad})|_e$, the differential of $\text{Ad} : H \rightarrow \text{GL}_{\mathfrak{h}}$ at the identity. We also write $[Y, X] := \text{ad}_Y(X) := \text{ad}(Y, X)$, called the **Lie bracket** of Y and X .

Lemma 2.35. We have the following facts:

- (1) $c_g(h_1 h_2) = c_g(h_1) c_g(h_2)$
- (2) $\text{Ad}_g([h_1, h_2]) = [\text{Ad}_g(h_1), \text{Ad}_g(h_2)]$
- (3) (Jacobi identity) $\text{ad}_Y([X_1, X_2]) = [\text{ad}_Y(X_1), X_2] + [X_1, \text{ad}_Y(X_2)]$.

Proof. The first one is obvious. Each next one is a differential of the previous one. $\square$

Remark 2.36. It is obvious from the definition that the Lie bracket is F -bilinear. We will later see that it is alternating.

We remark that the definitions of $\text{Lie}(H)$, Ad and ad can be done in the same way for algebraic groups. When $H = \mathbb{H}(F)$, the two definitions agree. In what follows we will do something non-algebraic. Suppose $U \subset H$ is an open subset with $e \in U$, and $c : U \rightarrow T_e U = \text{Lie}(H)$ is an analytic map such that $dc|_e = \text{id}_{\text{Lie}(H)}$. We consider

$$(3) \quad \log_{H,c}(h) := \lim_{i \rightarrow +\infty} p^{-i} c(h^{p^i})$$

provided that the limit exists in $T_e H$.

Definition 2.37. We say $h \in H$ is **topologically unipotent** if $\lim_{i \rightarrow +\infty} h^{p^i} = e$.

Lemma 2.38. Suppose h is topologically unipotent. The limit (3) exists for any U and c as above, and is independent of the choice of U and c . If $f : H \rightarrow H'$ is any Lie group homomorphism, then $(df)(\log(h)) = \log_{H'}(f(h))$ for any topologically unipotent $h \in H$. In particular $\text{Ad}_g(\log(h)) = \log(\text{Ad}_g(h))$.

We will only sketch Lemma 2.38. However, Exercise 7 provides a more informative proof (in my opinion) in the situations that we need.

Sketch of Lemma 2.38. We work in a local chart that matches e with $0 \in \mathcal{O}_n^F$. After shrinking U , we write $U = (\mathfrak{m}_F^N)^n \subset H$ and $\text{Lie}(H) = T_e U = F^n$. We have $c : (\mathfrak{m}_F^N)^n \rightarrow F^n$ is a power series with $dc|_0 = \text{id}_{F^n}$. This implies $c(x) = x + O(|x|^2)$. We have seen that multiplication map on $\mathcal{O}_F^n$ has the first order approximation $m(x, y) = x + y + O(\max(|x|, |y|)^2)$. In particular, $x^p = m(x, m(x, \dots)) = px + O(|x|^2)$. Therefore, in our local coordinate we have

$$|h^{p^{i+1}}| = |p| \cdot |h^{p^i}|$$

for $i \gg 0$ with $|h^{p^i}|$ sufficiently small, and

$$p^{-(i+1)} c(h^{p^{i+1}}) = p^{-(i+1)} \left(p \cdot c(h^{p^i}) + O(|h^{p^i}|^2) \right) = p^{-i} c(h^{p^i}) + p^{-(i+1)} \cdot O(|h^{p^i}|^2).$$

Hence $p^{-i}c(h^{p^i})$ is a Cauchy sequence that converges. For a different choice of c (and U), they again differ by $x + O(|x|)^2$ after shrinking, and similar techniques shows that they give the same limit. $\square$

Definition 2.39. We define $\log(h) := \log_{H,c}(h) \in T_e H$ thanks to Lemma 2.38.

Lemma 2.40. There exists an open subgroup $K \subset H$ such that every element in K is topologically unipotent, and such that $\log|_K : K \rightarrow \log(K) \subset \text{Lie}(H)$ is an isomorphism of analytic manifolds.

The proof is similar to that of Lemma 2.38. We skip it.

Definition 2.41. Let K be as in Lemma 2.40. We define $\exp : \log(K) \xrightarrow{\sim} K$ to be the inverse of $\log|_K$.

In fact, what is closer to our use is

Exercise 7. Show that there exist an open compact Lie subgroup $K \subset \text{GL}_n(F)$ and a lattice $\mathfrak{K} \subset \mathfrak{gl}_n$ such that

$$\log(h) = \sum_{i=1}^{\infty} \frac{(-1)^{i-1}(h - \text{id})^i}{i}$$

defines an isomorphism from K to $\mathfrak{K}$, and that

$$\exp(X) = \sum_{i=0}^{\infty} \frac{X^i}{i!}$$

defines the inverse of $\log$ from $\mathfrak{K}$ to K . Both series are computed in the space of $n \times n$ matrices in F . Show also that $\log(h^{-1}) = -\log(h)$ for $h \in K$.

The properties in Exercise 7 also holds for any conjugate of K . Since (3) already characterizes $\log$ uniquely, we have

Corollary 2.42. Let $\mathbb{H}$ be a linear algebraic group over F , and $H = \mathbb{H}(F)$. Then there exists an H -conjugation invariant neighborhood $e \in U \subset H$ and $0 \in \mathfrak{u} \subset \mathfrak{h}$ such that $\log$ maps U isomorphically to $\mathfrak{u}$, with inverse $\exp$, given by any embedding $\mathbb{H} \hookrightarrow \text{GL}_n$.

Proof. Apply Lemma 2.38 and Exercise 7 to (any) $\mathbb{H} \hookrightarrow \text{GL}_n$. $\square$

Example 2.43. Certainly, Corollary 2.42 holds whenever H is a closed Lie subgroup of some $\text{GL}_n(F)$. One might ask if there exists a p -adic Lie group that cannot be embedded into $\text{GL}_n(F)$ for any n . Such groups do exist. Examples can be found in <https://mathoverflow.net/questions/91789/non-linear-lie-group> by replacing the “ $\mathbb{R}$ ” in any answer by a p -adic field in appropriate ways. For example, consider

$$H = \left\{ \begin{bmatrix} 1 & a & b \\ & 1 & c \\ & & 1 \end{bmatrix} \mid a, b, c \in \mathbb{Q}_p \right\} \supset \Gamma = \left\{ \begin{bmatrix} 1 & & b \\ & 1 & \\ & & 1 \end{bmatrix} \mid b \in \mathbb{Z}_p \right\}.$$

Then H/Γ is a 2-dimensional p -adic Lie group, in fact with an abelian Lie algebra. We can verify that every element in H/Γ is topologically unipotent, but Corollary 2.42 does not hold for H/Γ .

Remark 2.44. If we replace F by $\mathbb{R}$, for real Lie groups we have

$$\exp(X) := \lim_{n \rightarrow +\infty} c\left(\frac{1}{n}X\right)^n \in G$$

for $X \in \text{Lie}(H)$. This generalizes the well-known fact for that for $G = \text{GL}_1(\mathbb{R}) = \mathbb{R}^\times$ we have $\exp(X) = \lim(1 + \frac{X}{n})^n$ for $X \in \mathfrak{g} = \mathbb{R}$. One can then define $\log$ (for real Lie group) to be the inverse of $\exp$.

Lemma 2.45. *When $G = \text{GL}_n$, we have $[X, Y] = XY - YX$.*

Corollary 2.46. *Let $\mathbb{H}$ be a linear algebraic group over F , and $H = \mathbb{H}(F)$. Let $X, Y \in \text{Lie}(H)$. Then the **Lie bracket** of X and Y satisfies*

$$[X, Y] = \lim_{\substack{a, b \in F \\ a, b \rightarrow 0}} \frac{\log(\exp(aX) \exp(bY) \exp(-aX) \exp(-bY))}{ab}$$

In particular, we have $[X, Y] = [Y, X]$. If $H \subset H'$ is a Lie subgroup, then for $X, Y \in \text{Lie}(H)$ we have that $[X, Y]$ is the same in $\text{Lie}(H')$ as in $\text{Lie}(H)$.

We also need

Definition 2.47. *An element $X \in \mathfrak{gl}_n(F)$ is called **semisimple** if it is diagonalizable after some finite extension. It is **nilpotent** if all eigenvalues = 0.*

*Let $\mathbb{G} \subset \text{GL}_n$ be a linear algebraic group. An element $X \in \mathfrak{g}$ is **semisimple** (resp. **nilpotent**) if it is so in $\mathfrak{gl}_n(F)$. A Jordan decomposition of $X \in \mathfrak{g}$ is $X = X_s + X_n$ such that $[X_s + X_n]$ where $X_s \in \mathfrak{g}$ is semisimple and $X_n \in \mathfrak{g}$ is nilpotent.*

We have the following analogue of Theorem 2.22; see also [Bor91, §4].

Theorem 2.48. *The notion of being semisimple or nilpotent does not depend on the embedding $\mathbb{G} \hookrightarrow \text{GL}_n$. Every element $\mathfrak{g}$ has a unique Jordan decomposition which also does not depend on the embedding.*

2.8. Measures, (generalized) functions and distributions. [CC: This subsection is added on March 24th, but we're probably never going to use it. Ok. Let's use it. The following is added on Apr. 5th.]

We fix X an F -analytic manifold.

Definition 2.49. *Let $\mathcal{C}_c^\infty(X) \subset \mathcal{C}^\infty(X)$ be the space of $\mathcal{C}$ -valued locally constant compact supported functions, resp. locally constant functions.*

Definition 2.50. *We say*

$$\mu : \{\text{open compact subset } J \subset X\} \rightarrow \mathcal{C}$$

*is a ($\mathcal{C}$ -valued) measure on X if $\mu(J_1 \sqcup J_2) = \mu(J_1) + \mu(J_2)$. Measures can be restricted to open subsets, and the **support** $\text{supp}(\mu)$ is the largest closed subset for which $\mu|_{X \setminus \text{supp}(\mu)} \equiv 0$.*

*We say μ is **locally constant** if, near every point, we can find a neighborhood of that point identified as $\mathcal{O}_F^n$ and a constant $c \in \mathcal{C}$ so that $\mu(x + (\mathfrak{m}_F^n)^n) = \frac{c}{q^{mn}}$ for any $x \in \mathcal{O}_F^n$.*

Denote by $M_c^\infty(X) \subset M^\infty(X)$ the space of locally constant compactly supported measures on X , resp. locally constant measures on X .

Remark 2.51. $M_c^\infty(X)$ is a $\mathcal{C}_c^\infty(X)$ -module, and $M^\infty(X)$ is a $\mathcal{C}_c^\infty(X)$ -module. Moreover $\mathcal{C}_c^\infty(X)$ takes $M^\infty(X)$ into $M_c^\infty(X)$. For $f \in \mathcal{C}^\infty(X)$ and $\mu \in M^\infty(X)$ such that $\text{supp}(f\mu)$ is compact, we will write $\int_X f\mu := (f\mu)(J)$ for any open compact $J \supset \text{supp}(f\mu)$.

Remark 2.52. Abstractly, $M_c^\infty(X) \cong \mathcal{C}_c^\infty(X)$ as a $\mathcal{C}_c^\infty(X)$ -module and $M^\infty(X) \cong \mathcal{C}^\infty(X)$ as a $\mathcal{C}^\infty$ -module; this can be seen by taking a *countable* open compact covering $\bigcup U_i$, making it disjoint by taking $\bigsqcup (U_i \setminus \bigcup_{j < i} U_j)$, and choosing isomorphisms on individual charts. Note however that $\mathcal{C}_c^\infty(X)$ is a non-unital algebra unless X is compact.

Definition 2.53. Let $D(X) := \text{Hom}_{\mathcal{C}}(\mathcal{C}_c^\infty(X), \mathcal{C})$ be the linear dual of $\mathcal{C}_c^\infty(X)$. We call it the space of **distributions** (on X).

By definition we have a natural embedding $M_c^\infty(X) \hookrightarrow D(X)$. Let us fix $U \subset X$ an open subset (therefore an open submanifold), and $Z = X \setminus U$.

Lemma 2.54. We have natural short exact sequences.

$$0 \rightarrow \mathcal{C}_c^\infty(U) \rightarrow \mathcal{C}_c^\infty(X) \rightarrow \mathcal{C}_c^\infty(Z) \rightarrow 0$$

$$0 \leftarrow D(U) \leftarrow D(X) \leftarrow D(Z) \rightarrow 0$$

The embedding $D(Z) \hookrightarrow D(X)$ identifies $D(Z)$ as the space of distributions on X with supported in Z .

Definition 2.55. Let $\mathcal{C}^{-\infty}(X) := \text{Hom}_{\mathcal{C}}(M_c^\infty(X), \mathcal{C})$ be the linear dual of $M_c^\infty(X)$. We call it the space of **generalized functions**.

Definition 2.56. We have a natural inclusion $\mathcal{C}^\infty(X) \hookrightarrow \mathcal{C}^{-\infty}(X)$. We say a $\phi \in \mathcal{C}^\infty(X)$ **represents** its image in $\mathcal{C}^{-\infty}(X)$.

Lemma 2.57. Let $U \subset X$ be an open subset, and $Z = X \setminus U$. We have natural maps $M_c^\infty(U) \hookrightarrow M_c^\infty(X)$ and $\mathcal{C}^{-\infty}(X) \twoheadrightarrow \mathcal{C}^{-\infty}(U)$.

Lemma 2.58. We have diagrams

$$\begin{array}{ccc} M_c^\infty(U) & \hookrightarrow & M_c^\infty(X) \\ \downarrow & & \downarrow \\ M^\infty(U) & \hookrightarrow & M^\infty(X) \\ \downarrow & \curvearrowright & \downarrow \\ D(U) & \twoheadleftarrow & D(X) \end{array} \quad , \quad \begin{array}{ccc} \mathcal{C}_c^\infty(U) & \hookrightarrow & \mathcal{C}_c^\infty(X) \\ \downarrow & & \downarrow \\ \mathcal{C}^\infty(U) & \hookrightarrow & \mathcal{C}^\infty(X) \\ \downarrow & \curvearrowright & \downarrow \\ \mathcal{C}^{-\infty}(U) & \twoheadleftarrow & \mathcal{C}^{-\infty}(X) \end{array}$$

that are commutative up to the sense that $\mathcal{C}^\infty(X) \twoheadrightarrow \mathcal{C}^\infty(U) \hookrightarrow \mathcal{C}^\infty(X)$ and $M^\infty(X) \twoheadrightarrow M^\infty(U) \hookrightarrow M^\infty(X)$ are certainly not the identity, but any two routes with the same source and target that don't include these two maps have the same composition.

Remark 2.59. It is tempted to define $M_c^\infty(Z) := M_c^\infty(X)/M_c^\infty(U)$. This however is **not** natural in the sense that it is not canonically defined in terms of Z ; see <https://gemini.google.com/share/f6a3ed0750bd>.

[CC: What used to be section 3 was moved to §5.1 on March 29th.]

3. ORBITAL INTEGRALS

This section partly follows [Kot05, §3, §6 and §17], with the major difference that Kottwitz only works with $\mathcal{C} = \mathbb{C}$, and we don't even assume $\text{char}(\mathcal{C}) = 0$. Every statement in [Kot05] can be made true with $\text{char}(\mathcal{C}) = 0$. **Not** every statement in [Kot05] is true when $\text{char}(\mathcal{C}) > 0$, so we need to be careful. A detailed study of what really happens when $\text{char}(\mathcal{C}) > 0$ is done in [VW01]; we will not dive into that.

3.1. Some motivation. Let $S^1 := \{z \in \mathbb{C} \mid |z| = 1\}$. For an abelian locally compact Hausdorff topological group H , it has a **Pontryagin dual** $\widehat{H} := \text{Hom}_{\text{cont}}(H, S^1)$, the group of continuous group homomorphisms equipped with the compact-open topology. The most important examples are in your calculus textbook:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{R} & \longrightarrow & S^1 \longrightarrow 1 \\ & & \wr \parallel & & \wr \parallel & & \wr \parallel \\ 1 & \longrightarrow & \text{Hom}_{\text{cont}}(S^1, S^1) & \longrightarrow & \text{Hom}_{\text{cont}}(\mathbb{R}, S^1) & \longrightarrow & \text{Hom}_{\text{cont}}(\mathbb{Z}, S^1) \longrightarrow 1 \end{array}$$

and in your algebraic number theory textbook:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathbb{Q} & \longrightarrow & \mathbb{A}_{\mathbb{Q}} & \longrightarrow & \mathbb{A}_{\mathbb{Q}}/\mathbb{Q} \longrightarrow 1 \\ & & \wr \parallel & & \wr \parallel & & \wr \parallel \\ 1 & \longrightarrow & \widehat{\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}} & \longrightarrow & \widehat{\mathbb{A}_{\mathbb{Q}}} & \longrightarrow & \widehat{\mathbb{Q}} \longrightarrow 1 \end{array}$$

and it has a p -adic counter-part:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathbb{Z}_p & \longrightarrow & \mathbb{Q}_p & \longrightarrow & \mathbb{Q}_p/\mathbb{Z}_p \longrightarrow 1 \\ & & \wr \parallel & & \wr \parallel & & \wr \parallel \\ 1 & \longrightarrow & \widehat{\mathbb{Q}_p/\mathbb{Z}_p} & \longrightarrow & \widehat{\mathbb{Q}_p} & \longrightarrow & \widehat{\mathbb{Z}_p} \longrightarrow 1 \end{array}$$

Consider $\psi : \mathbb{Z}[1/p] \rightarrow S^1$ by $n \mapsto e^{2\pi i n}$. This is continuous with respect to $|\cdot|_p$, and therefore extends to a continuous group homomorphism which we denote by $\psi_p : \mathbb{Q}_p \rightarrow S^1$. We define $\psi'_F : F \xrightarrow{\text{tr}_{F/\mathbb{Q}_p}} \mathbb{Q}_p \xrightarrow{\psi_p} S^1$. In particular, F is an abelian locally compact Hausdorff topological group and

$$\begin{array}{ccc} F & \cong & \widehat{F} := \text{Hom}_{\text{cont}}(F, S^1) \\ x & \mapsto & (y \mapsto \psi'_F(xy)) \end{array}$$

This is generalized to any vector space, e.g. $\mathfrak{g}$ as well as $\mathfrak{g}^* := \text{Hom}_F(\mathfrak{g}, F)$. We have

$$\begin{array}{ccc} \mathfrak{g}^* & \cong & \widehat{\mathfrak{g}^*} := \text{Hom}_{\text{cont}}(\mathfrak{g}, S^1) \\ X^* & \mapsto & (Y \mapsto \psi'_F(\langle X^*, Y \rangle)) \end{array}$$

By definition $\text{Hom}_{\text{cont}}(\mathfrak{g}, S^1)$ is the set of (continuous) characters of $\mathfrak{g}$ with values in S^1 . Now, let us try to pretend that $\mathfrak{g}$ and G are the same; at least near the identity they are isomorphic F -analytic manifolds via $\log$ and $\exp$. Then, we are hoping that characters of (representations of) G , maybe locally, look like elements in $\mathfrak{g}^*$.

Things cannot be this simple for an obvious reason: G is not abelian - not even locally, e.g. when $G = \text{SL}_2(F)$. In particular characters on G should **not** correspond

to a single element in $\mathfrak{g}^*$. Another hint comes already from the theory of characters of finite groups, where characters are compared with *class functions*. Therefore, we would like to study not one element $X^* \in \mathfrak{g}^*$, but rather a **coadjoint orbit** $O^* = \text{Ad}^*(G)(X^*) \subset \mathfrak{g}^*$.

3.2. Coadjoint orbits. For the rest of this section, we work over $\mathcal{C}$, any field of characteristic different from p . We are also going to work with $\mathfrak{g}^* := \text{Hom}_F(\mathfrak{g}, F)$. The action $\text{Ad} : G \rightarrow \text{Aut}_F(\mathfrak{g})$ and $\text{ad} : \mathfrak{g} \rightarrow \text{End}_F(\mathfrak{g})$ gives $\text{Ad}^* : G \rightarrow \text{Aut}_F(\mathfrak{g}^*)$ and $\text{ad}^* : \mathfrak{g} \rightarrow \text{End}_F(\mathfrak{g}^*)$, where $\text{Ad}^*(g) = (g^t)^{-1}$ and $\text{ad}^*(X) = -X^t$. They are called the **coadjoint action** and **coadjoint representation**. We however have

Theorem 3.1. *There exists a non-degenerate symmetric bilinear pairing $\mathfrak{g} \times \mathfrak{g} \rightarrow F$ that is $\text{Ad}(G)$ -invariant. Equivalently, there exists a self-adjoint isomorphism $\iota : \mathfrak{g}^* \cong \mathfrak{g}$ that is equivariant under G (for Ad^* and Ad).*

Moreover, Jordan decompositions do not depend on the choice of ι . That is, for any two G -equivariant isomorphisms $\iota_1, \iota_2 : \mathfrak{g}^* \xrightarrow{\sim} \mathfrak{g}$, if $X_s + X_n$ is a Jordan decomposition, then so is $(\iota_2)^{-1}(\iota_1(X_s)) + (\iota_2)^{-1}(\iota_1(X_n))$.

This can be deduced e.g. from Theorem 4.13 in <https://www.jmilne.org/math/CourseNotes/LAG.pdf>. As a consequence, it is ok if you fix ι and pretend $\mathfrak{g}^* = \mathfrak{g}$ the whole time. Anyhow, fix $X^* \in \mathfrak{g}^* = (\text{Lie}^* \mathbb{G})(F)$. Write $O^* = \text{Ad}^*(G)(X^*)$ and also $\mathbb{O}^* = \text{Ad}^*(\mathbb{G})(X^*)$. Let us put

Definition 3.2. *The centralizers of X^* are*

$$\begin{aligned} Z_{\mathbb{G}}(X^*) &= \{g \in \mathbb{G} \mid \text{Ad}^*(g)X^* = X^*\}. \\ Z_G(X^*) &:= \{g \in G \mid \text{Ad}^*(g)X^* = X^*\}. \\ Z_{\mathfrak{g}}(X^*) &:= \{Y \in \mathfrak{g} \mid \text{ad}^*(Y)X^* = X^*\}. \end{aligned}$$

Since F is perfect, by [Bor91, Proposition 6.7] we see that $\mathbb{O}^*$ is a locally closed⁵ smooth subvariety of $\text{Lie}^* \mathbb{G}$, and that the action map induces $\text{Lie} \mathbb{G} / \text{Lie} Z_{\mathbb{G}}(X^*) \cong T_{X^*} \mathbb{O}^*$. Passing to F -points, this says

$$(4) \quad (Y \mapsto \text{ad}^*(Y)X^*) : \mathfrak{g} \twoheadrightarrow T_{X^*}(\mathbb{O}^*(F)) \text{ with kernel } Z_{\mathfrak{g}}(X^*).$$

Note that O^* is a subset of $\mathbb{O}^*(F)$. The surjectivity (4) however shows that $(g \mapsto \text{Ad}^*(g)X^*) : G \twoheadrightarrow \mathbb{O}^*(F)$ has surjective differential and therefore its image, i.e. O^* is an open sub-manifold of $\mathbb{O}^*(F)$. The complement is a union of such orbits, and therefore open. That is $O^* \subset \mathbb{O}^*(F)$ is also closed. We remark that the implicit function theorem for $G \rightarrow O^*$ also shows

Lemma 3.3. *We have $\text{Lie} Z_G(X^*) = Z_{\mathfrak{g}}(X^*)$.*

Example 3.4. Let $\mathbb{G} = \text{SL}_2$ and $X^* = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then

$$\mathbb{O}^*(F) = \bigsqcup_{a \in F^\times / (F^\times)^2} \text{Ad}^*(G) \begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix}.$$

⁵Curiously, “locally closed” means an intersection of a Zariski closed and a Zariski open subset, not “closed locally in the ambient space.”

Exercise 8. Let $n \in \mathbb{Z}_{\geq 1}$. Suppose $\mathbb{G} = \mathrm{SL}_n$, $G = \mathrm{SL}_n(F)$ and X is the nilpotent matrix with $X_{ij} = 1$ when $j = i + 1$ and zero elsewhere. Let $\mathbb{O} = \mathrm{Ad}(\mathbb{G})X$. Describe directly (i.e. without using Galois cohomology, but in terms of F) how $\mathbb{O}(F)$ is a disjoint union of $\mathrm{Ad}(G)$ -orbits, and justify your description.

Remark 3.5. In general, there is a short exact sequence of pointed sets that looks like

$$1 \rightarrow Z_G(X^*) \rightarrow G \rightarrow \mathbb{O}^*(F) \rightarrow H^1(F, Z_{\mathbb{G}}(X^*)) \rightarrow H^1(F, \mathbb{G})$$

It can be proved that the **Galois cohomology** $H^1(F, \mathbb{G})$ for any linear algebraic group is finite. This implies that $\mathbb{O}^*(F)$ is always a finite disjoint union of $G/Z_G(X^*) = O^*$ for a finite number of $X^* \in \mathbb{O}^*(F)$, or $O^* \subset \mathbb{O}^*(F)$.

Remark 3.6. The “correct” definition of $Z_{\mathbb{G}}(X^*)$ should at least⁶ be the scheme-theoretic centralizer $\mathbb{G} \times_{\mathbb{O}^*} \{X^*\}$. The Lie algebra of the scheme-theoretic centralizer is by definition $Z_{\mathrm{Lie} \mathbb{G}}(X^*)$. Since $\mathrm{Lie}(Z_{\mathbb{G}}(X^*)) = Z_{\mathrm{Lie} \mathbb{G}}(X^*)$, we see that the scheme-theoretic centralizer $Z_{\mathbb{G}}(X^*)$ is smooth, and therefore geometrically reduced, and therefore equal to our variety-theoretic definition.

We have seen that $T_{X^*}O^* = \mathfrak{g}/Z_{\mathfrak{g}}(X^*)$. Observe that

$$\begin{array}{ccc} \omega_{X^*} : \mathfrak{g}/Z_{\mathfrak{g}}(X^*) \times \mathfrak{g}/Z_{\mathfrak{g}}(X^*) & \rightarrow & F \\ (Y, Z) & \mapsto & \langle [Y, Z], X^* \rangle \end{array}$$

is an anti-symmetric non-degenerate bilinear form; to see it is non-degenerate, we use that $\langle [Y, Z], X^* \rangle = \langle \mathrm{ad}(Y)Z, X^* \rangle = -\langle Z, \mathrm{ad}^*(Y)X^* \rangle$. If the last term is 0 for any $Z \in \mathfrak{g}$, then $\mathrm{ad}^*(Y)X^* = 0$, i.e. $Y \in Z_{\mathfrak{g}}(X^*)$. The very existence of ω_{X^*} shows that

Lemma 3.7. $\dim \mathbb{O}^* = \dim O^* = \dim \mathfrak{g}/Z_{\mathfrak{g}}(X^*)$ is even.

It is easy to see that ω_{X^*} for all $X^* \in O^*$ define an analytic 2-form ω_{O^*} on O^* .

Remark 3.8. The 2-form is evidently also algebraic. They are also G -invariant, meaning that they are preserved under the $\mathrm{Ad}^*(G)$ -action (resp. $\mathrm{Ad}^*(\mathbb{G})$ -action) on O^* . Lastly, they are closed, meaning that $d\omega_{O^*} = 0$, making O^* a symplectic manifold; see [CG10, Claim 1.1.6].

We won’t directly use that O^* is symplectic, and Cheng-Chiang is not aware if anything we will use implicitly share the same conceptual origin with this fact, but it is important in geometric representation theory.

Let us consider the top exterior power $\bigwedge^{\frac{1}{2} \dim O^*} \omega_{O^*}$, which is a nowhere-vanishing top form. Lemma 2.10 then gives us a $\mathbb{Z}[1/p]$ -valued measure on O^* . This allows us to integrate function, i.e. we have a $\mathcal{C}$ -linear functional

$$(5) \quad \begin{array}{ccc} I_{O^*} : \mathcal{C}_c^\infty(O^*) & \rightarrow & \mathcal{C} \\ f & \mapsto & q^{\frac{1}{2} \dim O^*} \cdot \int_{O^*} f |\omega_{O^*}| \end{array}$$

This feels like the best approximation to a class function; after all, there is no non-zero locally constant function on $\mathfrak{g}$ supported on O^* because $O^* \subset \mathfrak{g}^*$ is not an open subset. However it has a serious issue: The restriction of a function in $\mathcal{C}_c^\infty(\mathfrak{g}^*)$ to O^* does **not** always have compact support.

⁶Cheng-Chiang knows nothing about derived algebraic geometry.

Remark 3.9. Why the factor $q^{\frac{1}{2}\dim O^*}$? The ultimate mathematical reason is to be compatible with the result in [MW87], where I_{O^*} are related to important representation-theoretic quantities, namely dimensions of degenerate Whittaker models. What happens, in my opinion, is that for our later normalization in §4 that $\psi_F : F \rightarrow \mathbb{C}^\times$ is non-trivial on $\mathcal{O}_F$ but trivial on $\mathfrak{m}_F$, [MW87] suggests that we'd like $\int_{\mathcal{O}_F^n} dx_1 \dots dx_n = q^{n/2}$ because we'd like the preimage of a *Lagrangian*⁷ in $(\mathcal{O}_F/\mathfrak{m}_F)^n$ to have measure 1.

There is another point of view about this factor that is brought up in [Tsa20], that it is given by a weight 0 pure local system of perverse degree 0.

Example 3.10. Let $\mathbb{G} = \mathrm{GL}_2$, $G = \mathrm{GL}_2(F)$ and $X^* = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $\begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix} \in O^*$ for any $a \in F^\times$. In particular the zero matrix is in the closure of O^* , but not in O^* . Hence for any open subset $K \subset \mathfrak{g}^*$ with $0 \in K$, we see that $K \cap O^*$ is not closed in $\mathfrak{g}^*$, and therefore not compact.

An important result of Ranga Rao [RR72], also independently by Deligne, proves that

Theorem 3.11. *Suppose $\mathcal{C} = \mathbb{R}$ or $\mathbb{C}$. Then any $I_{O^*}(f)$ converges absolutely. That is, although $\mathrm{supp}(f|_{O^*})$ might not be compact, the integral $I_{O^*}(f|_{O^*})$ always converges absolutely.*

We will nevertheless talk about Theorem 3.11 from a point of view that involves no analysis on $\mathcal{C}$.

3.3. Basic geometry of $\mathrm{Ad}(G)$ -orbits. To understand what happens around Theorem 3.11, we need to understand the geometry of O^* . In Theorem 3.1, we have seen a G -equivariant isomorphism $\iota : \mathfrak{g}^* \xrightarrow{\sim} \mathfrak{g}$, and thus we can work on $\mathfrak{g}$. Let us first study the semisimple and nilpotent situation separately:

Proposition 3.12. *Let $X_s \in \mathfrak{g}$ be semisimple. Then $Z_{\mathbb{G}}(X_s)$ is a connected reductive group. After a finite base change, it is a Levi subgroup.*

Proof. Let $n \in \mathbb{Z}_{>0}$ be large enough so that $\exp(cX_s)$ for any $c \in \mathfrak{m}_F^n$ is defined. Since X_s is semisimple (diagonalizable over some finite extension in some $\mathfrak{gl}_n(F)$), all $\exp(cX_s)$ are also simultaneously diagonalizable over some finite extension. Let $\mathbb{T}$ be the identity component of the Zariski closure of $\{\exp(cX_s) \mid c \in \mathfrak{m}_F^n\}$. We claim that $\mathbb{T}$ is connected. Observe that $\mathbb{T}^\circ(F)$ is Lie subgroup of G such that $X_s = \frac{d\exp(cX_s)}{dc} \in \mathrm{Lie}(\mathbb{T}^\circ(F))$. Hence $\exp(cX_s) \in \mathbb{T}^\circ$ and thus $\mathbb{T}$ is the Zariski closure of $\mathbb{T}^\circ$ which means $\mathbb{T} = \mathbb{T}^\circ$.

This also gives $Z_{\mathbb{G}}(X_s) = Z_{\mathbb{G}}(\mathbb{T})$. It is therefore connected by [Bor91, Corollary 11.12]. That $Z_{\mathbb{G}}(X_s)$ is reductive is in [Bor91, §13.17, Corollary 2]. It is a Levi subgroup after some base change by [Bor91, Proposition 20.4]. $\square$

Remark 3.13. Proposition 3.12 implies, and gives a proof⁸, of the algebraic group version that $Z_{\mathbb{G}}(X_s)$ is connected for any field of characteristic 0. This is not true

⁷Meaning a Lagrangian subspace as in https://en.wikipedia.org/wiki/Symplectic_vector_space#Lagrangian_form.

⁸The other simple proof Cheng-Chiang knows uses real/complex Lie groups and is very similar.

for semisimple $s \in G$ in the group. A counter-example is $\mathbb{G} = \mathrm{SO}_V$ for $V = F^3$ with $(\vec{x}, \vec{y}) = x_1y_1 + x_2y_2 + x_3y_3$ and $s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ and $Z_{\mathbb{G}}(s) \cong \mathrm{O}_2$. The analogue of Proposition 3.12 is also not true over a field of (small) positive characteristic, e.g. for $\mathbb{G} = \mathrm{SO}_3/\mathbb{F}_2$ and $X_s = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Theorem 3.14. *Let X_s be semisimple. Then $\mathbb{O} = \mathrm{Ad}(\mathbb{G})X_s$ is Zariski closed, and $O = \mathrm{Ad}(G)X_s \subset \mathfrak{g}$ is a closed sub-manifold.*

Proof. The first assertion is [Bor91, Theorem 9.2]. The second assertion follows. $\square$

Next, we look at nilpotent elements.

Theorem 3.15. *Let $X_n \in \mathfrak{g}$ be nilpotent. Then there exists an algebraic group homomorphism $\lambda : \mathrm{GL}_1 \rightarrow \mathbb{G}$ such that*

$$\mathrm{Ad}(\lambda(t))X_n = t^2X_n$$

for any $t \in F^\times$.

Proof. We sketch the construction in [Car93, §5.5]. Elements $H \in \mathfrak{g}$ and $Y \in \mathfrak{g}$ are constructed so that

$$[X_n, Y] = H, [H, X_n] = 2X_n, [H, Y] = -2Y.$$

Then one constructs

$$\lambda(t) = \exp(-t^{-1}Y) \exp(tX_n) \exp(-t^{-1}Y) \exp(Y) \exp(-X_n) \exp(Y)$$

for any $t \in \bar{F}^\times$. This gives an algebraic map defined over F . $\square$

Remark 3.16. The construction may look a bit weird, but is in fact very natural and comes from the base case $G = \mathrm{SL}_2$ and $X_n = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $Y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ and $H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

In this case $\lambda(t) = \begin{bmatrix} t & 0 \\ 0 & t^{-1} \end{bmatrix}$.

Corollary 3.17. *Let $X_n \in \mathfrak{g}$ be nilpotent. Then for any $t \in F^\times$, we have $t^2X_n \in \mathrm{Ad}(G)X_n$.*

Remark 3.18. We can compare with the linear algebra fact, that a matrix A is conjugate to tA for some $t \in F^\times$ that is not a root of unity, or every $t \in F^\times$, if and only if A is nilpotent.

Corollary 3.19. *Let $X_n \in \mathfrak{g}$ be nilpotent. Then 0 is contained in the closure of $\mathrm{Ad}(G)X_n$. More generally, suppose $X = X_s + X_n$ is a Jordan decomposition. Then $\mathrm{Ad}(G)X_s$ is contained in the closure of $\mathrm{Ad}(G)X$.*

In particular, for general $X \in \mathfrak{g}$, the orbit $\mathrm{Ad}(g)X$ is closed if and only if X is semisimple. Same results are true for the algebraic group versions.

Proof. For X_n nilpotent, $\text{Ad}(G)X_n$ contains t^2X_n which converges to 0 as $|t| \rightarrow 0$. For general $X = X_s + X_n$, since $Z_{\mathbb{G}}(X_s)$ is reductive, we have that

$$t^2X_n \in \text{Ad}(Z_G(X_s))X_n \implies X_s + t^2X_n \in \text{Ad}(G)(X_s + X_n).$$

And thus X_s , or consequently $\text{Ad}(G)(X_s)$, is in the closure of $\text{Ad}(G)(X_s + X_n)$. The corresponding algebraic group version follows from the Lie group version directly. $\square$

It is also good to know that

Lemma 3.20. *We have $Z_G(X) \subset Z_G(X_s)$, $Z_G(X) \subset Z_G(X_n)$, $Z_{\mathfrak{g}}(X) \subset Z_{\mathfrak{g}}(X_s)$ and $Z_{\mathfrak{g}}(X) \subset Z_{\mathfrak{g}}(X_n)$.*

Proof. Suppose $\text{Ad}(g)X = X$, then $X = \text{Ad}(g)(X_s + X_n) = \text{Ad}(g)X_s + \text{Ad}(g)X_n$ and therefore $\text{Ad}(g)X_s = X_s$ and $\text{Ad}(g)X_n = X_n$. The result for the Lie algebra follows from Lemma 3.3 for X , X_s and X_n . $\square$

Definition 3.21. (i) Denote by $\mathfrak{g}/G$ (resp. $\mathfrak{g}^*/G$) the set of $\text{Ad}(G)$ -orbits in $\mathfrak{g}$ (resp. $\text{Ad}^*(G)$ -orbits in $\mathfrak{g}^*$).

(ii) Denote by $\mathfrak{g}^{\text{nil}} \subset \mathfrak{g}$ the subset of nilpotent elements. We call $\mathfrak{g}^{\text{nil}}$ the **nilpotent cone**. We have $\mathfrak{g}^{\text{nil}}/G \subset \mathfrak{g}/G$.

Remark 3.22. $\mathfrak{g}^{\text{nil}}$ is closed, and is a cone, i.e. X_n nilpotent $\implies tX_n$ nilpotent for $t \in F$; this is obvious with any embedding $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(F)$.

Theorem 3.23. *The set $\mathfrak{g}^{\text{nil}}/G \subset \mathfrak{g}/G$ of nilpotent orbits is finite. As a consequence, for any $O \in \mathfrak{g}/G$ we have $\{O' \in \mathfrak{g}/G \mid O' \leq O\}$ finite.*

Sketch. That the second sentence follows from the first sentence is Proposition 3.40 (where it will be obvious that the proof has nothing that could depends on this theorem or anything after).

For the first sentence, there are many proofs. Most of the proof Cheng-Chiang knows use Galois cohomology. One first proves that the set of $\mathbb{G}$ -orbits in $(\text{Lie } \mathbb{G})^{\text{nil}}$ is finite. This can be proved in many ways, including (a) using classification of nilpotent orbits [CM93, §3] similar to the proof of Theorem 3.15, (b) using the reductivity of $\mathfrak{g}$ and reduce to $\mathfrak{gl}_n$, or (c) by comparing with finite groups of Lie type. Once we know that the set of $\mathbb{G}$ -orbits in $(\text{Lie } \mathbb{G})^{\text{nil}}$ is finite, the rest follows from Remark 3.5.

There is another proof [DeB02] that only works with some additional assumption on p . It does not use the Galois cohomology but uses Bruhat-Tits theory instead, and gives an important description of $\mathfrak{g}^{\text{nil}}/G$. $\square$

3.4. Coinvariants. We work with $\mathfrak{g}^*$ once again, and transport all results from §3.3 using Theorem 3.1. We however begin with the remark that Corollary 3.19 implies the part of Theorem 3.1 that the notion of semisimple and nilpotent elements in $\mathfrak{g}^*$ does not depend on $\iota : \mathfrak{g}^* \cong \mathfrak{g}$.

Let $X^* \in \mathfrak{g}^*$ and $O^* := \text{Ad}^*(G)X^*$ be as before. We are ready to look at $I_{O^*} : \mathcal{C}_c^\infty(O^*) \rightarrow \mathcal{C}$.

Definition 3.24. *The coinvariant $\mathcal{C}_c^\infty(O^*)_G$ is*

$$\mathcal{C}_c^\infty(O^*) / \langle \text{Ad}^*(g)_* f - f \mid f \in \mathcal{C}_c^\infty(O^*), g \in G \rangle$$

Lemma 3.25. *I_{O^*} factors through $\mathcal{C}_c^\infty(O^*)_G$, and $\dim_{\mathcal{C}}(\mathcal{C}_c^\infty(O^*)_G) = 1$.*

[CC: The previous proof was wrong, and a correct (brand new) proof was updated on March 16th.]

Proof. The first assertion is obvious. To prove the second assertion, let us understand what O^* looks like locally. The map $(X^* \mapsto \text{Ad}^*(g)X^*) : G \twoheadrightarrow O^*$ is surjective on tangent spaces and induces a bijection $G/Z_G(X^*) \cong O^*$ (of sets). Since the group structure of G locally near e looks like $m(x, y) = x + y + O(|xy|)$, we may choose open compact subgroups $J_a \subset G$ with $J'_a = J_a \cap Z_G(X^*)$ as in (the proof of) Lemma 2.13, for sufficiently large a , such that locally $J_a \cong (\mathfrak{m}_F^a)^{\dim G} \supset (\mathfrak{m}_F^a)^{\dim Z_G(X^*)} \cong J'_a$, and such that $(\mathfrak{m}_F^a)^{\dim G - \dim Z_G(X^*)} \cong \text{Ad}^*(J_a)X^*$. (Implicit function theorem is being used again and again.)

Every function $f \in \mathcal{C}_c^\infty(\text{Ad}^*(J_a)X^*)$ is locally constant by some $\text{Ad}^*(J_b)$ for some big enough $b > a$, and thus is a function on $(\mathfrak{m}_F^a/\mathfrak{m}_F^b)^{\dim G - \dim Z_G(X^*)}$. The group $J_a/J_b \cong (\mathfrak{m}_F^a/\mathfrak{m}_F^b)^{\dim G}$ acts transitively on such functions. Taking the union for all $b > a$, this shows that $\dim_{\mathcal{C}} \mathcal{C}_c^\infty(\text{Ad}^*(J_a)X^*)_{J_a} = 1$. Since G acts transitively on O^* , and every $f \in \mathcal{C}_c^\infty(O^*)$ can be written as a linear combination of functions on small enough neighborhoods, this shows that $\dim \mathcal{C}_c^\infty(O^*)_G \leq 1$. That $\dim_{\mathcal{C}} \mathcal{C}_c^\infty(O^*)_G = 1$ then follows from that I_{O^*} is non-trivial. $\square$

To proceed, we need a partial order on $\mathfrak{g}/G$ and $\mathfrak{g}^*/G$:

Definition 3.26. (1) For $O_1, O_2 \in \mathfrak{g}/G$, we say $O_1 \geq O_2$ (and $O_2 \leq O_1$) if O_2 is contained in the closure of O_1 . We say $O_1 > O_2$ (and $O_2 < O_1$) if $O_1 \geq O_2$ and $O_1 \neq O_2$. Same definitions are given for $\mathfrak{g}^*/G$.

Remark 3.27. It is easy to see that $\geq$ defines a partial order on $\mathfrak{g}/G$. It also follows from definition that, for any $O \in \mathfrak{g}/G$ we have

$$\overline{O} := \text{closure of } O = \bigcup_{O' \leq O} O'.$$

Lemma 3.28. Suppose $O_1 > O_2$. Then $\dim O_1 > \dim O_2$.

Proof. Write $O_1 = \text{Ad}(G)X_1$ and $\mathbb{O}_1 = \text{Ad}(\mathbb{G})X_1$. Also $O_2 = \text{Ad}(G)X_2$ and $\mathbb{O}_2 = \text{Ad}(\mathbb{G})X_2$. We have seen that O_1 is Zariski dense in $\mathbb{O}_1$. That O_2 is contained in the closure of O_1 implies that $\mathbb{O}_2$ is in the (Zariski) closure of $\mathbb{O}_1$. Hence either $\mathbb{O}_2 = \mathbb{O}_1$, or $\dim O_2 = \dim \mathbb{O}_2 < \dim \mathbb{O}_1 = \dim O_1$. We show that $\mathbb{O}_2 = \mathbb{O}_1$ is impossible. Suppose otherwise, then we have seen that O_1 and O_2 are both closed in $\mathbb{O}_1(F) = \mathbb{O}_2(F)$, which is a contradiction. $\square$

Fix any O^* and consider

$$\overline{O^*} = \bigcup_{(O')^* \leq O^*} (O')^*, \quad \partial O^* = \bigcup_{(O')^* < O^*} (O')^*$$

Note that $O^* \subset \overline{O^*}$ is open as we have seen O^* is locally closed. We have a short exact sequence

$$0 \rightarrow \mathcal{C}_c^\infty(O^*) \rightarrow \mathcal{C}_c^\infty(\overline{O^*}) \rightarrow \mathcal{C}_c^\infty(\partial O^*) \rightarrow 0.$$

Here the first map is extension by zero using that $O^* \subset \overline{O^*}$ is open, and the second map is restriction, using that $\partial O^* \subset \overline{O^*}$ is closed.

Exercise 9. Show that for any p -adic manifold X and closed subsets $Z \subset Y \subset X$ we have a surjection $\mathcal{C}_c^\infty(Y) \twoheadrightarrow \mathcal{C}_c^\infty(Z)$. In particular $\mathcal{C}_c^\infty(\overline{O^*}) \rightarrow \mathcal{C}_c^\infty(\partial O^*)$ is surjective.

The short exact sequence induces a right exact sequence

$$(6) \quad \mathcal{C}_c^\infty(O^*)_G \rightarrow \mathcal{C}_c^\infty(\overline{O^*})_G \rightarrow \mathcal{C}_c^\infty(\partial O^*)_G \rightarrow 0$$

Corollary 3.29. When $\text{char}(\mathcal{C}) = 0$, the sequence (6) is furthermore exact on the left.

Proof. When $\mathcal{C} = \mathbb{C}$, this follows from the important Theorem 3.11, where absolute convergence is used to extend the functional I_{O^*} from $\mathcal{C}_c^\infty(O^*)_G$ to $\mathcal{C}_c^\infty(\overline{O^*})_G$ in a non-trivial way. The existence of such a functional implies that $\mathcal{C}_c^\infty(O^*)_G \rightarrow \mathcal{C}_c^\infty(\overline{O^*})_G$ is non-trivial, therefore injective, since $\dim_{\mathcal{C}}(\mathcal{C}_c^\infty(O^*)_G) = 1$.

That the sequence is exact for $\mathcal{C} = \mathbb{C}$ implies that it is exact for $\mathcal{C} = \mathbb{Q}$, and therefore exact whenever $\text{char}(\mathcal{C}) = 0$. $\square$

Fact 3.30. [VW01, §B] For any field $\mathcal{C}$ (of characteristic different from p), the sequence (6) is also exact on the left.

We see that, although the original statement and proof of Theorem 3.11 involves (real) analysis, Corollary 3.29 is completely algebraic. The curious situation is that Theorem 3.11 and Corollary 3.29 are *not* needed for our theory at this stage, as we will see in the next subsection.

Exercise 10. The **wave-front set** of $O \in \mathfrak{g}/G$ is defined to be

$$\{Y \in \mathfrak{g} \mid \exists t_i \in (F^\times)^2 \text{ and } X_i \in O \text{ such that } \lim t_i = 0 \text{ and } \lim t_i X_i = Y\}.$$

The determination of wave-front sets, for example for SO_V and Sp_V , is an open problem. (The answer is known, but rather non-trivial, when $F = \mathbb{R}$ and $(F^\times)^2 = \mathbb{R}_{>0}$.)

Suppose $\mathbb{G} = \text{GL}_n$. Show that the wave-front set of any $O \in \mathfrak{g}/G$ contains the closure of the nilpotent orbit $\text{Ad}(G)N \in \mathfrak{g}^{\text{nil}}/G$ that appears in its rational canonical form of O . That is, N is the strictly lower triangular part of the rational canonical form, as in https://en.wikipedia.org/wiki/Frobenius_normal_form#Example.

Exercise 11. (*) Show that in Exercise 10, the wave-front set actually **is** the closure of the indicated orbit.

3.5. Nilpotent orbital integrals, whenever they exist? Let us begin with a definition.

Definition 3.31. Let $E \subset \mathfrak{g}^*$ be any $\text{Ad}^*(G)$ -invariant subset. We denote by $D(E) := \text{Hom}_{\mathcal{C}}(\mathcal{C}_c^\infty(E), \mathcal{C})$ the space of $\mathcal{C}$ -linear functionals on $\mathcal{C}_c^\infty(E)$ (with no continuity condition). Suppose E is stabilized by $\text{Ad}^*(G)$. We denote by $D(E)^G$ the subspace of those functionals invariant under $\text{Ad}^*(g)$ for any $g \in G$. Equivalently

$$D(E)^G := \text{Hom}_{\mathcal{C}}(\mathcal{C}_c^\infty(E)_G, \mathcal{C}).$$

They will be called **invariant distributions** on E .

We fix an element $O^* \in (\mathfrak{g}^*)^{\text{nil}}$. The sequence (6) can be dualized to

$$(7) \quad 0 \rightarrow D(\partial O^*)^G \rightarrow D(\overline{O^*})^G \rightarrow D(O^*)^G$$

We have already defined $I_{O^*} \in D(O^*)^G$. For any function $f \in \mathcal{C}_c^\infty(O^*)$, and any $t \in F$, we denote by $f_{t^2} \in \mathcal{C}_c^\infty(O^*)$ given by $f_{t^2}(X) := f(t^2X)$. We have

Proposition 3.32. $I_{O^*}(f_{t^2}) = |t|^{-\dim O^*} I_{O^*}(f)$ for any $f \in \mathcal{C}_c^\infty(O^*)$ and $t \in F^\times$.

Proof. We have seen in Theorem 3.15 that $(X \mapsto t^2X)$ stabilizes O^* . The distribution I_{O^*} is defined by integrating the form $\bigwedge^{\frac{\dim O^*}{2}} \omega_{O^*}$. At $X^* \in O^*$, the 2-form ω_{X^*} is given by $\omega_{X^*}(Y, Z) = \langle [Y, Z], X^* \rangle$. In particular, (the push-forward of) the map $(X \mapsto t^2X)$ sends ω_{O^*} to $t^{-2}\omega_{O^*}$, and therefore $(X \mapsto t^2X)$ sends $\bigwedge^{\frac{\dim O^*}{2}} \omega_{O^*}$ to $t^{-\dim O^*} \cdot \bigwedge^{\frac{\dim O^*}{2}} \omega_{O^*}$.

The function f_{t^2} is the pullback of f under $(X \mapsto t^2X)$. Therefore

$$\int_{O^*} f_{t^2} \cdot |\omega| = \int_{O^*} f \cdot (X \mapsto t^2X)_* |\omega| = \int_{O^*} f \cdot |(X \mapsto t^2X)_* \omega| = |t|^{-\dim O^*} \int_{O^*} f |\omega|$$

which proves the Proposition. $\square$

Definition 3.33. Let $E \subset \mathfrak{g}^*$ be a $\text{Ad}^*(G)$ -invariant subset that is furthermore invariant under $(X \mapsto t^2X)$ for any $t \in F^\times$. For $k \in \mathbb{Z}$, we say $\delta \in D(E)^G$ is of **weight**⁹ k if

$$\delta(f_{t^2}) = |t|^k \cdot \delta(f)$$

for any $f \in \mathcal{C}_c^\infty(E)$ and $t \in F^\times$. Here $|t|^k \in q^\mathbb{Z}$ is mapped into $\mathcal{C}$ in the obvious way. We say $\delta \in D(E)^G$ is of **weights** W for a set W of integers if $\delta = \sum \delta_{k'}$ is a finite sum of elements of weight k' for some $k' \in W$. (Not all elements in W have to appear.)

We say a subspace of $D(E)^G$ is of some **weight(s)** if every element in that subspace is of those weights.

Remark 3.34. A non-zero element of weight k is of weight $k + a$ if and only if $q^a = 1$ in $\mathcal{C}$.

Proposition 3.35. Let $O^* \in (\mathfrak{g}^*)^{\text{nil}}/G$. The space $D(\overline{O^*})^G$ is of weights $\{k \in 2\mathbb{Z} \mid -\dim O^* \leq k \leq 0\}$. We also have $\dim_{\mathcal{C}} D(\overline{O^*})^G \leq \#\{O' \in (\mathfrak{g}^*)^{\text{nil}}/G \mid O' \leq O^*\}$.

Proof. We prove by induction on O^* , but before that we need to state a slightly different induction hypothesis. Consider any subset of $(\mathfrak{g}^*)^{\text{nil}}/G$ that is a lower set, i.e. $L \subset (\mathfrak{g}^*)^{\text{nil}}/G$ such that if $O^* \in L$ and $(O^*)' < O^*$, then $(O^*)' \in L$. Denote by

$$c(L) := \bigsqcup_{(O^*)' \in L} (O^*)' \subset \mathfrak{g}^*.$$

The proposition then follows from Lemma 3.36 below, applied to the lower set $L = \{(O^*)' \mid (O^*)' \leq O^*\}$. $\square$

Lemma 3.36. For any lower set $L \subset (\mathfrak{g}^*)^{\text{nil}}/G$, the space $D(c(L))^G$ is of weights $\{-\dim(O^*) \mid O^* \in L\}$ and $\dim_{\mathcal{C}} D(c(L))^G \leq \#L$.

Proof. We use induction on $\#L$. Suppose $L' \subset L$ is any subset that is also a lower set. Then $c(L \setminus L') \subset c(L)$ is closed, and we denote by $c'(L') := c(L) \setminus c(L \setminus L')$ the open

⁹I just invented this notion; it is not used in the community.

subset of $c(L)$. In fact $c(L \setminus L') = \partial c'(L')$. we have the following left exact sequence generalizing (7)

$$(8) \quad 0 \rightarrow D(c(L \setminus L'))^G \rightarrow D(c(L))^G \rightarrow D(c'(L'))^G.$$

When $L = \emptyset$ the assertion is trivial. Otherwise, let $O^* \in L$ be maximal and $L' := L \setminus \{O^*\}$. By induction the lemma holds for L' . The result then follows from (8), Lemma 3.25 and Proposition 3.32. $\square$

Corollary 3.37. *Let $O^* \in (\mathfrak{g}^*)^{\text{nil}}/G$. Suppose $q^a \neq 1$ in $\mathcal{C}$ for any $a \in \{k \in 2\mathbb{Z} \mid 2 \leq k \leq \dim O^*\}$. Then $D(\overline{O^*})$ has at most one line of weight $\dim O^*$. In this case, such a line exists if and only if (7) is furthermore right exact.*

Proof. This follows from (7) in which all the maps respect weights, and Lemma 3.36 applied to $L := \{(O')^* \in (\mathfrak{g}^*)^{\text{nil}}/G \mid (O')^* < O^*\}$. $\square$

Definition 3.38. *Suppose $q^a \neq 1$ in $\mathcal{C}$ for any $a \in \{k \in 2\mathbb{Z} \mid 2 \leq k \leq \dim O^*\}$. Then by abuse of language we denote by $I_{O^*} \in D(\overline{O^*})^G \subset D((\mathfrak{g}^*)^{\text{nil}})^G$ the unique element in $D(\overline{O^*})$ of weight $-\dim O^*$ that maps to I_{O^*} in (5), if it exists.*

The next corollary follows from induction and Corollary 3.37.

Corollary 3.39. *Suppose $q^a \neq 1$ in $\mathcal{C}$ for any even integer a with $0 < a \leq \dim O^*$ for some $O^* \in (\mathfrak{g}^*)^{\text{nil}}/G$. Then for any lower set $L \subset (\mathfrak{g}^*)^{\text{nil}}/G$, the space $D(c(L))^G$ has a basis given by $\{I_{O^*}\}$ in Definition 3.38 for those $O^* \in L$ for which I_{O^*} exists. In particular, $D((\mathfrak{g}^*)^{\text{nil}})^G$ has a basis given by $\{I_{O^*}\}$ for those $O^* \in (\mathfrak{g}^*)^{\text{nil}}/G$ for which I_{O^*} exists.*

In the end, by Corollary 3.29 and Fact 3.30 the element $I_{O^*} \in D(\overline{O^*})^G$ always exist. In fact, the cited proof of Fact 3.30 relies on Corollary 3.29, as some kind of lifting to characteristic 0, and both of them rely on the theorem of Ranga Rao and Deligne (Theorem 3.11). However, Proposition 3.35 and Corollary 3.39 will be what we really need for the character theory of p -adic reductive groups.

Exercise 12. (*) *Let p be any odd prime, $\mathcal{C} = \mathbb{Q}$ and $\mathbb{G} = \text{SL}_2$ over F . The Lie algebra $\mathfrak{sl}_2$ can be identified with its dual using $\langle A, B \rangle = \text{tr}(AB)$. Let $\mathfrak{g}(\mathcal{O}_F) = \mathfrak{sl}_2(\mathcal{O}_F) \subset \mathfrak{sl}_2(F)$ be those traceless 2×2 matrices in $\mathcal{O}_F$. Let $X^* := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \in (\mathfrak{g}^*)^{\text{nil}}$ and $O^* := \text{Ad}^*(G)X^*$. Show that*

$$I_{O^*}(1_{\mathfrak{g}(\mathcal{O}_F)}) = \frac{q}{2}.$$

The previous exercise shows that nilpotent orbital integrals as in Definition 3.38 sort of does not exist when $G = \text{SL}_2$ and $\text{char}(\mathcal{C}) = 2$. I thank Gemini Deep Think from Google Deepmind who found this example.

3.6. General orbital integrals. Let us work with $\mathfrak{g}$ for a moment, and discuss about reduction of a general element to its semisimple and nilpotent parts.

Proposition 3.40. *Let $X, Y \in \mathfrak{g}$. The following are equivalent:*

(i) *The element Y is contained in the closure of $\text{Ad}(G)X$, i.e.*

$$\text{Ad}(G)Y \leq_{\mathfrak{g}/G} \text{Ad}(G)X.$$

- (ii) *There exists $g \in G$ such that $\text{Ad}(g)Y_s = X_s$, and by putting $\mathbb{H} := Z_{\mathbb{G}}(X_s)$, $H := \mathbb{H}(F) = Z_G(X_s)$ and $\mathfrak{h} := \text{Lie } H$, we have that $\text{Ad}(g)Y_n$ is in the closure of $\text{Ad}(H)X_n$, i.e.*

$$\text{Ad}(H)(\text{Ad}(g)Y_n) \leq_{\mathfrak{g}/H} \text{Ad}(H)X_n.$$

Proof. Suppose (ii) holds. Then there exists $h_i \in H$ such that $\lim \text{Ad}(h_i)X_n = \text{Ad}(g)Y_n$. We have

$$\lim \text{Ad}(g^{-1}h_i)X = \text{Ad}(g^{-1})X_s + \text{Ad}(g^{-1})\text{Ad}(g)Y_n = Y_s + Y_n = Y.$$

Conversely, let $g_i \in G$ be such that $\lim \text{Ad}(g_i)X = Y$. Choose an embedding $\mathbb{G} \hookrightarrow \text{GL}_n$. The element Y and every $\text{Ad}(g_i)X$, viewed as $n \times n$ matrices, have the same characteristic polynomial P , and we have an equality of $n \times n$ matrices

$$Y_s = Q_P(Y), \quad \text{Ad}(g_i)X_s = Q_P(\text{Ad}(g_i)X)$$

where Q_P is an F -coefficient polynomial that depends only on P . Hence we have

$$\lim \text{Ad}(g_i)X_s = \lim g_i Q_P(X) g_i^{-1} = \lim Q_P(g_i X g_i^{-1}) = \lim Q_P(Y) = Y_s.$$

Since $\text{Ad}(G)X_s$ is closed, we have $Y_s \in \text{Ad}(G)X_s$, i.e. $\text{Ad}(g)Y_s = X_s$ for some g and

$$\lim \text{Ad}(gg_i)X_s = X_s.$$

Since $G \twoheadrightarrow \text{Ad}(G)X_s$ is surjective on the tangent space, we may choose $g'_i \in G$ with $\lim g'_i = e$ such that $\text{Ad}(g'_i)X_s = \text{Ad}(gg_i)X_s$, i.e. $(g'_i)^{-1}gg_i \in H$. We have

$$\begin{aligned} \lim \text{Ad}((g'_i)^{-1}gg_i)X &= \lim \text{Ad}(gg_i)X = \text{Ad}(g)Y \\ \implies \lim \text{Ad}((g'_i)^{-1}gg_i)X_n &= \lim \text{Ad}((g'_i)^{-1}gg_i)(X - X_s) \\ &= \text{Ad}(g)Y - \text{Ad}(g)Y_s = \text{Ad}(g)Y_n \end{aligned}$$

which implies (ii). $\square$

Now we are ready to say what happens in general. Let $X^* = X_s^* + X_n^* \in \mathfrak{g}$ be a Jordan decomposition. Let $\mathbb{H} := Z_{\mathbb{G}}(X_s^*)$ and $H = Z_G(X_s^*)$. We will show that

Proposition 3.41. *(Corollary 5.32) There is a canonical isomorphism $D(\overline{\text{Ad}(H)X_n^*})^H \cong D(\overline{\text{Ad}(G)X^*})^G$ sending $I_{X_s^*} = I_{\text{Ad}(H)X_s^*} \in D(\overline{\text{Ad}(H)X_n^*})^H$ to $I_{\text{Ad}(G)X_s^*} \in D(\overline{\text{Ad}(G)X^*})^G$.*

In particular, results in Corollary 3.37 and Corollary 3.39 applies to $D(\overline{\text{Ad}(G)X})^G$.

4. FOURIER TRANSFORMS

In §3.1, we had considered

$$(9) \quad \psi_p : \mathbb{Q}_p \rightarrow S^1$$

by extending $\psi = (n \mapsto e^{2\pi i n}) : \mathbb{Z}[1/p] \rightarrow S^1$, and $\psi'_F : F \xrightarrow{\text{tr}_{F/\mathbb{Q}_p}} \mathbb{Q}_p \xrightarrow{\psi_p} S^1$. We would like to replace S^1 by $\mathcal{C}^\times$. Recall that $\mathcal{C}$ is a field of characteristic different from p . In this section we assume that

Hypothesis 4.1. *For any $n \in \mathbb{Z}_{\geq 0}$, we have $\#\mu_{p^n}(\mathcal{C}) = \#\mu_{p^n}(\bar{\mathcal{C}}) = p^n$, where $\mu_{p^n}(\mathcal{C}) := \{x \in \mathcal{C} \mid x^{p^n} = 1\}$.*

Write also $\mu_{p^\infty}(\mathcal{C}) := \{x \in \mathcal{C} \mid x^{p^n} = 1 \text{ for some } n \in \mathbb{Z}_{\geq 0}\}$. In this case it is easy to see that we have abstract group isomorphisms

$$\mu_{p^\infty}(\mathcal{C}) \cong \mathbb{Z}[1/p]/\mathbb{Z} \cong \mathbb{Q}_p/\mathbb{Z}_p.$$

The field $k_F := \mathcal{O}_F/\mathfrak{m}_F$, as an abelian group, is p -torsion, and is contained in $F/\mathfrak{m}_F$ which is p^∞ -torsion just like $\mathbb{Z}[1/p]/\mathbb{Z}$. Using Zorn's lemma or some explicit $\mathbb{Q}_p$ -basis of F , we have

Lemma 4.2. *There exists an abelian group homomorphism $\psi_F : F/\mathfrak{m}_F \rightarrow \mu_{p^\infty}(\mathcal{C})$ that is non-trivial on $\mathcal{O}_F/\mathfrak{m}_F$.*

Exercise 13. *Prove Lemma 4.2, and furthermore that $\mathcal{O}_F^\times$ acts simply transitively on the set of such ψ_F .*

We now consider any such ψ_F , viewed as a locally constant function $\psi_F : F \rightarrow \mu_{p^\infty}(\mathcal{C}) \subset \mathcal{C}^\times$. [CC: Warning: in §3.1 we had defined another ψ_F , now relabeled as ψ'_F , that differs from such ψ_F by some scaling.] Let V be a finite-dimensional vector space over F . A lattice $\Lambda \subset V$ is a finitely generated $\mathcal{O}_F$ -submodule with $\text{rank}_{\mathcal{O}_F} \Lambda = \text{rank}_F V$, i.e. such that $V = F \cdot \Lambda$. (Note that $\mathcal{O}_F$ is a PID.) Let $V^* := \text{Hom}_F(V, F)$ be the linear dual.

Definition 4.3. *Let $\Lambda \subset V$ be a lattice. We write*

$$\Lambda^\perp := \{X^* \in V^* \mid \langle X^*, Y \rangle \in \mathfrak{m}_F, \forall Y \in \Lambda\}.$$

Remark 4.4. $\Lambda^\perp$ is obviously a lattice. $(\Lambda^\perp)^\perp = \Lambda \in V = (V^*)^*$; this can be seen by choosing a basis and reduce to the case $V = F$, in which case $((\mathfrak{m}_F)^n)^\perp = (\mathfrak{m}_F)^{1-n}$. Also $\Lambda_1 \subset \Lambda_2 \iff \Lambda_1^\perp \supset \Lambda_2^\perp$.

Consider a measure $\nu : \{\Lambda \subset V \text{ lattice}\} \rightarrow p^\mathbb{Z}$ with $\nu(\Lambda) := [\Lambda : \Lambda_0] \nu(\Lambda_0)$ for **any** choice of Λ_0 and $\nu(\Lambda_0) \in p^\mathbb{Z}$. Let M_V be the $\mathbb{Z}[1/p]$ -module generated by ν . We have that $M_c^\infty(V) := \mathcal{C}_c^\infty(V) \otimes_{\mathbb{Z}[1/p]} M_V$ where $M_c^\infty(V)$ is as in Definition 2.50.

Let $\nu^\perp : \{\Lambda^* \subset V^* \text{ lattice}\} \rightarrow \mathbb{Z}[1/p]$ be such that $\nu(\Lambda) \cdot \nu^\perp(\Lambda^\perp) = 1$ for any lattice $\Lambda \subset V$.

Definition 4.5. *For $X^* \in V^*$ and $Y \in V$, we write*

$$\psi_{X^*}(Y) := \psi(\langle X^*, Y \rangle) \in \mathcal{C}^\infty(V).$$

Definition 4.6. *Let $\mu \in M_c^\infty(V)$. We define its Fourier transform $\hat{\mu} \in \mathcal{C}_c^\infty(V^*)$ by*

$$\hat{\mu}(X^*) := \int_V \psi_{X^*} \cdot \mu$$

Equivalently, for $f \in \mathcal{C}_c^\infty(V)$ we define its Fourier transform (with respect to ν) by

$$\hat{f} := \widehat{(f \otimes \nu)}, \text{ i.e. } \hat{f}(X^*) := \int_V \psi_{X^*}(Y) \cdot f(Y) \cdot \nu(Y)$$

Remark 4.7. The element $\widehat{(f \otimes \nu)} \otimes \nu^\perp \in M_c^\infty(V^*)$ is canonical and does not depend the choice of ν , because $(p\nu)^\perp = p^{-1} \cdot \nu^\perp$. Therefore, canonically speaking Fourier transforms send functions to measures and measures to functions. Some author will define “half-measures” so that Fourier transforms act on half-measures.

Exercise 14. Let $f \in \mathcal{C}_c^\infty(V)$. Show that with respect to ν and $\nu^\perp$ we have $\hat{\hat{f}}(X) = f(-X)$. In particular $(\mu \mapsto \hat{\mu})$ or $(f \mapsto \hat{f})$ is an isomorphism of vector spaces, with $\hat{\hat{f}} = f$.

We write $\check{f} := \hat{\hat{f}}$, so that $\check{f} = f$. Same for $\check{\mu} := \hat{\hat{\mu}}$.

[CC: The following lemma is supposed to be part of the above exercise. Ouch.]

Lemma 4.8. For $f \in \mathcal{C}_c^\infty(V)$ and $\mu \in M_c^\infty(V)$, we have

$$\int_V f \cdot \mu = \int_V \hat{f} \cdot \check{\mu}.$$

From now on we fix ν and $\nu^\perp$, and abbreviate things like $\int_{Z \in V} f(Z) \nu(Z)$ to $\int_V f(Z) dZ$.

Definition 4.9. We denote by

- (1) $\mathcal{C}_c(V^*/\Lambda^\perp) \subset \mathcal{C}_c^\infty(V^*)$ the subspace of those functions such that $f(X^*) = f(X^* + Y^*)$ for $X^* \in V^*$, $Y^* \in \Lambda^\perp$.
- (2) $\mathcal{C}^\infty(\Lambda) \subset \mathcal{C}^\infty(V)$ the subspace of those functions supported on Λ , and $M^\infty(\Lambda) \subset M_c^\infty(V)$ is given by $M^\infty(\Lambda) = \mathcal{C}^\infty(\Lambda) \otimes_{\mathbb{Z}[1/p]} M_V$.

Lemma 4.10. For any $Z \in V \setminus \Lambda$ we have

$$\int_{\Lambda^\perp} \psi_{Y^*}(Z) dY^* = 0$$

Proof. Since $\Lambda = (\Lambda^\perp)^\perp$, there exists $W^* \in \Lambda^\perp$ such that $\langle W^*, Z \rangle \notin \mathfrak{m}_F$. Choose $a \in \mathcal{O}_F$ such that $\psi_F(a) \neq 1$. Multiplying W^* by an element in $\mathcal{O}_F$ we may suppose $\langle W^*, Z \rangle = a \implies \psi_{W^*}(Z) = \psi(\langle W^*, Z \rangle) \neq 1$. Note that $W^* + \Lambda^\perp = \Lambda^\perp$. Hence

$$\int_{\Lambda^\perp} \psi_{Y^*}(Z) dY^* = \int_{\Lambda^\perp} \psi_{Y^* + W^*}(Z) dY^* = \psi_{W^*}(Z) \cdot \int_{\Lambda^\perp} \psi_{Y^*}(Z) dY^*$$

which implies the lemma. □

Corollary 4.11. We have a commutative diagram

$$\begin{array}{ccc} M_c^\infty(V) & \xrightarrow[\sim]{\mu \mapsto \hat{\mu}} & \mathcal{C}_c^\infty(V^*) \\ \downarrow \mu \mapsto \mu|_\Lambda & & \downarrow f \mapsto [1_{\Lambda^\perp}] * f \\ M^\infty(\Lambda) & \xrightarrow[\sim]{\mu \mapsto \hat{\mu}} & \mathcal{C}_c(V^*/\Lambda^\perp) \end{array}$$

where for $f \in \mathcal{C}_c^\infty(V^*)$ we define $[1_{\Lambda^\perp}] * f$ by

$$([1_{\Lambda^\perp}] * f)(X^*) := \int_{Y^* \in \Lambda^\perp} f(X^* + Y^*) \nu_{\Lambda^\perp}(Y^*)$$

where $\nu_{\Lambda^\perp}$ is the measure for which $\nu_{\Lambda^\perp}(\Lambda^\perp) = 1$.

Proof. That the Fourier transform is an isomorphism appears in Exercise 14; we'll prove that the diagram commutes. Let $\mu = f \otimes \nu$. Again hiding all ν and $\nu^\perp$ we have

$$\begin{aligned}
& ([1_{\Lambda^\perp}] * \hat{f})(X^*) = \int_{\Lambda^\perp} \hat{f}(X^* + Y^*) dY^* \\
&= \int_{\Lambda^\perp} \int_V \psi_{X^*+Y^*}(Z) f(Z) (Y^*) dZ dY^* = \int_V \int_{\Lambda^\perp} \psi_{X^*+Y^*}(Z) f(Z) dY^* dZ \\
&= \int_{\Lambda} \int_{\Lambda^\perp} \psi_{X^*+Y^*}(Z) f(Z) dY^* dZ + \int_{V \setminus \Lambda} \int_{\Lambda^\perp} \psi_{X^*+Y^*}(Z) f(Z) dY^* dZ \\
&\stackrel{*}{=} \int_{\Lambda} \psi_{X^*}(Z) f(Z) dZ + 0 = \widehat{f|_{\Lambda}}(X^*)
\end{aligned}$$

where $\stackrel{*}{=}$ follows from Lemma 4.10. $\square$

Definition 4.12. For a distribution $\delta \in D(V^*) = \text{Hom}_{\mathcal{C}}(\mathcal{C}_c^\infty(V^*), \mathcal{C})$. We define a generalized function $\hat{\delta} \in \mathcal{C}^{-\infty}(V) = \text{Hom}_{\mathcal{C}}(M_c^\infty(V), \mathcal{C})$ by

$$\hat{\delta}(\mu) = \delta(\hat{\mu}).$$

We now apply the mechanism of Fourier transforms to the case when $V = \mathfrak{g}$, and state the first version of our main theorem

Theorem 4.13. Assume Hypothesis 4.1 and fix an ψ_F as in Lemma 4.2.

- (1) For any distribution $\delta \in D(\mathfrak{g}^*)^G$ such that $\text{supp}(\delta) \in \text{Ad}(G)\Lambda^*$ for some lattice $\Lambda^* \subset \mathfrak{g}^*$, there exist a distribution $\delta_n \in D((\mathfrak{g}^*)^{\text{nil}})^G$ and a lattice $\Lambda \subset \mathfrak{g}$ such that

$$\widehat{\delta}|_{\Lambda} = \widehat{\delta}_n|_{\Lambda},$$

Equivalently (by Corollary 4.11), for any $f \in \mathcal{C}_c^\infty(\mathfrak{g}^*/\Lambda^\perp)$ we have

$$\delta(f) = \delta_n(f).$$

- (2) Let π be an irreducible admissible $\mathcal{C}$ -representation of G . Then there exist a distribution $\delta_n \in D((\mathfrak{g}^*)^{\text{nil}})^G$ and a lattice $\Lambda \subset \mathfrak{g}$ such that $\exp : \Lambda \xrightarrow{\sim} \exp(\Lambda)$ is an isomorphism, and

$$\Theta_\pi|_{\exp(\Lambda)} \stackrel{*}{=} \widehat{\delta}_n|_{\Lambda}$$

where $\stackrel{*}{=}$ is understood via an isomorphism $M_c^\infty(\Lambda) \cong M_c^\infty(\exp(\Lambda))$ induced by $\exp$.

Remark 4.14. A standard example for the Lie algebra part is $\delta = I_{O_s^*}$ for $O_s^* = \text{Ad}^*(G)X_s^*$ for some $X_s^* \in \mathfrak{g}^*$ semisimple; $\text{supp}(I_{O_s^*}) = O_s^* = \overline{O_s^*}$.

Remark 4.15. Theorem 4.13 describes $\hat{\delta}$ and Θ_π near the identity. In Theorem C' we will generalize this to arbitrary (semisimple) element.

Remark 4.16. By Corollary 3.39, as long as $q^a \neq 1$ for some a (e.g. when $\text{char}(\mathcal{C}) = 0$), the distribution δ_n has the form $\delta_n = \sum c_i I_{O_i^*}$ for a finite number of $c_i \in \mathcal{C}$ and $O_i^* \in (\mathfrak{g}^*)^{\text{nil}}/G$.

Remark 4.17. By Exercise 13, the validity of the theorem does not depend on the choice of ψ_F ; a different choice amount to scaling δ_n by an element in $\mathcal{O}_F^\times$.

5. THE STATEMENT OF LOCAL CHARACTER EXPANSIONS

5.1. Reductive $\implies$ unimodular. In this section we see the first place (among the several) where reductivity of $\mathbb{G}$ is used. Let $\mathbb{H}$ be any linear algebraic group, $H = \mathbb{H}(F)$ and $\mathfrak{h} := \text{Lie } H$.

Traditionally, Lie brackets are defined for vector fields. Indeed, the tangent space $\text{Lie } \mathbb{H} = T_e \mathbb{H}$ can be identified via left translation ($l_h : g \mapsto hg$) to $T_g \mathbb{H}$ by any $h \in \mathbb{H}(\bar{F})$. Hence $\text{Lie } \mathbb{H}$ can be identified with the space of left-invariant algebraic vector fields on $\mathbb{H}$. Likewise $\mathfrak{h}$ can be identified with the space of left-invariant analytic vector fields on H . In both the algebraic and analytic case, the Lie bracket agree with the commutator in the sense of derivations¹⁰.

Define $\mathfrak{h}^*$ as the dual vector space, but as an affine space over F . Then $\mathfrak{h}^*$ can be identified with the space of left-invariant analytic 1-forms on H . Consequently, $\bigwedge^{\text{top}}(\mathfrak{h}^*)$ is the space of left-invariant analytic top forms on H . Note that $\bigwedge^{\text{top}}(\mathfrak{h}^*)$ is the dual space of $\bigwedge^{\text{top}}(\mathfrak{h})$.

Lemma 2.10 allows us to integrate any such top form and get $|\omega|$ that assigns every open compact subset of G a positive value (“measure”) in $\mathbb{Z}[1/p]$. By construction $|\omega|$ is left-invariant, meaning that $|\omega|(gE) = |\omega|(E)$ for any $g \in G$. We are then wondering if $|\omega|$ is also right-invariant, or equivalently if $|\omega|(gEg^{-1}) = |\omega|(E)$ for any $g \in G$. This leads to

Definition 5.1. *The modulus character is $\Delta_H = \det(\text{Ad}) : H \rightarrow F^\times$ is the action of H on $\bigwedge^{\text{top}}(\mathfrak{h}^*)$ through $\text{Ad} : H \rightarrow \text{Aut}_F(\mathfrak{h}) = \text{Aut}_F(\mathfrak{h}^*)$.*

By definition Δ_H measures whether left-invariant top forms are right-invariant. The goal of this section is to prove that

Theorem 5.2. *Let $\mathbb{G}$ be a reductive group over F and $G = \mathbb{G}(F)$. Then $|\Delta_G| : G \rightarrow q^\mathbb{Z}$ is trivial. In this case we say G is **unimodular**. If $\mathbb{G}$ is also connected then in fact Δ_G is trivial.*

Example 5.3. It may takes more than one minute, but not too hard to check that Theorem 5.2 is true for $\mathbb{G} = \text{GL}_V, \text{SL}_V, \text{O}_V, \text{SO}_V$ and Sp_V . Therefore, although we will have two pages of algebraic groups to prove Theorem 5.2, it is not needed if one is restricting to such subgroups.

We review some definitions.

Definition 5.4. *Let $\mathbb{H}$ be a linear algebraic group. The **identity component** (also **neutral component**) $\mathbb{H}^\circ$ is the connected component of the variety $\mathbb{H}$ that contains $e \in \mathbb{H}(F)$. The **component group** is $\mathbb{G}/\mathbb{G}^\circ$ which is a linear algebraic group of dimension 0.*

Definition 5.5. *The **center** of $\mathbb{H}$ is given by $Z(\mathbb{H})(\bar{F}) = \{g \in \mathbb{H}(\bar{F}) \mid gh = hg, \forall h \in \mathbb{H}(\bar{F})\}$. The **connected center** is $Z(\mathbb{H})^\circ$, the identity component of $Z(\mathbb{H})$.*

Corollary 5.6. *Suppose that $\mathbb{H}$ is connected. Let $H = \mathbb{H}(F)$. We have $Z(\mathbb{H})(\bar{F}) = \{g \in \mathbb{H}(\bar{F}) \mid gh = hg, \forall h \in H\}$. Thus $Z(\mathbb{H})$ is a closed subgroup defined over F . We have $Z(\mathbb{H})(F) = Z(H)$.*

¹⁰See https://chatgpt.com/s/t_69a95e881e608191a335067caf887d2c for more detail.

Proof. Suppose $g \in Z(\mathbb{H})(\bar{F})$ is such that $c_g(h) = h$ for every $h \in H$. We have to prove that c_g is trivial on $\mathbb{H}$. This follows immediately from Lemma 2.16. Therefore, $Z(\mathbb{H})$ is defined by algebraic equations defined over F (by Noetherian property, actually a finite number of such). The last sentence then follows from definition. $\square$

Remark 5.7. We warn that Definition 5.5 would be problematic (and “incorrect” from the modern point of view), if $\text{char}(F)$ was positive; the center of the abstract group $\text{SL}_2(\bar{\mathbb{F}}_2)$ is trivial, but $Z(\text{SL}_2/\mathbb{F}_2) = \mu_2/\mathbb{F}_2$ is not supposed to be (geometrically) reduced; as a scheme, we have $\mu_2(\mathbb{F}_2[\epsilon]/\epsilon^2) = \{x \in \mathbb{F}_2[\epsilon]/\epsilon^2 \mid x^2 = 1\} = \{1, 1 + \epsilon\}$ which is non-trivial.

Definition 5.8. *The derived subgroup $\mathbb{H}^{\text{der}} \subset \mathbb{H}$ is the Zariski closure of the subgroup generated by $(\mathbb{H}(\bar{F}), \mathbb{H}(\bar{F}))$.*

Example 5.9. Write $H = \mathbb{H}(F)$. Denote by $Z(H)$ the usual center, and H^{der} the closure of the commutator (H, H) ; it can be proved that H^{der} is always a closed Lie subgroup, though we won’t need it.

We have the following table for those in Example 2.20:

$\mathbb{H}$	$Z(\mathbb{H})$	$Z(\mathbb{H})^\circ$	$\mathbb{H}^{\text{der}}$	H	$Z(H)$	H^{der}
GL_n	GL_1	GL_1	SL_n	$\text{GL}_n(F)$	$F^\times$	$\text{SL}_n(F)$
SL_n	μ_n	$\{e\}$	SL_n	$\text{SL}_n(F)$	$\mu_n(F)$	$\text{SL}_n(F)$
B_2	GL_1	GL_1	$\mathbb{G}_a$	$B_2(F)$	$F^\times$	$\mathbb{G}_a(F)$
$\mathbb{G}_a$	$\mathbb{G}_a$	$\mathbb{G}_a$	$\{e\}$	$\mathbb{G}_a(F) = F$	F	$\{e\}$
O_V	μ_2	$\{e\}$	SO_V	$\text{O}_{V(F)}$	$\mu_2(F) = \{\pm 1\}$	$*$
SO_V	$\mu_{\text{gcd}(2, \dim V)}$	$\{e\}$	SO_V	$\text{SO}_{V(F)}$	$\mu_{\text{gcd}(2, \dim V)}(F)$	$*$
Sp_V	μ_2	$\{e\}$	Sp_V	$\text{Sp}_{V(F)}$	$\{\pm 1\}$	$\text{Sp}_{V(F)}$

In the case of SO_V , it is assumed that $\dim V \neq 2$ (otherwise SO_V is a torus and thus abelian). Also

$$\mu_n := \{x \in \text{GL}_1(\bar{F}) \mid x^n = 1\}$$

is a closed subgroup of GL_1 of dimension 0, and $\mu_n(F) = \{x \in F^\times \mid x^n = 1\}$. Lastly,

$$* = \text{im}(\text{Spin}_V(F) \rightarrow \text{SO}_V(F)) = \ker(\text{SO}_V(F) \xrightarrow{\text{spinor norm}} F^\times / (F^\times)^2 = H^1(F, \mu_2))$$

In particular $H^{\text{der}} \subset \mathbb{H}^{\text{der}}(F)$ could be different.

Proposition 5.10. *Suppose $\mathbb{G}$ be a connected reductive group. Then we have $\mathbb{G} = Z(\mathbb{G})^\circ \cdot \mathbb{G}^{\text{der}}$. This means, by definition, that any element in $g \in G = \mathbb{G}(\bar{F})$ can be written as $g = zg'$ for $z \in Z(\mathbb{G})^\circ(\bar{F})$ and $g' \in \mathbb{G}^{\text{der}}(\bar{F})$.*

Sketch. The semisimple case is treated in [Spr09, 8.1.5. Theorem]. To deal with the general case, observe that $\mathbb{G}$ is generated by a maximal torus $\mathbb{T}$ and all root subgroups (with respect to $\mathbb{T}$). The proof of [Spr09, 8.1.5. Theorem] shows that $\mathbb{G}^{\text{der}}$ contains all root subgroups and $\alpha^\vee(\mathbb{G}_m)$ for all coroots $\alpha^\vee$. One verifies that $Z(\mathbb{G})^\circ$ contains (in fact, generated by) $\lambda(\mathbb{G}_m)$ for those $\lambda \in X_*(\mathbb{T})$ perpendicular to all roots. The collection of such λ and $\alpha^\vee$ generate $X_*(\mathbb{T}) \otimes \mathbb{Q}$, and thus the torus they generate is $\mathbb{T}$. This proves the proposition. $\square$

Example 5.11. The analogous statement $G = Z(G) \cdot G^{\text{der}}$ for F -points is **not** true. We have $\text{GL}_2(F) \supsetneq F^\times \cdot \text{SL}_2(F)$. (Why?)

Corollary 5.12. *Suppose $\mathbb{G}$ is connected reductive. Then $\text{Ad} : \mathbb{G} \rightarrow \text{GL}_{\text{Lie } \mathbb{G}}$ has image in $\text{SL}_{\text{Lie } \mathbb{G}}$.*

Proof. By definition $\text{Ad}(Z(\mathbb{G})) = \{e\}$. Also $\text{Ad}(\mathbb{G}^{\text{der}}(\bar{F})) \subset (\text{GL}_{\text{Lie}(\mathbb{G})})^{\text{der}}(\bar{F}) = \text{SL}_{\text{Lie}(\mathbb{G})}(\bar{F})$. Hence the corollary follows from Proposition 5.10. $\square$

Remark 5.13. We remark that $\mathbb{G}$ is geometrically reduced, so the (scheme-theoretic) image is geometrically reduced. In this case it is appropriate to study the image at the variety level, i.e. using $\text{Ad}(\mathbb{G}(\bar{F})) \subset \text{GL}_{\mathbb{G}}(\bar{F})$.

Remark 5.14. As a non-example, when $\mathbb{H} = B_2$ as in Example 2.20, the action of $\text{Ad}\left(\begin{bmatrix} a & 1 \\ 0 & 1 \end{bmatrix}\right)$ has determinant a .

Proof of Theorem 5.2. Firstly, when $\mathbb{G}$ is connected reductive, Corollary 5.12 says $\det(\text{Ad}) : \mathbb{G} \rightarrow \text{GL}_1 = \text{GL}_{\wedge^{\text{top}}(\text{Lie}^* \mathbb{G})}$ is trivial. This is the algebraic version of Δ_G ; passing to F -points, we see that $\Delta_G : G \rightarrow F^\times$ is trivial.

Now suppose $\mathbb{G}$ is only reductive. Write¹¹ $G^\circ := \mathbb{G}^\circ(F)$. Then $|\Delta_G| : G \rightarrow q^\mathbb{Z}$ is trivial on G° because $\text{Lie } G = (\text{Lie } \mathbb{G})(F) = (\text{Lie } \mathbb{G}^\circ)(F) = \text{Lie } G^\circ$. But $G/G^\circ \hookrightarrow \pi_0(G)(F)$ and the latter is a finite group. Therefore, $\Delta_G : G/G^\circ \rightarrow q^\mathbb{Z}$ factors through a finite group, and is trivial as $q^\mathbb{Z}$ is torsion free. $\square$

In the rest of this section let $\mathbb{G}$ be reductive, but not necessarily connected. Theorem 5.2 implies that, in fact, every left invariant top form on G is **also** right-invariant. And therefore any of them gives an **bi-invariant** measure $|\omega|$; this is often called a **Haar measure**.

Note that $|\omega|$ for non-zero ω can be restricted to a non-trivial element $\nu_\omega \in M_G$; indeed, to every open compact subgroup $J \in \Omega(G)$ it assigns a value $\nu(J) = |\omega|(J) \in \mathbb{Z}[1/p]$, and it is not hard to check the property in Definition 1.4. We have the following result that a priori has nothing to do with measures.

Corollary 5.15. *Let $J \in \Omega(G)$ and $g \in G$. Then $[J : gJg^{-1}] = 1$, i.e.*

$$[J : J \cap gJg^{-1}] = [gJg^{-1} : J \cap gJg^{-1}]$$

Proof. Choose any $\omega \neq 0$ as above. We have

$$[J : J \cap gJg^{-1}] = \frac{\nu_\omega(J)}{\nu_\omega(J \cap gJg^{-1})} = \frac{\nu_\omega(gJg^{-1})}{\nu_\omega(J \cap gJg^{-1})} = [gJg^{-1} : J \cap gJg^{-1}]$$

This proves the corollary. $\square$

Reversing the logic, one can see that Corollary 5.15 is equivalent to Theorem 5.2. An equivalent form of Theorem 5.2 is that ω is not only left-invariant, but bi-invariant. Recall that $\mathcal{C}$ is a field with $\text{char}(\mathcal{C}) \neq p$. We have

Lemma 5.16. *Any non-zero element $\omega_e \in \wedge^{\text{top}}(\mathfrak{g}^*)$ gives a nowhere vanishing bi-invariant top form ω_G on G , and $|\omega|$ gives an isomorphism $(f \mapsto f|\omega|) : \mathcal{C}_c^\infty(G) \xrightarrow{\sim} M_c^\infty(G)$.*

¹¹Warning: this is an abuse of language; G° is not a connected component of G . In fact it is **not** determined by G as a p -adic Lie group. The definition really makes use of the algebraic geometry.

Likewise, we have a top form $\omega_{\mathfrak{g}}$ on $\mathfrak{g}$ that is both translation-invariant and $\text{Ad}(G)$ -invariant. At the identity we have $\omega_G|_e = \omega_{\mathfrak{g}}|_0$ under $T_e G \cong T_0 \mathfrak{g} = \mathfrak{g}$. Recall that in Corollary 2.42 we have an invariant neighborhoods $e \in U \subset G$ and $0 \in \mathfrak{u} \in \mathfrak{g}$ that corresponds under the logarithm/exponential map. Since $d(\log)|_e = \text{id}_{\mathfrak{g}}$, by continuity we have $\log_* |\omega_G| = |\omega_{\mathfrak{g}}|$ on some neighborhood $0 \in \Lambda \subset \mathfrak{u}$ and therefore on $\text{Ad}(g)\Lambda$ for any $g \in G$. By replacing $\mathfrak{u}$ with $\text{Ad}(G)\Lambda$ and U with $\exp(\text{Ad}(G)\Lambda)$ we have

Corollary 5.17. *There exist G -conjugation invariant neighborhoods $e \in U \subset G$ and $0 \in \mathfrak{u} \in \mathfrak{g}$ such that $\log$ maps U isomorphically to $\mathfrak{u}$, with inverse $\exp$, such that $\log_* |\omega_G| = |\omega_{\mathfrak{g}}|$ for any ω as above.*

5.2. G/H and integration. In this subsection, suppose G is a general Lie group over F , and $H \subset G$ is a closed Lie subgroup. Choose a subspace $\mathfrak{p} \subset \mathfrak{g}$ that is complementary to $\mathfrak{h}$. Choose a sufficiently small lattice $\Lambda_{\mathfrak{p}} \subset \mathfrak{p}$ and $\Lambda_{\mathfrak{h}}$ such that (i) $(X, Y) \mapsto \exp(X)\exp(Y)$ is an isomorphism from $\Lambda_{\mathfrak{p}} \times \Lambda_{\mathfrak{h}}$ to some open compact subgroup $J \subset G$, and (ii) $J \cap H = \exp(\Lambda_{\mathfrak{h}})$. Then we have a bijection $\Lambda_{\mathfrak{p}} \cong J/J \cap H \cong JH/H$, and $\Lambda_{\mathfrak{p}}$ can be used as a coordinate chart for gJH/H for any $g \in G$. We have

Lemma 5.18. *This makes G/H an F -analytic manifold, that does not depend on the choices above (i.e. any two set of choices give isomorphic manifolds). The left action $G \times G/H \rightarrow G/H$ is analytic. We have a canonical identification $T_{eH}(G/H) = \mathfrak{g}/\mathfrak{h}$.*

[Ser06, Part II, IV-5] has a more general construction and proof. Such G/H is called a **homogeneous space**. Now suppose we'd like to integrate on G/H , and hope for $\omega_{G/H}$ a (left-) G -invariant nowhere vanishing top form on G/H . In particular it is (left-) H -invariant. Observe that

$$(10) \quad \bigwedge^{\text{top}}(\mathfrak{g}) = \bigwedge^{\text{top}}(\mathfrak{h}) \otimes \bigwedge^{\text{top}}(\mathfrak{g}/\mathfrak{h}).$$

The left-action of G/H agrees with $\text{Ad}(H)$ on G/H , and from there one sees that H acts $T_e(G/H) = \mathfrak{g}/\mathfrak{h}$ by $\text{Ad}(H)$. By (10), the existence of $\omega_{G/H}$ implies that $\Delta_H = \Delta_G|_H$.

The converse is also true: For any $g \in G$, the map $\pi : G \rightarrow G/H$ induces a surjection $d\pi_g : T_g G \twoheadrightarrow T_{gH}(G/H)$. The kernel of $d\pi_g$ is exactly the space of left translations by g of vectors in $\mathfrak{h} = T_e H$. Let ω_H be a non-zero left-invariant top form on H . From ω_H , we can have a collection $((\omega_H)_g \in \bigwedge^{\text{top}}(\ker(d\pi_g))^* \mid g \in G)$, i.e. a section of the corresponding vector bundle on H . Let ω_G be a non-zero left-invariant top form on G . By (10), we have a collection $((\omega_G/\omega_H)_g \in \bigwedge^{\text{top}}(\text{coker}(d\pi_g))^* \mid g \in G)$. For each $g \in G$, $(\omega_G/\omega_H)_g$ can be realized as in $\bigwedge^{\text{top}}(T_{gH}^*(G/H))$. If $\Delta_G|_H = \Delta_H$, then for any $h \in H$ we have $(\omega_G/\omega_H)_g = (\omega_G/\omega_H)_{gh}$ via right translation by h thanks to (10). Therefore ω_G/ω_H defines a nowhere vanishing top form on G/H . We have more or less shown that

Lemma 5.19. *The following are equivalent:*

- (1) $\Delta_G|_H = \Delta_H$.
- (2) There exists $\omega_{G/H}$ a (left-) G -invariant nowhere vanishing top form on G/H .
- (3) In the above construction $(\omega_G/\omega_H)_g = (\omega_G/\omega_H)_{gh}$ for any $g \in G$, $h \in H$, and induces $\omega_{G/H} := \omega_G/\omega_H$.

When the equivalent condition holds, the top form $\omega_{G/H}$ on G/H is analytic, and all such top forms are contained in a 1-dimensional space.

The proof of the following lemma is similar (with extra bookkeeping that we will skip):

Lemma 5.20. *The following are equivalent:*

- (1) $|\Delta_G|_H = |\Delta_H|$.
- (2) There exists $|\omega_{G/H}|$ a non-trivial (left-) G -invariant measure on G/H (that takes any open compact subset and gives a value in $\mathcal{C}$).
- (3) In the above construction $|(\omega_G/\omega_H)_g| = |(\omega_G/\omega_H)_{gh}|$ for any $g \in G$, $h \in H$, and induces $|\omega_{G/H}| := |\omega_G|/|\omega_H|$.

All such measures are contained in a 1-dimensional space over $\mathcal{C}$.

Exercise 15. Suppose G is a p -adic Lie group and H a closed Lie subgroup such that $\Delta_G = 1$ and $\Delta_H = 1$. Choose measures as in Lemma 5.19.

- (1) For any open compact subgroup $J \subset G$ and any $g \in G$, we have

$$|\omega_{G/H}|(JgH/H) \cdot |\omega_H|(g^{-1}Jg \cap H) = |\omega_G|(J).$$

- (2) Denote the measures by dg , dh and $d(gH)$. For any $f \in \mathcal{C}_c^\infty(G)$ we have

$$\int_G f(g)dg = \int_{G/H} \int_H f(gh)dh \cdot d(gH).$$

When $G = \mathbb{G}(F)$ where $\mathbb{G}$ is connected reductive and $H = Z_G(X)$ for any $X \in \mathfrak{g}^* \cong \mathfrak{g}$, in Remark 3.8 we see that $G/H \cong O^*$ has a 2-form whose top wedge ω_{O^*} is a nowhere vanishing top-form. In particular Lemma 5.19 gives $\Delta_H = \Delta_G|_H = 1$. Note that $\mathbb{H} = Z_{\mathbb{G}}(X)$ needs not be reductive when X is not semisimple, so this is otherwise non-obvious.

5.3. Semisimple descent. Firstly, let us give a definition. Suppose X is an F -analytic manifold. Let H be a Lie group over F that acts on X from the right freely. Denote by X/H the quotient set. Suppose at any $x \in X$ we can find a compact neighborhood U_X and open compact sub-manifold $U_Y \subset U_X$ containing x as well as an open compact subgroup J_H such that the action map $U_Y \times J_H \rightarrow U_X$ is an analytic isomorphism, and induces an injection $U_Y \hookrightarrow U_X/J_H \subset X/H$. In particular, this induces a tangent space isomorphism $T_x(U_Y) \cong T_x(U_X)/\mathfrak{h}$. Using U_Y as coordinate charts on X/H , this gives X/H the structure of an F -analytic manifold. We will call it a **quotient manifold**. This generalizes the construction of G/H . Note that $\mathcal{C}_c^\infty(X/H)$ is by definition the subspace of $\mathcal{C}^\infty(X)$ that is right H -invariant and is compactly supported mod H .

We will work with both the group setting and the Lie algebra setting. The group setting is slightly more involved, so we will deal with it and the same method works for the Lie algebra. Let $s \in G$ be semisimple. Let $\mathbb{H} := Z_{\mathbb{G}}(s)$, which is reductive [Bor91, p. 178, Proposition 13.19] though possibly disconnected. Let $H = \mathbb{H}(F)$.

Let

$$H' := \{h' \in H \mid Z_{\mathfrak{g}}(h') \subset \mathfrak{h}\}.$$

Let H acts on $G \times H'$ by $h.(g, h') = (gh^{-1}, hh'h^{-1})$. Denote by $G \times^H H'$ the quotient manifold. We have the following lemma

Lemma 5.21. *The subset $H' \subset H$ is open, and the map*

$$\begin{aligned} c_{G,H} : G \times^H H' &\rightarrow G \\ (g, h') &\mapsto \text{Ad}(g)h' \end{aligned}$$

is a covering map, i.e. it induces an isomorphism on tangent spaces. In particular $\text{Ad}(G)H' \subset G$ is open.

Proof. We have from definition that $\mathfrak{h} = Z_{\mathfrak{g}}(s)$. We have an $\text{Ad}(H)$ -invariant complement $\mathfrak{p} \subset \mathfrak{g}$ to $\mathfrak{h}$ given by $\mathfrak{p} = \text{im}(\text{Ad}(s) - \text{id}_{\mathfrak{g}})$ using that $\text{Ad}(s)$ is semisimple. That $Z_{\mathfrak{g}}(h') \subset \mathfrak{h}$ is thus equivalent to that h' acts on $\mathfrak{p}$ with no eigenvalue 1. In particular $H' \subset H$ is open.

The differential of $c_{G,H}$ at (g, h') is given by $(X, Y) \mapsto \text{Ad}(g)(Y + (\text{Ad}(h')^{-1} - \text{id}_{\mathfrak{g}})X)$. That $Z_G(h') \subset H \implies Z_{\mathfrak{g}}(h') \subset \mathfrak{h}$. Equivalently, the kernel of $\text{Ad}(h') - \text{id}_{\mathfrak{g}}$ has zero intersection with $\mathfrak{p}$, which is equivalent to that $\text{im}(\text{Ad}(h')^{-1} - \text{id}_{\mathfrak{g}}) \supset \mathfrak{p}$. This shows that the differential of $c_{G,H}$ is surjective, and thus is an isomorphism by dimension equality. $\square$

Remark 5.22. Note that $G \times^H H' \rightarrow \text{Ad}(G)H'$ is G -equivariant with respect to the left- G -action on $G \times^H H'$ and the conjugacy action on $\text{Ad}(G)H'$.

Remark 5.23. Every fiber of the map $c_{G,H}$ is finite, but the map might not be proper (inverse of compact needs not be compact); the simplest (counter-)example is proba-

bly $\mathbb{G} = \text{GL}_3$ and $s = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 2 \end{bmatrix}$. On the other hand, $c_{G,H}$ has an apparent algebraic analogue, that is étale and quasi-finite but not necessarily finite. This suggests the need of the next proposition.

Proposition 5.24. *There exists an open and closed H -invariant neighborhood $U' \subset H'$ containing s such that*

$$\begin{aligned} c_{G,H}|_{G \times^H U'} : G \times^H U' &\rightarrow \text{Ad}(G)U' \\ (g, h') &\mapsto \text{Ad}(g)h' \end{aligned}$$

is an isomorphism.

Proof. By Lemma 5.21, it suffices to make $c_{G,H}|_{G \times^H U'}$ injective. That is, whenever $\text{Ad}(g_1)h'_1 = \text{Ad}(g_2)h'_2$ for $h'_1, h'_2 \in U'$, we would like $g_1H = g_2H$. Multiplying by g_2^{-1} it suffices to consider $g_2 = e$. Choose an embedding $\mathbb{G} \hookrightarrow \text{GL}_n$. We would like to prove that there exists an open and closed $Z_{\text{GL}_n(F)}(s)$ -invariant subset $U'_{\text{GL}_n} \subset Z_{\text{GL}_n(F)}(s)$, such that whenever $x_1, x_2 \in U'_{\text{GL}_n}$ and $g \in \text{GL}_n(F)$ are such that $\text{Ad}(g)x_1 = x_2$, we have $g \in Z_{\text{GL}_n(F)}(s)$.

By furthermore embedding $\text{GL}_n(F) \hookrightarrow \text{GL}_n(F')$ for some finite extension, we may assume that the characteristic polynomial of $s \in \text{GL}_n(F)$ splits completely. That is, $s \in \text{GL}_n(F)$ acts on F^n in the way that there exists a decomposition $F^n = \bigoplus_i V_i$ where s acts on V_i via mutually distinct eigenvalues $\lambda_i \in F^\times$. In particular s acts on V_i with the characteristic polynomial $(\lambda - \lambda_i)^{\dim V_i}$. We choose sufficiently small open and closed neighborhoods C_i of $(\lambda - \lambda_i)^{\dim V_i}$ in the space of monic polynomials of degree V_i , so that for each $i \neq j$, the roots in C_i and the roots in C_j are disjoint;

such neighborhoods exist by looking at the resultants¹² [CC: I wrongly stated in the lecture that all nearby polynomials have roots in F . That is incorrect already for $(\lambda - 1)^2$. The above correct argument is added on Apr. 3rd.]

The subgroup $Z_{\mathrm{GL}_n(F)}(s)$ is exactly the subgroup that stabilizes each V_i , and we choose U'_{GL_n} the subset that acts on each V_i by a characteristic polynomial in C_i . For any $x_1, x_2 \in U'_{\mathrm{GL}_n}$ and $g \in \mathrm{GL}_n(F)$ such that $\mathrm{Ad}(g)x_1 = x_2$, the element g must send each V_i into the same V_i , i.e. $g \in Z_{\mathrm{GL}_n(F)}(s)$ and we are done. $\square$

Remark 5.25. Proposition 5.24 does not hold for a general étale map; see <https://gemini.google.com/share/f854316437bd>.

Fix U' as in Proposition 5.24, and Haar measures (bi-invariant measures) on G and H as in previous subsections. Consider any $a \in \mathcal{C}_c^\infty(G)$ with $\int_G a(g)dg = 1$. For any $f \in \mathcal{C}_c^\infty(U')$ we have the function $a \boxtimes f \in \mathcal{C}_c^\infty(G \times U')$. Integrating over H , we define $a \boxtimes^H f \in \mathcal{C}_c^\infty(G \times^H U')$, and denote by $f_a \in \mathcal{C}_c^\infty(\mathrm{Ad}(G)U')$ the function corresponding to $a \boxtimes^H f$ under $c_{G,H}|_{G \times^H U'}$. In other words, for any $g \in G$, $h' \in U'$ we define

$$(11) \quad f_a(gh'g^{-1}) = (a \boxtimes^H f)(g, h') = \int_H a(gh^{-1})f(hh'h^{-1})dh$$

We have

Theorem 5.26. *The map $(f \mapsto f_a)$ induces an isomorphism*

$$c_{[a]} : M_c^\infty(U')_H \xrightarrow{\sim} M_c^\infty(\mathrm{Ad}(G)U')_G$$

that is independent of the choice of a .

Proof of Theorem 5.26. Firstly, previously we fix Haar measures dg and dh and work with functions. However we see that $c_{[a]}$ scales with the choice of $\frac{dg}{dh}$, and therefore should be a canonical map from $M_c^\infty(U')_H \xrightarrow{\sim} M_c^\infty(\mathrm{Ad}(G)U')_G$. In the following we will fix dg and dh again and pretend $M_c^\infty(U')_H = \mathcal{C}_c^\infty(U')_H$, etc..

Suppose we have two different $a, a' \in \mathcal{C}_c^\infty(G)$ with $\int_G a = \int_G a' = 1$. In particular $\int_G a - a' = 0$. Choose an open compact pro- p subgroup $J \subset G$ so that both a and a' are right- J -invariant. Then we have a finite sum $a - a' = \sum_i c_i(1_{g_i J} - 1_J)$ for some $c_i \in \mathcal{C}$. When we plug in $1_{g_i J} - 1_J$ to (11), we get $\mathrm{Ad}(g_i)_* f_{1_J} - f_{1_J}$, and thus zero in $\mathcal{C}_c^\infty(\mathrm{Ad}(G)U')_G$. This shows that the map $c_{[a]} : \mathcal{C}_c^\infty(U') \rightarrow \mathcal{C}_c^\infty(\mathrm{Ad}(G)U')_G$ indeed does not depend on the choice of a . From there it is easy to see that it factors through $\mathcal{C}_c^\infty(U')_H$.

We describe the inverse of $c_{[a]} : \mathcal{C}_c^\infty(U')_H \rightarrow \mathcal{C}_c^\infty(\mathrm{Ad}(G)U')_G$. There is a projection map $\mathrm{pr}_1 : G \times^H U' \rightarrow G/H$. Let $f^G \in \mathcal{C}_c^\infty(\mathrm{Ad}(G)U')$. The set $\mathrm{pr}_1(c_{G,H}^{-1}(\mathrm{supp}(f^G))) \subset G/H$ is compact, and thus contained in a finite disjoint union $\bigcup Jg_i H$ for an open compact pro- p subgroup $J \subset G$, for some choice g_i of representatives of the double cosets in $J \backslash G/H$.

[CC: A mistake on the definition of b (that it is not supposed to be H -right-invariant) was corrected on Apr. 6th.] Let $b \in \mathcal{C}_c^\infty(G)$ be given by

$$b(x) = \begin{cases} \frac{1}{|\omega_H|(g^{-1}Jg \cap H)} & , \text{ if } x \in \bigcup Jg_i \\ 0 & , \text{ otherwise} \end{cases}$$

¹²<https://en.wikipedia.org/wiki/Resultant#Zeros>.

We have $\int_H b(gh)dh = 1$ for every $gH \in \text{pr}_1(c_{G,H}^{-1}(\text{supp}(f^G))) \subset G/H$. We define a linear map $\int_{G/H}^{[b]}(f^G) : \mathcal{C}_c^\infty(\text{Ad}(G)U') \rightarrow \mathcal{C}_c^\infty(U')$ (that depends on b) by

$$\left(\int_{G/H}^{[b]}(f^G)\right)(h') := \int_G f^G(gh'g^{-1})b(g)dg.$$

We claim that passing to coinvariants we have $\int_{G/H}^{[b]} = (c_{[a]})^{-1}$. Begin with $f \in \mathcal{C}_c^\infty(U')_H$. Let us compute

$$\begin{aligned} \int_{G/H}^{[b]}(c_{[a]}(f))(h') &= \int_G c_{[a]}(f)(gh'g^{-1})b(g)dg = \int_G \left(\int_H a(gh^{-1})f(hh'h^{-1})dh \right) b(g)dg \\ &= \int_H \left(\int_G a(gh^{-1})b(g)dg \right) f(hh'h^{-1})dh. \end{aligned}$$

Observe that the “coefficient” $\left(\int_G a(gh^{-1})b(g)dg\right)$ satisfies

$$\begin{aligned} \int_H \left(\int_G a(gh^{-1})b(g)dg \right) dh &= \int_H \left(\int_G a(g)b(gh)dg \right) dh \\ &= \int_G a(g) \left(\int_H b(gh)dh \right) dg = \int_G a(g)dg = 1. \end{aligned}$$

From this we can see $\int_{G/H}^{[b]}(c_{[a]}(f)) = f$ in $\mathcal{C}_c^\infty(U')_H$.

It remains to show that $c_{[a]}$ maps surjectively to $\mathcal{C}_c^\infty(\text{Ad}(G)U')_G$. By the local charts on $G \times^H U'$, every locally constant compactly supported function $f^G \in \mathcal{C}_c^\infty(G \times^H U')$ can be written as a finite sum of H -integrations of $c_i(1_J g_i \boxtimes 1_{V_i})$ for some open compact pro- p subgroup $J \subset G$ and open compact subset $V_i \subset U'$. Note that we pick J small enough for all i . Then $f^G = f_a$ with $a = \frac{1}{\int_J dg} 1_J$ and $f = \int_J dg \cdot (\sum_i c_i 1_{V_i}) \in \mathcal{C}_c^\infty(U')$. $\square$

Remark 5.27. The definition in Theorem 5.26 has a normalization problem; when we choose fix our measures dg and dh , the manifold $G \times U'$ has a product measure and $G \times^H U'$ has its quotient measure. This quotient measure does **not** correspond (under Proposition 5.24) to the restriction of dg to $\text{Ad}(G)U'$. By choosing U' small enough, we may however assume that the the former measure is the latter measure times a constant $n(s) \in \mathcal{C}^\times$ that depend on s . The canonical map should then be $n(s) \cdot c_{[a]}$.

Remark 5.28. Let us give an conceptual explanation of Theorem 5.26. We had essentially proved the following

- (1) We have a canonical isomorphism

$$M_c^\infty(G) \otimes_{M_c^\infty(H)} M_c^\infty(U') \xrightarrow{\sim} M_c^\infty(\text{Ad}(G)U')$$

defined by an appropriate integration over H , so that the $c_{[a]}(f)$ corresponds to $a \otimes f$. In fact, the map naturally factors through $\text{ind}_H^G M_c^\infty(U')$, the space of locally constant H -equivariant $M_c^\infty(U')$ -valued functions on G that is compactly supported mod H .

(2) We have an isomorphism of coinvariants

$$M_c^\infty(U')_H \cong (M_c^\infty(G) \otimes_{M_c^\infty(H)} M_c^\infty(U'))_G$$

given by $f \mapsto a \otimes f$ as above. The inverse is by integration over G/H .

Exercise 16. (*) Prove indeed that for any smooth H -representation ρ we have

(1) Consider the right H -action on G . There is a G -equivariant isomorphism

$$M_c^\infty(G) \otimes_{M_c^\infty(H)} \rho \xrightarrow{\sim} \text{ind}_H^G \rho$$

where $M_c^\infty(H)$ acts on $M_c^\infty(G)$ by right convolution, and $\text{ind}_H^G \rho$ is the space of locally constant H -equivariant $M_c^\infty(U')$ -valued functions on G that is compactly supported mod H .

(2) We have an isomorphism of coinvariants

$$\rho_H \cong (\text{ind}_H^G \rho)_G.$$

Let us also state the Lie algebra version of Theorem 5.26. The proof is identical.

Theorem 5.29. Let $X_s \in \mathfrak{g}$ be semisimple and $H = Z_G(X_s)$. There exists an $\text{Ad}(H)$ -invariant neighborhood

$$U' \subset \{X' \in \mathfrak{h} \mid Z_G(X') \subset H\}.$$

containing X_s , such that $G \times^H U' \cong \text{Ad}(G)U'$ is an isomorphism. We have a natural isomorphism

$$M_c^\infty(U')_H \xrightarrow{\sim} M_c^\infty(\text{Ad}(G)U')_G.$$

Taking $\mathcal{C}$ -linear dual of Theorem 5.26 and Theorem 5.29, in light of Remark 5.27, we have

Corollary 5.30. In both the group setting and the Lie algebra setting, we have a natural isomorphism

$$\mathcal{C}^{-\infty}(U')^H \cong \mathcal{C}^{-\infty}(\text{Ad}(G)U')^G$$

of the space of invariant generalized functions.

Corollary 5.31. Let $O \subset U'$ be any H -orbit (in the group or Lie algebra). With choices of Haar measures we have a commutative diagram

$$\begin{array}{ccc} D(U')^H & \xrightarrow{\sim} & D(\text{Ad}(G)U')^G \\ \uparrow & & \uparrow \\ D(\overline{O})^H & \xrightarrow{\sim} & D(\overline{\text{Ad}(G)O})^G \\ \downarrow & & \downarrow \\ D(O)^H & \xrightarrow{\sim} & D(\text{Ad}(G)O)^G \end{array}$$

Sketch. An orbit $O \subset U'$ gives a locally closed subset $G \times O \subset G \times U'$ that is invariant under the H -action that we have studied, and thus descend to $G \times^H O$. The correspondence $O \leftrightarrow G \times^H O$ is a bijection between H -orbits in U' and G -orbits in $G \times^H U'$. Since $G \times U' \rightarrow G \times^H U'$ is an open map, this shows that the correspondence also preserve the partial order relation studied in Proposition 3.40

(we only did the Lie algebra case but the same result with the same proof works for the group case). In particular we have $\text{Ad}(G)\overline{O} = \overline{\text{Ad}(G)O}$. Knowing this, the rest follows the construction and Exercise 9. $\square$

The proof of Theorem 5.29 can also be made canonical with $\mathfrak{g}^*$. In that case Corollary 5.31 leads to the following corollary that we skip the lengthy check.

Corollary 5.32. *Let $X^* \in \mathfrak{g}^*$ be with semisimple part X_s^* and $H = Z_G(X_s^*)$. Then there are canonical¹³ commuting isomorphisms*

$$\begin{array}{ccc} D(\overline{\text{Ad}(H)X^*})^H & \xrightarrow{\sim} & D(\overline{\text{Ad}(G)X^*})^G \\ \downarrow & & \downarrow \\ D(\text{Ad}(H)X^*)^H & \xrightarrow{\sim} & D(\text{Ad}(G)X^*)^G \end{array}$$

where the bottom arrow matches $I_{\text{Ad}(H)X^*}$ with $I_{\text{Ad}(G)X^*}$ as defined in (5). Under the condition of Definition 3.38, the element $I_{\text{Ad}(G)X^*}$ has a canonical lift in $D(\overline{\text{Ad}(G)X^*})^G$ that matches with (the canonical lift of) $I_{\text{Ad}(H)X^*}$.

5.4. The statement of Theorem C'. Now we are ready to generalize Theorem 4.13.

Theorem C'. *Assume Hypothesis 4.1 and fix an ψ_F as in Lemma 4.2.*

- (1) *Fix $X_s \in \mathfrak{g}$ semisimple and let $H = Z_G(X_s)$. For any invariant distribution $\delta \in D(\mathfrak{g}^*)^G$ such that $\text{supp}(\delta) \in \text{Ad}(G)\Lambda^*$ for some lattice $\Lambda^* \subset \mathfrak{g}^*$, there exist a distribution $\delta_n \in D((\mathfrak{h}^*)^{\text{nil}})^H \subset D((\mathfrak{h}^*)^{\text{nil}})^H$ and a lattice $\Lambda \subset \mathfrak{h}$ such that*

$$c_{[a]}^* \left(\widehat{\delta}|_{\text{Ad}(G)(X_s + \Lambda)} \right) = \widehat{\delta}_n|_{X_s + \text{Ad}(H)\Lambda},$$

Equivalently, for any $f \in C_c^\infty(\mathfrak{h}^/\Lambda^\perp)$ we have*

$$\widehat{\delta} \left(c_{[a]} \left(\widehat{f \cdot \psi_{X_s}} \right) \right) = \widehat{\delta}_n \left(\widehat{f \cdot \psi_{X_s}} \right) = \widehat{\delta}_n(\widehat{f}) = \delta_n(f \cdot \psi_{X_s}) = \delta_n(f).$$

- (2) *Fix $s \in G$ semisimple and let $H = Z_G(s)$. Let π be an irreducible admissible $\mathcal{C}$ -representation of G . Then there exist a distribution $\delta_n \in D((\mathfrak{h}^{\text{nil}})^*)^G$ and a lattice $\Lambda \subset \mathfrak{h}$ such that $\exp : \Lambda \xrightarrow{\sim} \exp(\Lambda)$ is an isomorphism, and*

$$c_{[a]}^* \left(\Theta_\pi|_{\text{Ad}(G)(s \cdot \exp(\Lambda))} \right) \stackrel{*}{=} \widehat{\delta}_n|_{\text{Ad}(H)\Lambda}$$

where $\stackrel{}{=}$ is understood via an isomorphism*

$$M_c^\infty(\text{Ad}(H)\Lambda) \cong M_c^\infty(\text{Ad}(H)\exp(\Lambda)) \cong M_c^\infty(s \cdot \text{Ad}(H)\exp(\Lambda)).$$

The majority of the rest of the course will be devoted to proving Theorem C'. We remark that Theorem C', together with Theorem 5.26 and Corollary 3.39, implies Theorem C. The theorem might be a bit difficult to understand, so let us consider a consequence which will be eventually be equivalent to Theorem C' itself if Theorem B holds (we will however use Theorem C' to prove Theorem B).

[CC: The following is added on Apr. 3rd.]

¹³That is, it does not depend on the choice of Haar measures nor that of $\iota : \mathfrak{g}^* \cong \mathfrak{g}$.

Definition 5.34. Let $U \subset G$ be an open subset. A function $\phi \in \mathcal{C}_c^\infty(U)$ **represents a generalized function $\tilde{\phi}$ on U** (see also Definition 2.56), more precisely the functional on $M_c^\infty(U)$, given by

$$\tilde{\phi} : \mu \mapsto \int_U \phi \mu.$$

We view this as an embedding $\mathcal{C}_c^\infty(U) \hookrightarrow \mathcal{C}^{-\infty}(U)$.

Let $U \subset V \subset G$ be open subsets. We say a generalized function $\phi' \in \mathcal{C}^{-\infty}(V)$ is **locally constant on U** if ϕ' is represented by ϕ on U for some $\phi \in \mathcal{C}_c^\infty(U)$.

Proposition 5.35. Let U' be as in Theorem 5.26. Let $\phi' \in \mathcal{C}^{-\infty}(\text{Ad}(G)U')^G$ and $(\phi')^a \in \mathcal{C}^{-\infty}(U')^H$ correspond under Corollary 5.30. For $V' \in U'$ that is open and $\text{Ad}(H)$ -invariant, and $\phi \in \mathcal{C}^\infty(\text{Ad}(G)V')^G$, the following are equivalent

- (1) $\phi'|_{\text{Ad}(G)V'}$ is represented by ϕ .
- (2) $(\phi')^a|_{V'}$ is represented by $\phi|_{V'}$.

Proof. By definition, we have to verify for any $f \in \mathcal{C}_c^\infty(U')_H$ and $f_a \in \mathcal{C}_c^\infty(\text{Ad}(G)U')_G$ (canonically, locally constant compactly supported measures on H and G) that

$$\langle f_a|_{\text{Ad}(G)V'}, \phi \rangle = \langle f|_{V'}, \phi|_{V'} \rangle$$

Since ϕ is $\text{Ad}(G)$ -invariant, we have

$$\phi(gh'g^{-1}) = \phi(h'), \quad \forall g \in G, h' \in V'.$$

Let us denote by $\phi_{G \times^H V'}$ the pullback of ϕ under $G \times^H V' \cong \text{Ad}(G)V'$ and $\phi_{G \times V'}$ its further pullback. Then $\phi_{G \times V'} = \text{pr}_1^* \phi|_{V'}$. We thus have

$$\langle f|_{V'}, \phi|_{V'} \rangle = \langle a \boxtimes f, \phi_{G \times V'} \rangle = \langle a \boxtimes^H f, \phi_{G \times^H V'} \rangle = \langle f_a, \phi \rangle$$

This proves the proposition. $\square$

Corollary 5.36. Let H be in Theorem C', either in the group or Lie algebra case. Suppose $\hat{\delta}_n \in \mathcal{C}^{-\infty}(\mathfrak{h})$ is locally constant on an open subset $V' \subset U'$ where U' is as in Proposition 5.24 or Theorem 5.29. Then Θ_π (resp. $\hat{\delta}$) is locally constant on $\text{Ad}(G)V'$, with the value given by $\Theta_\pi(gh'g^{-1}) = \hat{\delta}_n(\log h')$ (resp. $\hat{\delta}(\text{Ad}(g)h') = \hat{\delta}_n(\log h')$).

Corollary 5.37. [Theorem A] Suppose $s \in G$ (resp. $X_s \in \mathfrak{g}$) such that $\mathfrak{h} := Z_{\mathfrak{g}}(s)$ (resp. $\mathfrak{h} := Z_{\mathfrak{g}}(X_s)$) contains no non-zero nilpotent element. Then the generalized functions Θ_π (resp. $\hat{\delta}$) in Theorem C' are locally constant near s (resp. X_s).

Proof. This follows from Corollary 5.36, that $\mathfrak{h}^* \cong \mathfrak{h}$, and the fact that the Fourier transform of a distribution supported at $0 \in \mathfrak{h}^*$ is represented by a constant function. To see the fact, let $\delta \in D(\mathfrak{h}^*)$ be such that $\text{supp}(\delta) \subset \{0\}$. Then $\delta(f) = c \cdot f(0)$ for some $c \in \mathbb{C}$, and consequently $\hat{\delta}(f) = \delta(\hat{f}) = \hat{f}(0) = \int_{\mathfrak{h}} f(X) dX$, i.e. δ is represented by the constant function c . $\square$

Example 5.38. Suppose G is compact. Then every irreducible smooth representation of G is actually a representation of G/N for some finite index open subgroup N , and therefore its character is constant on every (open) N -coset.

On the other hand, as G is compact, $\mathfrak{g}$ contains no non-zero nilpotent element; otherwise, any $0 \neq X \in \mathfrak{g}^{\text{nil}}$ will give an unbounded $\{\exp(tX) \mid t \in F\} \subset G \subset$

$\mathrm{GL}_n(F)$. Therefore Corollary 5.37 re-confirms that Θ_π is locally constant everywhere, i.e. a locally constant function on G .

6. LOCAL INTEGRABILITY

The goal of this section is to prove Theorem B, conditional on Theorem C'. In this section we suppose $\mathcal{C} = \mathbb{C}$ is the field of complex numbers (see Remark 6.28 for some general cases). We will denote by $\phi \in C_c^\infty(\mathfrak{g}^{\mathrm{rs}})$ (instead of $C_c^\infty(-)$) to emphasize a $\mathbb{C}$ -valued test function, etc..

All the material of this section appear in [CC: arXiv link to be inserted].

6.1. Absolute convergence.

Definition 6.1. Let X be a p -adic manifold and X' an open subset.

- (1) We say ϕ is **locally- L^1** or **locally integrable near** $x \in X$, if $\int_U |\phi(y)| dy < +\infty$ for sufficiently small neighborhood $U \subset X$ of x . It is said to be just **locally- L^1** if it is so at every $x \in X$.
- (2) Suppose ϕ is locally- L^1 . Then for $\mu \in M_c^\infty(X)$ we define $\langle \phi, \mu \rangle = \int_X \phi \mu$ using absolute convergence.
- (3) We say ϕ **represents** a generalized function $\Phi \in C^{-\infty}(X)$ if ϕ defines Φ , i.e. for any $\mu \in M_c^\infty(X)$ we have $\langle \Phi, \mu \rangle = \langle \phi, \mu \rangle$. We say ϕ **represents** Φ near $x \in X$ if $\phi|_U$ represents $\Phi|_U$ for some neighborhood of x .

Lemma 6.2. If ϕ_1 and ϕ_2 both represents some Φ , then $\phi_1 = \phi_2$.

Proof. Suppose $\phi_1(x) \neq \phi_2(x)$ take different constant values in some neighborhood. Then $\langle \phi_1, \mu \rangle \neq \langle \phi_2, \mu \rangle$ for some non-zero positive μ supported within such a small neighborhood. $\square$

Proposition 6.3. We adapt the setting of Proposition 5.35. The following are equivalent:

- (1) ϕ' is represented by ϕ near $s \in G$.
- (2) $(\phi')^a$ is represented by $\phi|_{V'}$ near $s \in H$.

Proof. Note that since ϕ' and ϕ are $\mathrm{Ad}(G)$ -invariant, ϕ' is represented by ϕ near $s \in G$ iff it is so near any $s' \in \mathrm{Ad}(G)s$. The same proof then works, with the classical notion of absolute convergence inserted everywhere. $\square$

Corollary 6.4. Assume Theorem C'. Suppose that for any reductive group $\mathbb{G}'$ with $\dim \mathbb{G}' \leq \dim \mathbb{G}$ and any $\delta_n \in D((\mathfrak{g}'^*)^{\mathrm{nil}})^{G'}$, we have $\widehat{\delta}_n$ represented by a locally- L^1 function. Then the same is true for Θ_π for an irreducible admissible complex representation π .

Proof. The assert is local; it suffices to prove it near any $g \in G$. Theorem C' asserts that near a semisimple element $s \in G$, the generalized function Θ_π corresponds to a $\widehat{\delta}_n$ in the sense of Proposition 6.3, where $\delta_n \in D((\mathfrak{h}^*)^{\mathrm{nil}})^H$ with $H = Z_G(s)$. The proposition therefore implies that the corollary is true near any semisimple element.

For a general element $g = su$, the element s is contained in the closure of $\mathrm{Ad}(g)(su)$ because the identity $e_H = \exp_H(0)$ is contained in the closure of $\mathrm{Ad}(H)u = \exp(\mathrm{Ad}(H)\log(u))$. Hence a neighborhood of s contains a neighborhood of some conjugate of su . Since everything is $\mathrm{Ad}(G)$ -invariant, the corollary is also true near su and we are done. $\square$

Corollary 6.5. Assume Theorem C'. Suppose for any reductive group $\mathbb{G}'$ with $\dim \mathbb{G}' < \dim \mathbb{G}$ any $\delta_n \in D((\mathfrak{g}'^*)^{\text{nil}})^{G'}$, we have that $\widehat{\delta}_n$ is represented by a locally- L^1 function. Then for any $\delta_n \in D((\mathfrak{g}^*)^{\text{nil}})^G$, we have that $\widehat{\delta}_n$ is represented by a locally- L^1 function near any $X \in \mathfrak{g}$ such that $Z_{\mathfrak{g}}(X_s) \subsetneq \mathfrak{g}$ for its semisimple part X_s , i.e. that $X_s \notin Z(\mathfrak{g})$.

Proof. The assert is again local, and we apply Theorem C' to δ_n and X_s to reduce to $\mathbb{G}' := Z_{\mathbb{G}}(X_s)$ as in the proof of Corollary 6.4. $\square$

6.2. Maximal tori. Recall that

Definition 6.6. (1) A **maximal F -torus** $\mathbb{T} \subset \mathbb{G}$ is a connected commutative closed algebraic subgroup defined over F , such that every $t \in \mathbb{T}(\bar{F})$ is semisimple, and that $Z_{\mathbb{G}}(\mathbb{T}) = \mathbb{T}$.
 (2) A **(rational) maximal torus** $T \subset G$ is a closed Lie subgroup of the form $T = \mathbb{T}(F)$.
 (3) A **Cartan subalgebra** $\mathfrak{t} \subset \mathfrak{g}$ is a commutative Lie subalgebra consisting of semisimple elements, such that $Z_{\mathfrak{g}}(\mathfrak{t}) = \mathfrak{t}$.

Fact 6.7. (1) The correspondence $\mathbb{T} \leftrightarrow T \leftrightarrow \mathfrak{t}$ with $T = \mathbb{T}(F)$ and $\mathfrak{t} = \text{Lie } T$ is a bijection between maximal F -tori, rational maximal tori, and Cartan subalgebras. We will sometimes casually called refer to any of these by a maximal torus.
 (2) By [Bor91, 11.3 Corollary], all maximal F -tori (and thus rational maximal tori, Cartan subalgebras) in $\mathbb{G}$ are conjugate under $\mathbb{G}(\bar{F})$. They are **not** conjugate under $G = \mathbb{G}(F)$ already when $\mathbb{G} = \text{SL}_2$. [CC: This wrongly said GL_2 and was corrected on Apr. 15th.] In particular they all have the same dimensions $\text{rank}_{\bar{F}} \mathbb{G}$, which is called the **absolute rank** of $\mathbb{G}$.
 (3) [CC: Added on Apr. 15th.] Up to G -conjugate, there is only a finite number of maximal F -tori. This is because that, relative to any fixed maximal F -torus $\mathbb{T}$, the set of G -conjugacy classes maximal F -tori (including that of $\mathbb{T}$ itself) is in bijection with the Galois cohomology $\ker(H^1(F, N_{\mathbb{G}}(\mathbb{T})) \rightarrow H^1(F, \mathbb{G}))$, and this is finite because the Galois cohomology of any linear algebraic group over F is finite.

Exercise 17. Prove that $\text{Ad}(G)$ -conjugacy classes of maximal F -tori of GL_n/F are in 1-1 correspondence with isomorphism classes of n -dimensional étale commutative algebras (i.e. products of separable field extensions) over F . Here two étale algebras E_1 and E_2 are isomorphic if there is an isomorphism $E_1 \xrightarrow{\sim} E_2$ that restricts to the identity on F .

We thank Chih-Kai Liao for correcting an earlier mistake in the following exercise.

Exercise 18. (*) For any n -dimensional étale commutative F -algebra E , define $C_{E/F} \subset \{\pm 1\} \subset F^\times$ to be the set

$$\{\det(\sigma) \text{ as an } F\text{-linear map on } E \mid \sigma \in \text{Gal}(E/F)\}.$$

Prove that there exists a bijection between

- (1) $\text{SL}_n(F)$ -conjugacy classes of rational maximal tori of $\text{SL}_n(F)$.

- (2) The set $\{[E], \alpha\}$ where E is an isomorphism class of n -dimensional étale commutative F -algebras as above, and $\alpha \in F^\times / N_{E/F}(E^\times) \cdot C_{E/F}$.

such that a rational maximal torus in $\mathrm{SL}_n(F)$ is compact iff the corresponding étale algebra is a field.

Definition 6.8. For convenience we denote by $\mathrm{rank} G := \mathrm{rank}_{\bar{F}} G$, and call it the **rank** of G .

Definition 6.9. An element $g \in G$ is called **regular semisimple** iff the following equivalent conditions hold

- (1) $Z_G(g)^\circ$ is a maximal torus.
- (2) $Z_{\mathfrak{g}}(g)$ is a Cartan subalgebra.

An element $X \in \mathfrak{g}$ is called **regular semisimple** iff the following equivalent conditions hold

- (1) $Z_G(X)$ is a maximal torus.
- (2) $Z_G(X)$ is a maximal torus.
- (3) $Z_{\mathfrak{g}}(X)$ is a Cartan subalgebra.

We denote by $\mathfrak{g}^{\mathrm{rs}}$ (resp. $\mathfrak{g}^{\mathrm{rs}}$) the subset of regular semisimple elements in G (resp. $\mathfrak{g}$).

Lemma 6.10. Definition 6.9 is well-defined. The subset $\mathfrak{g}^{\mathrm{rs}}$ (resp. $\mathfrak{g}^{\mathrm{rs}}$) is open dense in G (resp. $\mathfrak{g}$). Moreover, a semisimple $s \in G^{\mathrm{ss}}$ (resp. $X_s \in \mathfrak{g}^{\mathrm{ss}}$) is regular semisimple iff the $\dim_F \ker(\mathrm{Ad}(s) - \mathrm{id}_{\mathfrak{g}}) = \mathrm{rank}_{\bar{F}} G$ (resp. $\dim_F \ker(\mathrm{ad}(X_s)) = \mathrm{rank}_{\bar{F}} G$).

For the standard proof we refer to <https://gemini.google.com/share/1a49fb909956>.
[CC: Maybe I should write it ...]

Definition 6.11. The **Harish-Chandra discriminant** is the function $D_G : G \rightarrow F$ given by

$$D_G(g) = \text{degree}(\dim G - \mathrm{rank} G) \text{ coefficient of } \det(\mathrm{Ad}(g) - (\lambda + 1)\mathrm{id}_{\mathfrak{g}}).$$

And $D_{\mathfrak{g}} : \mathfrak{g} \rightarrow F$ is

$$D_{\mathfrak{g}}(X) = \text{degree}(\dim G - \mathrm{rank} G) \text{ coefficient of } \det(\mathrm{ad}(X) - \lambda \mathrm{id}_{\mathfrak{g}}).$$

We [CC: should, thanks to Gemini ...] have seen that $\mathfrak{g}^{\mathrm{rs}} = \{g \in G \mid D_G(g) \neq 0\}$ and $\mathfrak{g}^{\mathrm{rs}} = \{X \in \mathfrak{g} \mid D_{\mathfrak{g}}(X) \neq 0\}$.

6.3. Weyl integration formula. Let $X \in \mathfrak{g}^{\mathrm{rs}}$ and $T = Z_G(X)$ be a maximal torus, and $\mathfrak{t} = \mathrm{Lie} T$. In the language of Lemma 5.21, the subset $\mathfrak{t}^{\mathrm{rs}} = \mathfrak{g}^{\mathrm{rs}} \cap \mathfrak{t}$ plays the role of H' , and the map

$$\begin{aligned} c_{G,T} : (G/T) \times \mathfrak{t}^{\mathrm{r}} &= G \times^T \mathfrak{t}^{\mathrm{rs}} \rightarrow \mathfrak{g}^{\mathrm{rs}} \\ (g, Y) &\mapsto \mathrm{Ad}(g)Y \end{aligned}$$

is a covering map. In particular it maps surjectively to an open subset of $\mathfrak{g}^{\mathrm{rs}}$ containing X itself.

Lemma 6.12. The image $\mathrm{im}(c_{G,T})$ is those $X \in \mathfrak{g}^{\mathrm{rs}}$ such that $Z_G(X)$ is a conjugate of T , i.e. of the form $\mathrm{Ad}(g)T$, $g \in G$. It is a Galois covering map onto its image, with Galois group $W_T := N_G(T)/T$ acting on $(G/T) \times \mathfrak{t}^{\mathrm{r}}$ by

$$n.(g, Y) = (gn^{-1}, \mathrm{Ad}(n)Y).$$

Proof. Only the last assertion requires a proof. We have to show that W_T acts simply transitively on each fiber. By definition $T = Z_G(Y)$ for $Y \in \mathfrak{t}^r$. Hence the action of $N_G(T)$ has kernel exactly T . For $(g, Y) \cdot (g', Y') \in G \times \mathfrak{t}^r$ with $\text{Ad}(g)Y = \text{Ad}(g')Y' \Leftrightarrow \text{Ad}(g^{-1}g')Y' = Y$, we have $\text{Ad}(g^{-1}g')Z_G(Y') = Z_G(Y) \implies \text{Ad}(g^{-1}g)T = T \implies g^{-1}g' \in N_G(T)$. Hence $(g', Y') = n \cdot (g, Y)$. This proves the transitivity and proves the lemma. $\square$

Remark 6.13. Lemma 6.12 essentially says

$$\text{im}(c_{G,T}) \cong ((G/T) \times \mathfrak{t}^r) / (N_G(T)/T) = G \times^{N_G(T)} \mathfrak{t}^r.$$

Lemma 6.14. *Consider a nowhere vanishing bi-invariant top form ω_G on G which corresponds to a translation invariant top form $\omega_{\mathfrak{g}}$ on $\mathfrak{g}$. We also consider ω_T on T and $\omega_{\mathfrak{t}}$ on $\mathfrak{t}$. We have thus the form $\omega_G \wedge \omega_{\mathfrak{t}}|_{\mathfrak{t}^r}$ on $G \times \mathfrak{t}^r$ and the quotient $\frac{\omega_G}{\omega_T} \wedge \omega_{\mathfrak{t}}|_{\mathfrak{t}^r}$ on $(G/T) \times \mathfrak{t}^r$. We have*

$$c_{G,T}^*(\omega_{\mathfrak{g}}|_{\mathfrak{g}^{rs}}) = \text{pr}_2^*(D_{\mathfrak{g}}|_{\mathfrak{t}^r}) \cdot \frac{\omega_G}{\omega_T} \wedge \omega_{\mathfrak{t}}|_{\mathfrak{t}^r}$$

Proof. This is a direct calculation of the Jacobian of $c_{G/T}$. Since everything is G -invariant, it suffices to compute at $(e, Y) \in G \times \mathfrak{t}^r$. In this case the Jacobian is $\det(\text{ad}(Y) : \mathfrak{g}/\mathfrak{t} \rightarrow \mathfrak{g}/\mathfrak{t})$ which is $D_{\mathfrak{g}}(Y)$. $\square$

Denote ω_G , $\omega_{\mathfrak{g}}$, ω_T and $\omega_{\mathfrak{t}}$ by dg , dX , dt and dY , respectively.

Corollary 6.15. (Weyl integration formula) *For $f \in C_c^\infty(\mathfrak{g}^{rs})$ we have*

$$\int_{\mathfrak{g}^{rs}} f(X) dX = \sum_T \frac{1}{|W_T|} \int_{\mathfrak{t}^r} |D_{\mathfrak{g}}(Y)| \left(\int_{G/T} f(\text{Ad}(g)Y) \frac{dg}{dt} \right) dY.$$

6.4. Reduction to maximal tori.

Proposition 6.16. *Let $\phi \in C^\infty(\mathfrak{g}^{rs})$ and $X \in \mathfrak{g}^{ss}$ be semisimple. The following are equivalent*

- (1) ϕ is locally- L^1 near X on $\mathfrak{g}$.
- (2) $\phi \cdot f$ is locally- L^1 near X on $\mathfrak{g}$ for some $f \in C_c^\infty(\mathfrak{g})$ with $f(X) \neq 0$.
- (3) $\phi \cdot f$ is locally- L^1 near X on $\mathfrak{g}$ for every $f \in C_c^\infty(\mathfrak{g})$.
- (4) For some $f \in C_c^\infty(\mathfrak{g})$ with $f(X) \neq 0$, and every maximal torus $T \subset Z_G(X)$, the function on $\mathfrak{t}^r$ given by

$$(12) \quad Y \mapsto D_{\mathfrak{g}}(Y) \cdot \int_{G/T} (\phi \cdot f)(\text{Ad}(g)Y) \frac{dg}{dt}$$

is locally- L^1 near X on $\mathfrak{t}$.

- (5) The last condition is true for every $f \in C_c^\infty(\mathfrak{g})$.

Suppose the above equivalent condition holds for all $X \in \mathfrak{g}^{ss}$, then the following are equivalent for $\Phi \in C^{-\infty}(\mathfrak{g}^{rs})$:

- (1) Φ is represented by ϕ .
- (2) Φ is represented by

$$f \mapsto \sum_T \frac{1}{|W_T|} \int_{\mathfrak{t}^r} |D_{\mathfrak{g}}(Y)| \left(\int_{G/T} (\phi \cdot f)(\text{Ad}(g)Y) \frac{dg}{dt} \right) dY.$$

Proof. This follows Corollary 6.15 applies to $\phi \cdot f$ and the standard theory of absolute convergence. Note that $\int_{G/T} \phi(\text{Ad}(g)Y)dg$ is in general not well-defined; “truncating” with f is necessary. $\square$

Recall that using Theorem C' we had proved that $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}}$ is represented by a locally constant function. It is customary to denote the value of function at $X \in \mathfrak{g}^{\text{rs}}$ by $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}}(X)$.

Corollary 6.17. (1) The function $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}} \in C^\infty(\mathfrak{g}^{\text{rs}})^G$ is locally- L^1 near $X \in \mathfrak{g}^{\text{ss}}$ iff for some, or every $f \in C_c^\infty(\mathfrak{g})$ with $f(X) \neq 0$, and every maximal torus $T \subset Z_G(X_s)$, the function on $\mathfrak{t}^r$ given by

$$(13) \quad Y \mapsto |D_{\mathfrak{g}}(Y)| \cdot \widehat{\delta}_n(Y) \cdot \int_{G/T} f(\text{Ad}(g)Y) \frac{dg}{dt}$$

is locally constant near X .

(2) Suppose the equivalent conditions in the last item holds. Then $\widehat{\delta}_n$ is represented by $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}}$ iff it is represented by

$$(14) \quad f \mapsto \sum_T \frac{1}{|W_T|} \int_{\mathfrak{t}^r} |D_{\mathfrak{g}}(Y)| \cdot \widehat{\delta}_n(Y) \left(\int_{G/T} f(\text{Ad}(g)Y) \frac{dg}{dt} \right) dY.$$

Proof. This follows immediately from Proposition 6.16, using that $\widehat{\delta}_n$ is $\text{Ad}(G)$ -invariant. $\square$

6.5. Scaling behaviours. We had seen in Corollary 3.39 that δ_n , which comes from Theorem C' and we have little idea what it is (still do in frontier research), is a linear combination of I_{O^*} . We need a scaling property about it. Recall for $f \in C_c^\infty(\mathfrak{g})$ and $t \in F^\times$, we define $f_{t^2} \in C_c^\infty(\mathfrak{g})$ by $f_{t^2}(X) = f(t^2X)$.

Lemma 6.18. The generalized function $\widehat{I_{O^*}}$ has weight $-\dim O^* + 2 \dim G$, i.e.

$$\widehat{I_{O^*}}(f_{t^2}) = |t|^{2 \dim G - \dim O^*} \cdot \widehat{I_{O^*}}(f), \quad \forall t \in F^\times, f \in C_c^\infty(\mathfrak{g}).$$

Proof. This follows from that $\widehat{f_{t^2}} = t^{2 \dim G} \cdot (\widehat{f})_{t^{-2}}$, Definition 3.38 and Corollary 3.39. $\square$

Lemma 6.19. The function $Y \mapsto \widehat{I_{O^*}}(Y)$ has weight $-\dim O^*$, in the sense that

$$\widehat{I_{O^*}}(t^2Y) = |t|^{-\dim O^*} \widehat{I_{O^*}}(Y), \quad \forall t \in F^\times, Y \in \mathfrak{t}^r.$$

Proof. The value $\widehat{I_{O^*}}(Y)$ is given by

$$\widehat{I_{O^*}}(Y) = \frac{\widehat{I_{O^*}}(1_{Y+\Lambda})}{\int_{\mathfrak{g}} 1_{Y+\Lambda}(X) dX}$$

for a sufficiently small lattice Λ . The lemma therefore follows from Lemma 6.18, that $(1_{Y+\Lambda})_{t^2} = 1_{t^{-2}Y+t^{-2}\Lambda}$, and that $\int_{\mathfrak{g}} 1_{t^{-2}Y+t^{-2}\Lambda}(X) dX = |t|^{-2 \dim G} \cdot \int_{\mathfrak{g}} 1_{Y+\Lambda}(X) dX$ $\square$

Example 6.20. The function t_π in Exercise 4 satisfies

$$t_\pi(\exp(t^2 Y)) = |t|^{-2} \cdot t_\pi(Y)$$

for $t \in F^\times$, $|t| \leq 1$, and $Y \in p^2 \mathfrak{gl}_2(\mathbb{Z}_p)$. Meanwhile, the only non-zero orbit in $\mathfrak{gl}_2^*$ has dimension 2.

Exercise 19. Let $\mathcal{C} = \mathbb{C}$, prove directly without using Theorem C' that the function t_π in Exercise 4 is locally- L^1 .

Exercise 20. (*) Again $\mathcal{C} = \mathbb{C}$ and we look at the representation $C_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ of $\mathrm{GL}_2(\mathbb{Q}_p)$. Prove (as in Theorem 4.13(2), but without using it or Theorem C') locally near the identity we have $\Theta_\pi \stackrel{*}{=} \widehat{I_{O^*}}$ where O^* is the unique non-zero nilpotent orbit in $\mathfrak{gl}_2(\mathbb{Q}_p)$.

Exercise 21. Suppose $\mathcal{C} = \mathbb{F}_2$ and p is odd, and we look at the representation $C_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ of $\mathrm{GL}_2(\mathbb{Q}_p)$. Show that its character as in Definition 1.10 is a non-zero generalized function.

(On the other hand, the function t_π as in Exercise 4 is identically 0 mod 2, i.e. the character is identically zero on G^{rs} .)

Therefore, by induction we may assume that (13) is locally- L^1 near any $X \in \mathfrak{t}$ with $Z_{\mathfrak{g}}(X) \subsetneq \mathfrak{g}$. We have seen in Proposition 5.10 that $\mathbb{G} = \mathbb{G}^{\mathrm{der}} \cdot Z(\mathbb{G})^\circ$. The proof also shows $\mathfrak{g} = \mathfrak{g}^{\mathrm{der}} \oplus Z(\mathfrak{g})$ where $\mathfrak{g}^{\mathrm{der}} = [\mathfrak{g}, \mathfrak{g}] = (\mathrm{Lie} \mathbb{G}^{\mathrm{der}})(F)$. It is routine to verify that $\widehat{\delta}_n|_{\mathfrak{g}^{\mathrm{rs}}}$ comes from $\mathfrak{g}^{\mathrm{der}}$, i.e. we can write a $\delta_n^{\mathrm{der}} \in D(((\mathfrak{g}^{\mathrm{der}})^*)^{\mathrm{nil}})^{G^{\mathrm{der}}}$ such that $\widehat{\delta}_n(Y^{\mathrm{der}}, Z) = \widehat{\delta}_n^{\mathrm{der}}(Y^{\mathrm{der}})$. Therefore we can reduce the problem from $\mathbb{G}$ to $\mathbb{G}^{\mathrm{der}}$, and therefore assuming that $Z(\mathfrak{g}) = 0$. In particular, it remains to show that (13) is locally- L^1 near 0.

Now we need a theorem closely related to Theorem 4.13(1).

Theorem 6.21. (Shalika germ expansion, strong form) There exists small enough lattice $\Lambda_{\mathrm{small}} \subset \mathfrak{g}$ and large enough lattice Λ_{big} satisfying the following property: For any $d \in D(\mathfrak{g})^G$ with $\mathrm{supp}(\Gamma) \subset \mathrm{Ad}(G)\Lambda_{\mathrm{small}}$, there exists a unique $d_n \in D(\mathfrak{g}^{\mathrm{nil}})^G$ such that

$$d(f) = d_n(f), \quad \forall f \in C_c(\mathfrak{g}/\Lambda_{\mathrm{big}}).$$

Remark 6.22. By an isomorphism $\mathfrak{g}^* \cong \mathfrak{g}$, Corollary 3.39 implies that $d_n = \sum_O \Gamma_O(d) I_O$ for $O \in \mathfrak{g}^{\mathrm{nil}}/G$. The distribution d_n or the functions $\Gamma_d(O)$ is called the **Shalika germ expansion** of d .

Remark 6.23. Note that Theorem 4.13 contains the weaker statement which asserts the existence of Λ_{big} a priori depending on δ . The original theorem of Shalika [Sha72, Theorem 2.1.1] asserted, for the special case $d = I_{\mathrm{Ad}(G)X}$ for $X \in \Lambda_{\mathrm{small}} \cap \mathfrak{g}^{\mathrm{rs}}$, the existence of such Λ_{small} a priori depending on f ; originally the $\Gamma_O(d) = \Gamma_O(X)$ were viewed as germs for X near the identity. In fact the special case is what we need for this section. [CC: We will prove the weaker version and sketch or prove (depending on time) the stronger Theorem 6.21 in a later section. That is hard.]

Remark 6.24. Given any choices of Λ_{small} and Λ_{big} , we can scale them both by any $a \in F^\times$. In particular, by choosing Λ_{big} arbitrarily large we can also make Λ_{small} arbitrarily large. The map $d \mapsto d_n$ is therefore canonical and independent of the choice of lattices.

Let $I_Y^{G/T} \in D(\mathfrak{g})^G$ be given by $I_Y^{G/T}(f) := \int_{G/T} f(\text{Ad}(g)Y) dY$ for $f \in C_c^\infty(\mathfrak{g})$. Let

$$(I_Y^{G/T})_n = \sum_{O \in \mathfrak{g}^{\text{nil}}/G} \Gamma_O(Y) I_O$$

be its Shalika germ expansion. Suppose $\phi \in C^\infty(\mathfrak{g}^{\text{rs}})^G$ is $\text{Ad}(G)$ -invariant. Then in (13) we are looking at

$$Y \mapsto D_{\mathfrak{g}}(Y) \cdot \widehat{\delta}_n(Y) \cdot \sum_O \Gamma_O(Y) I_O$$

for $Y \in \mathfrak{t}^*$ sufficiently close to 0.

Proposition 6.25. $\Gamma_O(Y)$, as a function of $Y \in \mathfrak{t}^*$, has weight $\{-\dim O \mid O \in \mathfrak{g}^{\text{nil}}/G\}$, i.e.

$$\Gamma_O(t^2 Y) = |t|^{-\dim O} \Gamma_O(Y), \quad \forall t \in F^\times, Y \in \mathfrak{t}^*.$$

Proof. Let f_{t^2} be the function given by $f_{t^2}(X) = f(t^2 X)$. Then from definition we have

$$I_{t^2 Y}^{G/T}(f) = I_Y^{G/T}(f_{t^2}).$$

Therefore, for all $0 < |t| \leq 1$, $Y \in \mathfrak{g}^{\text{rs}} \cap \Lambda_{\text{small}}$ and $f \in C_c(\mathfrak{g}/\Lambda_{\text{big}})$, we have

$$\begin{aligned} \sum_O \Gamma_O(t^2 Y) I_O(f) &= (I_{t^2 Y}^{G/T})_n(f) = I_{t^2 Y}^{G/T}(f) \\ &\parallel \\ \sum_O \Gamma_O(Y) I_O(f_{t^2}) &= (I_Y^{G/T})_n(f_{t^2}) = I_Y^{G/T}(f_{t^2}) \end{aligned}$$

However, by Corollary 3.37, for all $0 < |t| \leq 1$ the term $I_O(f_{t^2})$ has weight $-\dim O$. This proves the proposition if the data $\{(\sum_O \Gamma_O(Y) I_O(f))_{f \in C_c(\mathfrak{g}/\Lambda_{\text{big}})}\}$ determines $\Gamma_O(Y)$, i.e. there is no non-trivial linear combination of I_O that is trivial on all $f \in C_c(\mathfrak{g}/\Lambda_{\text{big}})$. Assume otherwise, then by scaling the assumed linear combination is trivial on all $f \in C_c^\infty(\mathfrak{g})$, contradicting to the definition of distributions. This finishes the proof. $\square$

6.6. Proof of Theorem B.

Theorem 6.26. Suppose Theorem C' holds for $\mathbb{G}$ and all reductive groups of smaller dimension. Then $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}} \in C^\infty(\mathfrak{g}^{\text{rs}})$ is locally- L^1 on $\mathfrak{g}$.

Proof. We had explained that we may assume $\mathbb{G}$ semisimple. By Corollary 6.17, it suffices to prove that

$$(15) \quad Y \mapsto D_{\mathfrak{g}}(Y) \cdot \widehat{\delta}_n(Y) \cdot \int_{G/T} f(\text{Ad}(g)Y) \frac{dg}{dt}$$

is locally- L^1 near 0 on $\mathfrak{t}$, and by induction on $\dim \mathbb{G}$ we know by the other direction of Proposition 6.16 that (15) is locally- L^1 near any non-zero element. Choose any lattice $\Lambda_{\mathfrak{t}} \subset \mathfrak{t}$ and $t \in F^\times$ with $0 < |t| < 1$. By compactness of $\Lambda_{\mathfrak{t}} \backslash t^2 \Lambda_{\mathfrak{t}}$ we have that (15) is L^1 on $\Lambda_{\mathfrak{t}} \backslash t^2 \Lambda_{\mathfrak{t}}$.

Let us compare (15) on $\Lambda_{\mathfrak{t}} \backslash t^2 \Lambda_{\mathfrak{t}}$ and $t^{2i} \Lambda_{\mathfrak{t}} \backslash t^{2i+2} \Lambda_{\mathfrak{t}}$ for any $i \in \mathbb{Z}_{\geq 0}$. We have

$$D_{\mathfrak{g}}(t^{2i} Y) = |t|^{(\dim G - \dim T) \cdot 2i}$$

By Lemma 6.19 we have

$$\widehat{\delta}_n(t^{2i}Y) \text{ is a polynomial in } |t|^{-2i} \text{ with degree } \leq \max(\dim O)$$

By Proposition 6.25 we have

$$(I_{t^{2i}Y}^{G/T})_n(f) \text{ is a polynomial in } |t|^{-2i} \text{ with degree } \leq \max(\dim O)$$

Also the measure of $\Lambda_t \backslash t^2 \Lambda_t$ and $t^{2i} \Lambda_t \backslash t^{2i+2} \Lambda_t$ differ by $|t|^{\dim T}$. Therefore, the L^1 -norm of (15) on Λ_t is a sum of terms that form a geometric series with ratio $\leq |t|^{2 \dim G - 2 \max \dim O}$. We have $\dim O < \dim G$; for $O = \text{Ad}(G)X_n$ we have $\dim O = \dim G - \dim Z_{\mathfrak{g}}(X_n) \leq \max(0, \dim G - 1)$ since either $X_n = 0$ or $X_n \in Z_{\mathfrak{g}}(X_n)$. Hence the geometric series converge, and (15) is locally- L^1 as asserted. $\square$

Theorem 6.27. *Suppose Theorem C' holds for $\mathbb{G}$ and all reductive groups of smaller dimension. Then $\widehat{\delta}_n$ is represented by $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}}$.*

Proof. Note that by Theorem 6.26, the function $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}} \in C^\infty(\mathfrak{g})$ does represent some generalized function, and we have to prove that this generalized function is indeed $\widehat{\delta}_n$. Again we may reduce to the case when $\mathbb{G}$ is semisimple. By induction and Corollary 6.17, we know that $\widehat{\delta}_n$ is represented by $\widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}}$ near every $X \in \mathfrak{g}$ with semisimple part $X_s \neq 0$. In other words, the difference of $\widehat{\delta}_n - \widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}}$ is supported on $\mathfrak{g}^{\text{nil}}$. By Lemma 6.18, the generalized function $\widehat{\delta}_n - \widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}}$ has weights $\{-\dim O^* + 2 \dim G\}$, while by Corollary 3.39 any $\text{Ad}(G)$ -invariant generalized function supported on $\mathfrak{g}$ has weight $\{-\dim O\}$. We have seen that $\dim G > \dim O$ (and also $\dim O^*$). Hence the weights never agree and therefore $\widehat{\delta}_n - \widehat{\delta}_n|_{\mathfrak{g}^{\text{rs}}} = 0$. $\square$

Exercise 22. (**) Suppose $\mathcal{C} = \mathbb{C}$. The **wave-front set** of an irreducible admissible $\mathbb{C}$ -representation is the support $\text{supp}(\delta_n) \subset (\mathfrak{g}^*)^{\text{nil}}$ where δ_n is as in Theorem C'(2) applied to the case $s = e$ (i.e. Theorem 4.13(2)). Likewise, the **wave-front set**¹⁴ of $\widehat{\delta}$ in Theorem C'(1) applied to the case $X_s = 0$ (i.e. Theorem 4.13(1)) is the support $\text{supp}(\delta_n) \subset (\mathfrak{g}^*)^{\text{nil}}$.

Suppose $\mathbb{G} = \text{GL}_n$. Consider any $X \in \mathfrak{gl}_n(F)$. Let $I_{\text{Ad}(G)X}$ be as in Theorem 3.11 or equivalently Corollary 5.32. Show that the wave-front set of $\widehat{I_{\text{Ad}(G)X}}$ is equal to the wave-front set of $\text{Ad}(G)X$ defined in Exercise 10.

(That this holds for arbitrary $\mathbb{G}$ when p is sufficiently large is an unpublished result of DeBacker and myself. I do not know if this is true for arbitrary $\mathbb{G}$ when p is small.)

6.7. Additional Remarks.

Remark 6.28. Our proof shows that the only “convergence” needed is the convergence of a bunch of geometric series, which is of course completely algebraic. Suppose we don't work with $\mathbb{C}$, but our general field $\mathcal{C}$ satisfying Hypothesis 4.1 and moreover that

- (1) We have $q^{2n} \neq 1$ in $\mathcal{C}$ for every $0 < n \leq \dim G$.
- (2) $\text{char}(\mathcal{C})$ does not divide any $|W_T|$.

¹⁴These are the p -adic analogue of that in https://en.wikipedia.org/wiki/Wave_front_set.

[CC: There is a better way to do this: I'll take some time to add it.] In this case, (14) defines inductively (on $\dim G$) the generalized function $\widehat{\delta}_n$ everywhere locally. That is, by induction we assume the problem is addressed for smaller groups. For any test function $f \in \mathcal{C}_c^\infty(\mathfrak{g})$ (technically, a test measure instead), the integral $I_Y^{G/T}(f)$ $X_i \in \mathfrak{g}^{\text{ss}}$. If $Z_G(X_i) \subsetneq G$, then by induction $\widehat{\delta}_n(f_i)$ is defined. Otherwise, we use (14) to pass to a finite number of $\mathfrak{t}$, and we need to define the “integral” over some $\Lambda_{\mathfrak{t}} \subset \mathfrak{t}$. We define it using the geometric series of integrals over $t^{2i} \setminus t^{2i+2} \Lambda_{\mathfrak{t}}$. To do that, we need a generalization of Shalika germ expansion as in [Kot05, Theorem 17.6].

Remark 6.29. Everything in §6 works with the more general setup that $\mathbb{G}$ is reductive, but not connected. The only thing we need is the notion of regular semisimple elements in $\mathbb{G}(F)$ and that they are dense. See [Kno, Moh03, Proposition 3.8].

7. LOCAL CHARACTER EXPANSIONS FOR $G = \text{GL}_2$

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